



A Review on Spectroscopic Techniques for Analysis of Nanomaterials and Biomaterials

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Abstract

Spectroscopy, the study of the interaction between electromagnetic radiation and matter, is a versatile and powerful analytical technique used in various scientific disciplines. This review provides a comprehensive overview of spectroscopy, covering its principles, instrumentation, techniques, applications, and recent advancements. Spectroscopy encompasses a wide range of methods, each offering unique insights into the structure, composition, dynamics, and properties of materials. At its core, spectroscopy relies on the measurement of the intensity and wavelength (or frequency) of electromagnetic radiation absorbed, emitted, or scattered by a sample. The interaction between light and matter can reveal valuable information about the chemical, physical, and electronic properties of substances, including molecular structure, electronic transitions, vibrational modes, and magnetic interactions. Thus, spectroscopy remains a cornerstone of scientific research and technological innovation, offering unparalleled capabilities for understanding the properties and behavior of matter across the electromagnetic spectrum. Continued advancements in spectroscopic instrumentation, techniques, and applications promise to further enhance our ability to explore and exploit the mysteries of the universe at the molecular and atomic levels.

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1. Introduction

Spectroscopy is a scientific technique used to study the interaction between matter and electromagnetic radiation across various wavelengths of the electromagnetic spectrum. It involves the measurement and analysis of how light is absorbed, emitted, or scattered by substances, providing valuable information about their composition, structure, and properties. By examining the unique patterns of absorption,

emission, or scattering of light by different materials, spectroscopy allows researchers to identify substances, determine their chemical composition, understand molecular structures, and investigate molecular dynamics. Spectroscopic techniques are widely used in diverse fields such as chemistry, physics, biology, materials science, environmental science, and astronomy for research, analysis, diagnostics, and exploration.

Therefore, we understand that spectroscopic techniques are indispensable tools for analyzing nanomaterials and biomaterials, offering valuable insights into their composition, structure, properties, and interactions. Ultraviolet (UV)-Visible spectroscopy measures the absorption of ultraviolet, visible, and near-infrared light, providing information about electronic transitions and optical properties. Fluorescence spectroscopy characterizes materials based on their emission of light upon excitation, aiding in the study of fluorescent nanoparticles and biomolecular interactions. Raman spectroscopy analyzes light scattering to reveal molecular vibrations and surface properties, facilitating the identification

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of functional groups and crystallinity. X-ray photoelectron spectroscopy (XPS) provides details about elemental composition and chemical state by measuring the kinetic energy of emitted electrons. Fourier transform infrared spectroscopy (FT-IR) identifies molecular vibrations and functional groups, aiding in bio-molecular characterization and surface analysis. Nuclear magnetic resonance spectroscopy (NMR) offers insights into atomic environments, chemical bonding, and molecular dynamics, facilitating structural elucidation and material characterization. Surface-enhanced Raman spectroscopy (SERS) enhances Raman signals, enabling sensitive detection of trace analytes and surface interactions. These spectroscopic techniques collectively would allow researchers to comprehensively understand nanomaterials and biomaterials, advancing fields such as materials science, chemistry, biology, and medicine.

Obviously, electromagnetic radiation forms visible light, but a tiny part may be considered as a simple harmonic wave propagated from a source and traveling in monochromatic lines except when reflected. This property is related to a physical displacement of a stretched string vibrating in the atmosphere during the passage of sound waves that are interconnected with electric and magnetic fields. Any simple harmonic wave has properties of the sine wave, defined by $y = A \sin \Theta$. Here, y is the displacement with the maximum value, *i.e.*, amplitude A , and θ is the angle varying from 0 to 360. The various areas of the electromagnetic spectrum have different properties, and therefore, regions in the electromagnetic spectrum are illustrated below:

a) Radiofrequency region: This region ranges from 3×10^6 to 3×10^{10} Hz, *i.e.*, 10 m to 1 cm wavelength. The change in energy arises from the reversal of spin of a nucleus or electron

in the order of 0.001-10 J/m. In this range, Nuclear magnetic resonance and Electron spin resonance work.

b) Microwave region: This region ranges from 3×10^{10} to 3×10^{12} Hz, *i.e.*, 1 cm to 100 micrometer wavelength. Rotational phenomenon in spectroscopic studies takes place in this range.

c) Infra-red region: This region ranges from 3×10^{12} to 3×10^{14} Hz, *i.e.*, 100 nm to 1 mm wavelength. From a chemical point of view, this is one of the most important regions in the electromagnetic spectrum. The separation between levels is some 10^4 joules/mole.

d) Visible and ultraviolet region: This region ranges from 3×10^{14} to 3×10^{16} Hz, *i.e.*, 1 mm to 100 nm. In this region, electronic transition takes place. The separation between the energies of valence electrons is some hundreds of kilojoules per mole.

e) X-ray region: This region lies in the range from 3×10^{16} to 3×10^{18} Hz, *i.e.*, 10 nm to 100 pico-meter in wavelength. The energy changes in this region of inner electrons of an atom or a molecule maybe 10 thousand kilojoules.

f) γ -ray region: This region lies in the range from 3×10^{18} to 3×10^{20} Hz, *i.e.*, 100 pico-meter to 1 pico-meter in wavelength. The energy changes involve the rearrangement of nuclear particles, which have energies of 10^9 to 10^{11} joules per gram atom.

g) Radiofrequency region: Nucleus and electrons are considered tiny charged particles, and it follows that their spin is associated with a tiny dipole moment. The reversal of this dipole consequent upon the spin reversal can interact with the magnetic field of electromagnetic radiation at the appropriate frequency. Thus, all such spin reversal produces an absorption and emission spectrum.

h) Microwave region: If a molecule has an electronegativity difference, it has a permanent dipole moment that bears positive and negative charges on the atoms, such as HCl. Here, hydrogen bears a partial positive charge, and chlorine bears a partial negative charge and is considered to have a permanent dipole moment. Also, the N_2 and O_2 are homo-nuclear molecules, do not bear any electronegativity difference, and ultimately have no permanent dipole moment, *i.e.*, zero dipole moment. If we consider rotation in the molecules having permanent dipole moment, we observe that pure rotation takes place, and the center of gravity of the molecule does not change. All the molecules having permanent dipole moments are microwave-active. If there is no dipole, as in the case of N_2 or O_2 , no interaction can take place, and the molecule is microwave inactive. This imposes a limitation on the applicability of microwave spectroscopy.

i) The visible and ultra-violet region: The excitation of a valence electron involves the movement of electronic charges in the molecule. The consequent change in the electric dipole gives rise to a spectrum by its interaction with the undulatory electric field of radiation. So far we have seen that spectroscopy deals with the interaction of electromagnetic radiation with matter. In most cases, the light is derived from

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the theory of classical electromagnetism. In classical electromagnetic theory light is described as coupled electric and magnetic field propagating through the path as a traveling waves. However, this wave theory is not sufficient to explain the properties of light at very low intensity. For the justification of properties of light at very low intensity, another theory called quantum theory was proposed. Quantum theory states that light consists of discrete packets of energy called photons. However, neither a classical wave model nor a classical particle model could correctly describe the nature of light. Therefore, the nature of light can be explained with wave-particle duality. The wave particle duality is shared by all the primary constituents of instance electrons have both wave particle duality in nature. Since the mid-20th century, a more comprehensive theory of light, known as quantum electrodynamics, is considered complete. Quantum electrodynamics combines the ideas of electromagnetism, quantum mechanics, *etc.*

1.1 Theories of light from historical perspectives

1.1.1 Ray theory in the ancient world

We are quite aware that mirrors, concave lenses, and convex lenses have been used by philosophers since ancient times. The conceptual hurdle of distinguishing the human perception of visual effects from the physical nature of light hampered the development of theories of light. Contemplation of the mechanism of vision dominated these early studies. Pythagoras was a great thinker and philosopher. He believed that sight is due to visual rays emanating from the eyes and striking objects. Empedocles was a great Greek philosopher. He believed that vision is due to the emission of light by the object and the eyes. Epicurus was another thinker and philosopher. He proposed that light is emitted from a source other than the eye, and vision is produced when light is reflected off the object and enters the eye. Euclid was another thinker, philosopher, and mathematician. He proposed the concept of reflection and discussed the propagation of light waves in straight lines. Ptolemy explained the phenomenon of refraction of light. The speed of light in air or vacuum is 3×10^8 m/s. This is the fastest speed an object can travel in the universe. If we compare this with the speed of sound, then we see the speed of sound is only 300m/s. This is why we see lightning first and then sound during a thunderstorm.

1.1.2 Light as electromagnetic waves

Wave can be defined as any disturbance made in the medium that propagates through space. Most of the waves require some of the supporting media for propagation. The time dependence of displacement at any single point in the space often oscillates about some equilibrium position. For example, sound waves travel through the medium of air, and disturbance is the small collective displacement of air molecules. Therefore, it has properties of waves such as ocean waves, and light waves have crests and troughs. The distance between two successive crests or troughs is known as wavelength. Frequency can be defined

as the number of waves passing from a point in one second. The velocity of light is the product of the wavelength of light and frequency of light. The different colors of light have different frequencies and wavelengths. Light can pass straight lines through some objects but not all. The objects or materials through which light can pass are known as transparent materials. The objects or materials that do not allow light to pass through them are known as opaque and partially known as translucent materials.

1.2 Historical developments in spectroscopic techniques

The history of spectroscopic techniques dates back to the early 19th century, with significant advancements made over the years in understanding the interaction between matter and electromagnetic radiation. Here a brief overview of the key milestones in the history of spectroscopic technique is furnished as below.

1.2.1 Prism spectroscopy (17-18th century)

The earliest observations of spectroscopic phenomena can be traced back to experiments conducted by Isaac Newton in the late 17th century. He demonstrated that white light could be separated into its component colors using a prism, revealing the spectrum of visible light. A prism is an opaque triangular device that is made of plastic or glass. When light passes through the prism, the light bends due to a phenomenon called refraction. Refraction can be defined as the phenomenon of bending of light when it passes from one medium into another medium. An ordinary triangular prism can separate white light into its constituent colors called spectrum. Light is emitted from a source such as a vapor lamp. A slit selects the thin strip of light which passes through the collimator, where it gets parallelized. The aligned light then passes through the prism which is refracted twice.

1.2.2 Spectral analysis (19th century)

In the early 19th century, Joseph von Fraunhofer made significant contributions to spectroscopy by systematically studying the absorption lines in the solar spectrum. He also discovered hundreds of dark lines,^[1] known as Fraunhofer lines, which are caused by the absorption of specific wavelengths of light by elements in the Sun's atmosphere. Gustav Kirchhoff and Robert Bunsen further advanced the field of spectroscopy in the mid-19th century. They developed the technique of flame emission spectroscopy, which involves heating samples in a flame to produce characteristic emission spectra. Their work laid the foundation for quantitative analysis of chemical elements based on their spectral signatures.^[2-8]

1.2.3 Quantum mechanics and atomic spectra (Early 20th century)

The development of quantum mechanics in the early 20th century revolutionized our understanding of atomic structure and spectral lines. Scientists such as Niels Bohr, Max Planck,

and Werner Heisenberg formulated theories to explain the discrete energy levels of atoms and the quantized nature of electromagnetic radiation. The Bohr model of the atom, proposed by Niels Bohr in 1913, provided a theoretical framework for understanding atomic spectra and the transitions of electrons between energy levels.^[9]

1.2.4 Modern spectroscopic techniques (20th-21st century)

Throughout the 20th and 21st centuries, spectroscopic techniques have continued to evolve with technological advancements and instrumentation. Developing techniques such as infrared (IR) spectroscopy, Raman spectroscopy, nuclear magnetic resonance spectroscopy, and X-ray spectroscopy has expanded the range of applications in fields such as chemistry, physics, biology, materials science, and medicine. Innovations in detectors, lasers, optics, and computational methods have enabled researchers to achieve higher sensitivity, resolution, and versatility in spectroscopic measurements.^[10,11]

1.3 Researcher's contribution in the development of spectroscopy

The development of spectroscopic techniques involves contributions from numerous scientists and researchers over several centuries. Thus, it is difficult to mention the scientist who developed the methods. However, an effort has been made to identify some important personalities with the lion's share in the design and development of spectroscopic methods.

1.3.1 Sir Isaac Newton (1642-1727)

The fact is that Newton was not the inventor of the spectroscopic technique, but he conducted groundbreaking experiments in optics, including his famous prism experiments demonstrating the dispersion of white light into its component colors. Sir Isaac Newton proposed the idea of developing spectroscopy based on his concept of optic experiment.^[12]

1.3.2 Gustav Kirchhoff (1824-1887) and Robert Bunsen (1811-1899)

These two essential scientists pioneered the development of flame spectroscopy in the mid-19th century. Their work led to the development of the Bunsen burner and using flame tests to identify and quantify elements based on their characteristic emission spectra.

1.3.3 Lord Rayleigh (1842-1919)

Rayleigh is another important personality who made significant contributions to understanding scattering phenomena, including Rayleigh scattering. Rayleigh scattering effectively explains the blue color of the sky. His work laid the foundation for the study of light scattering in spectroscopy.^[13]

1.3.4 Niels Bohr (1885-1962)

The famous Bohr's atomic model was proposed in 1913 by

Neils Bohr.^[14] His work provided a theoretical framework for understanding atomic spectra and the quantized nature of energy levels in atoms. His work laid the foundation for quantum mechanics and spectroscopic studies of atomic and molecular systems.

1.3.5 Albert Einstein (1879-1955)

Einstein's theory of the photoelectric effect, published in 1905, helped explain the interaction of light with matter at the atomic level. For this critical discovery, he was awarded the Nobel Prize in Physics.^[15] This landmark discovery laid the foundation of modern spectroscopy.

1.3.6 Jaroslav Heyrovský (1890-1967)

In 1922, J. Heyrovský invented the polarography technique based on the measurement of current flowing through an electrochemical cell. This laid the foundation for electrochemical spectroscopic techniques.^[16]

These are just a few examples of the many scientists who have contributed to the invention and development of spectroscopic techniques over the centuries. Spectroscopy continues to evolve with technological advancements and remains a vibrant field of scientific research with applications across numerous disciplines.

Thus, the history of spectroscopic techniques reflects the gradual progression from simple observations of spectral phenomena to sophisticated methods for analyzing the composition, structure, and properties of matter across a wide range of disciplines.

1.4 Classification and types of major spectroscopic techniques

The spectroscopic techniques can be classified as optical spectroscopy, electron spectroscopy, magnetic resonance spectroscopy, mass spectroscopy, thermal spectroscopy, vibrational spectroscopy, optical emission spectroscopy, *etc.*

As we have discussed above, there are various types of spectroscopy. The instrumentations and principles of working vary accordingly. However, all spectroscopic techniques have some basic instrumentation, which is common. Here, an attempt is made to discuss typical instrumentation in the spectroscopic techniques:

a) Absorption instrumentation: The radiation from a white source is directed by some guiding device onto the sample, from which it passes through an analyzer, which selects the frequency that reaches the detector at any given time. The signal from the analyzer passes to a recorder, which is synchronized with the analyzer to produce a trace of absorbance as the frequency varies. A modulator is often placed between the sample and the analyzer. The role of the modulator is to manipulate the properties of light or other electromagnetic radiation to exact specific information from a sample. The types of modulators are as follows: (1) Optical modulator: This type of modulator can modify the properties of light, such as intensity, phase, or polarization. (2) Frequency

Modulator: This modulator can regulate the intensity of the radiation to optimize the signal-to-noise ratio. (3) Polarization Control: Here, the modulators adjust the polarization state of light to study sample properties. (4) Frequency Scanning: the modulators sweep the frequency or wavelength of the radiation to acquire spectral information. (5) Signal Enhancement: the modulators amplify or suppress specific signals to improve sensitivity. Thus, modulators play a vital role in spectroscopy by controlling the properties of radiation, enabling the extraction of specific information from samples, and enhancing the overall performance of spectroscopic techniques. In the microwave and radiofrequency (RF) regions, it is possible to construct a monochromatic source whose emission frequency can be varied over the range.

b) Emission instrument: Emission instrumentation in spectroscopy refers to the device and systems used to measure the emission spectra of atoms, molecules, or ions. The components of emission spectra are as follows: (1) Source: Some suitable excitation source to energize the sample, such as a laser, lamp, or plasma, is required. (2) Monochromator: The role of a monochromator is to select a specific wavelength or frequency of radiation. (3) Sample Chamber: The sample chamber is the place to house the sample under study. This is often provided with temperature and pressure control. (4) Detector: This is the device that measures the emitted radiation, *i.e.*, photomultiplier tubes (PTMs) or charge-coupled devices (CCDs), and (5) Amplifier and Processor: This enhances and processes the signal.

Many instruments are developed based on the emission mechanism, such as atomic emission spectroscopy, optical emission spectroscopy, inductively coupled plasma emission spectroscopy, laser-induced breakdown spectroscopy, and fluorescence spectroscopy.

2. Optical spectroscopy

The principle of optical spectroscopy revolves around the interaction between matter and electromagnetic radiation within the optical range of the electromagnetic spectrum, typically spanning from ultraviolet to visible and near-infrared wavelengths. This interaction involves light absorption, emission, or light scattering by atoms, molecules, or materials. This leads to characteristic spectral features that can be analyzed to extract information about the sample's composition, structure, and properties.^[17-20]

At its core, optical spectroscopy relies on the quantized nature of energy levels in atoms and molecules. Once electromagnetic radiation interacts with matter, it induces a transition between two different energy levels, resulting in the absorption or emission spectra. This phenomenon is governed by the laws of quantum mechanics, particularly the selection rules that dictate which transitions are allowed or forbidden based on the quantum numbers of the states involved.

In optical spectroscopy, the absorption of light by a sample is typically described by Beer-Lambert's law, which states that the absorbance of light is proportional to the concentration of

the absorbing species, the path length of the light through the sample, and the molar absorptivity of the species at a given wavelength. By measuring the absorbance of light at different wavelengths, one can obtain an absorption spectrum that reveals the characteristic absorption bands or peaks associated with specific molecular or electronic transitions.

In emission spectroscopy, the intensity of light emitted is measured. This light, which may occur instantaneously or with a delay, contains information about the energy levels, electronic states, and chemical environment of the emitting species. There are two main types of spectroscopy: fluorescence and Phosphorescence spectroscopy. These two spectroscopic techniques are common types of emission spectroscopy, distinguished by their respective mechanisms of excited state relaxation and lifetime.

The Scattering spectroscopic technique examines the scattered light from interacting with particles or structures within the sample. Raman spectroscopy is a well-known example of scattering spectroscopic techniques. It analyzes the inelastic scattering of the photon by molecular vibration and provides valuable information about chemical bonding, molecular structure, crystallinity, *etc.*

The principle of optical spectroscopy lies in the analysis of the interaction between light and matter, enabling the characterization and analysis of samples across a wide range of scientific disciplines, including chemistry, physics, biology, materials science, and environmental science.^[21]

2.1 Ultraviolet-visible spectroscopy

Ultraviolet-visible (UV-Vis) spectroscopy has a rich historical development, beginning with Sir Isaac Newton's pioneering scientific contribution to the dispersion of light through prisms in the 17th century. However, the modern era of UV-Vis spectroscopy began in the early 20th century with the development of electronic components and advancements in optics.

In 1930, the first commercial UV-Vis spectrophotometer was launched by the Beckman Instruments Company, marking a significant milestone in the field. Throughout the mid-20th century, UV-Vis spectroscopy became an important scientific tool widely in various scientific disciplines, including chemistry, biochemistry, materials science, veterinary medicine, pharmaceutical sciences, and environmental science, due to its versatility and utility in analyzing the electronic transitions of molecules. The discovery of Beer-Lambert's law in the 19th century laid the theoretical foundation for quantitative analysis using UV-Vis spectroscopy, allowing researchers to determine the concentration of analytes in solution based on their UV or visible light absorption. This principle is still widely used in various applications such as pharmaceutical analysis, environmental monitoring, and quality control in manufacturing processes.

Modern, sophisticated UV-Vis spectrophotometers are manufactured using advanced technologies, including

computer-controlled systems, solid-state light sources, and high-resolution detectors. The modern instrument shows enhanced sensitivity, accuracy, and automation capabilities, allowing for rapid and precise measurements across a wide range of wavelengths. In addition, modern UV-Vis spectrophotometers often feature software packages for data analysis, spectral processing, and instrument calibration, enhancing their usability and versatility.

The scope of UV-Vis spectroscopy continues to expand with advancements in nanotechnology, materials science, and biotechnology, driving innovation and interdisciplinary research. UV-Vis spectroscopy is now used not only for qualitative and quantitative analysis but also for studying molecular interactions, characterizing nanoparticles, monitoring chemical reactions in real time, and exploring the electronic properties of advanced materials.

2.1.1 Instrumentation of UV-Vis spectroscopy

UV-Vis spectroscopy instrumentation typically consists of several key components designed to generate, manipulate, and detect UV and visible light, as well as analyze the resulting spectra. Here's an overview of the instrumentation commonly found in UV-Vis spectrophotometers, and the schematic diagram is shown in Fig. 1.

a) Light source: Light sources are crucial components for spectrometers and spectrophotometers as they provide the necessary energy needed for excitation that spectral transition requires. UV-Vis spectrophotometers are facilitated with a light source that emits UV and visible light over a specific wavelength range. The commonly used sources of light are tungsten-halogen lamps for visible light and deuterium lamps for UV light. Some instruments may also incorporate xenon arc lamps or light-emitting diodes (LEDs) for specific applications or extended wavelength coverage. (1) Deuterium Lamp: A gas discharge lamp that emits a continuous spectrum of light in near ultraviolet region from 150 nm to 400 nm. It is a stable source in the ultraviolet range but requires a complex power supply. (2) Tungsten or Halogen Lamp: A lamp that emits light over the entire visible range and part of the UV spectrum. Tungsten lamps are similar to high-intensity lamps used for domestic purposes. Halogen lamps are cheap and have a long service life. Apart from these two sources of light, there may be some other types of sources of light, such as Xenon, Mercury, *etc.*

b) Monochromator: The monochromator is a crucial component that selects a narrow range of wavelengths from the broadband light source. It typically consists of diffraction gratings or prisms that disperse light into its component wavelengths, allowing the user to select the desired wavelength or wavelength range for analysis. Two types of UV-Vis spectrophotometers are available: single monochromator and double monochromator. The monochromator is placed between the light source and the sample. Monochromators are tunable optical bandgap filters, typically based on diffraction gratings, used for various

applications such as spectroscopy.

c) Sample compartment: The sample compartment houses the sample to be analyzed. The sample containers are designed to hold various types of sample containers, such as cuvettes, microplates, or solid sample holders. The compartment may include temperature control features to maintain sample stability during analysis. A cuvette is a type of sample holder for liquid samples. The material the cuvette is made from will also have a UV-Vis absorbance cut. Glass will absorb all of light higher in energy starting at about 300 nm, so if the sample absorbs in the UV, quartz will be more practical as the absorbance cut off is around 160 nm for quartz. In the sample compartment, the sample is positioned to allow the beam from the monochromator to pass through the sample. The liquid sample would typically be held in the cuvette with a known fixed path length for absorbance measurement. A cuvette is a rectangular liquid holder made from glass, quartz, and plastic, which can transmit UV-Vis radiations. Solid samples can be placed for sample transmission measurement. It can be measured at various angles of incidence. Light can also be taken out of the sample compartment using fiber optics. Fiber optics are helpful when measuring large, hot, cold, radioactive, and dangerous samples. The fiber optics can take light from the spectrophotometer through a fiber optic probe to measure the solution outside the sample compartment.

d) Detector: The detector measures the intensity of light transmitted through or reflected from the sample across the selected wavelength range. Thus, detector can be scientifically considered as a light intensity sensor. To detect the intensity of light Photodiode arrays, or charge-coupled devices are commonly used as detectors in UV-Vis spectrophotometers. These detectors convert the optical signal into an electrical signal that can be processed and analyzed by the instrument's electronics.

e) Optical system: The optical system consists of lenses, mirrors, and filters that direct and focus light onto the sample and detector. It is designed to minimize optical aberrations and ensure accurate and reproducible measurements. The types of optical systems can be Single Beam, Double Beam, and Split Beam.

f) Electronics and data processing: Modern UV-Vis spectrophotometers are equipped with sophisticated electronics and software for data acquisition, processing, and analysis. These systems control instrument functions, record spectral data, perform baseline correction, and calculate sample concentrations or other analytical parameters based on established algorithms. The electronic components are Detectors. The data processing technique includes (1) signal averaging, which reduces noise by averaging multiple scans; (2) smoothing, which removes high-frequency noise; (3) baseline corrections, which subtracts the background signal; (4) data normalization, which scales data to compare spectra, (5) peak peaking which identifies and quantifies spectral peaks. The Data analysis software includes (1) Spectroscopy software such as Origin, Excel and Grams, *etc.*, (2)

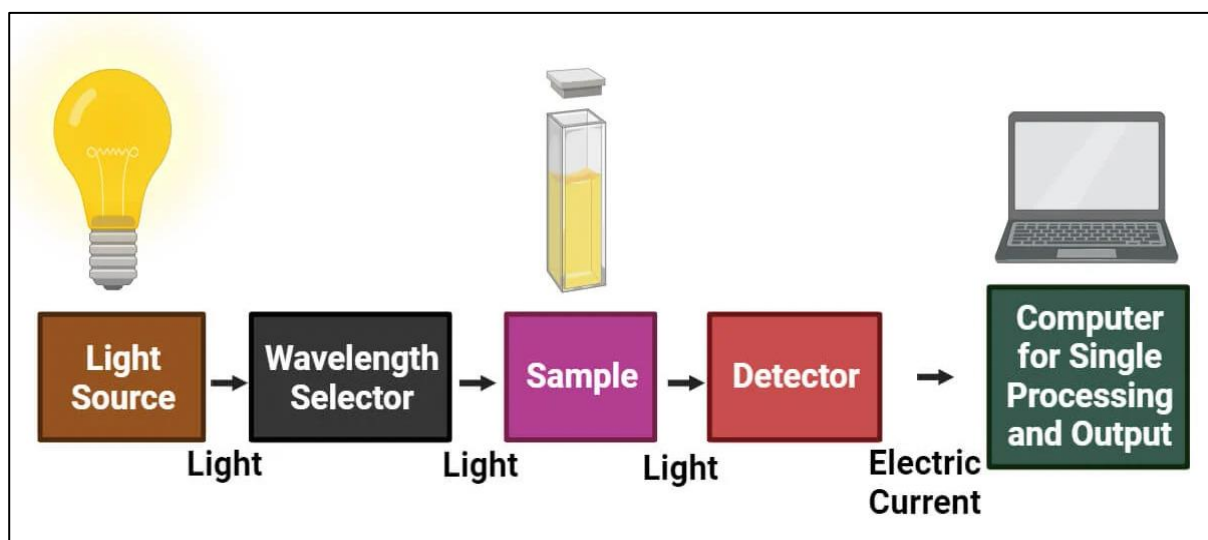


Fig. 1: A pictorial representation of UV-Vis Spectroscopy.^[22] (Copyright 2023 Kingfa scientific and technological Co. Ltd.)

Chemometric software such as Soft Independent Modeling of Class Analogy (SIMCA) and MATrix LABoratory (MATLAB), (3) Instrument-specific software such as Shimadzu UV Probe. Data processing algorithms such as (1) Fast Fourier transform, which enhances spectral resolution; (2) Principle component analysis, which reduces data dimensionality; (3) Partial least square regression; (4) Artificial neural network: predictive modeling.

g) User interface: UV-Vis spectrophotometers feature user-friendly interfaces for instrument operation and data analysis. These may include touchscreen displays, software applications, or computer interfaces that allow users to control instrument parameters, select measurement modes, and visualize spectral data in real time.

2.1.2 Principle of UV-Vis spectroscopy

UV-Vis spectroscopy is based on the principle of measuring the absorption or transmission of UV and visible light by molecules or materials. When light passes through or is reflected from a sample, the amount of light absorbed is the difference between the incident radiation and transmitted radiation. The amount of light absorbed is expressed as absorbance. Transmittance, or light that passes through a sample, is usually given in terms of fraction 1 or as a percentage. For most applications, absorbance values are used since the relationship between absorbance and both concentration and path length is usually linear. The principle involves several key components:

a) Electromagnetic radiation: UV-Vis spectroscopy utilizes electromagnetic radiation in the UV and visible regions of the electromagnetic spectrum. UV radiation has shorter wavelengths and higher energy than visible light, while visible light has longer wavelengths and lower power. UV and visible light can interact with molecules, causing electronic transitions.

b) Electronic transitions: Molecules absorb UV and visible light through electronic transitions, where electrons move

from lower energy (ground) states to higher energy (excited) states. The energy difference between the ground and excited states corresponds to specific wavelengths of light, which are absorbed by the molecule. The absorbed energy promotes electrons to higher energy levels, producing electronic excitation.

c) Absorption spectra: UV-Vis spectroscopy measures the absorption of light by a sample across a range of wavelengths. The absorption spectrum represents the intensity of absorbed light as a function of wavelength. Each molecule or compound has characteristic absorption bands in the UV-Vis spectrum, which arise from specific electronic transitions associated with molecular structure and bonding.

d) Beer-Lambert law: The Beer-Lambert law describes the relationship between the absorbance of light by a sample, the concentration of the absorbing species, the path length of the light through the sample, and the molar absorptivity of the species at a given wavelength. According to this law, absorbance (A) is proportional to the concentration (C) and path length (l), as expressed by the equation $A = \epsilon cl$, where ϵ is the molar absorptivity (also known as the extinction coefficient) of the species. The mathematical equation for Beer-Lambert's Law is mentioned below: "When a beam of monochromatic light is passed through a homogeneous absorbing medium, then the rate of decrease of intensity of radiation with the thickness of absorbing medium is directly proportional to the intensity of radiation as well as the concentration of the solution."

$$-\frac{dI}{dx} = kIc$$

$$-\frac{dI}{I} = kdx$$

Integrating the above equations, we get,

$$-\int_{I_0}^I \frac{dI}{I} = kc \int_0^x dx$$

$$-\log_e \frac{I}{I_0} = kcx$$

$$\log_e \frac{I}{I_0} = -kcx$$

$$I = I_0 e^{-kcx} \quad (1)$$

Equation (1) is known as Beer – Lambert's equation. Beer Lambert's law can also be mathematically represented in another way, as mentioned in equation (2).

$$A = \varepsilon cl \quad (2)$$

where A is the absorbance, ε is the molar absorptivity or extinction coefficient ($L/mol \times cm$), c is the concentration of the absorbing species (mol/L), and l is the path length through which the light passes (cm).

(e) Sample preparation parameters affected: For Beer-Lambert's Law to hold, the concentration should be within a range where the absorbance is linearly related to concentration. Too high a concentration can lead to deviations due to chemical interactions or scattering. Thus, accurate dilutions are often necessary to bring the sample within the optimal range. This involves preparing stock solutions and performing serial dilutions. Another parameter, path length, is defined by the width of the cuvette used in the spectrophotometer. Standard cuvettes have a path length of 1 cm, but depending on the concentration and absorbance range, different path lengths might be chosen. Ensuring that the cuvette is properly placed in the spectrophotometer is crucial to avoid any misalignment that could affect the path length. Wavelength Selection is also an important parameter, so choosing the correct wavelength at which the absorbance measurement is taken is crucial. This is usually the wavelength where the analyte has maximum absorbance (λ_{max}). The molar absorptivity should be determined using standards with known concentrations to ensure measurement accuracy.

(f) Practical considerations in sample preparation: The solvent used in sample preparation should not absorb significantly at the measurement wavelength. It should also be compatible so as not to cause any chemical reactions or precipitations. The sample should be free of particulates, bubbles, or turbidity, as these can scatter light and lead to erroneous absorbance readings. Filtration or centrifugation might be necessary. Sufficient volume is required to fill the cuvette adequately, ensuring consistent path length and interaction with the light beam. The sample should be thoroughly mixed to ensure homogeneity, preventing concentration gradients that could affect absorbance readings. Some analytes have absorbance that is temperature-dependent. Maintaining a constant temperature during measurements is important for accuracy.

2.1.3 Applications of UV-Vis spectroscopy

UV-Vis spectroscopy finds various applications across various scientific disciplines and industries due to its versatility, sensitivity, and ease of use. Some key applications are as described below:

a) Quantitative analysis: UV-Vis spectroscopy is widely used for quantitative analysis of analytes in solution. Beer-Lambert's law relates the absorbance of light to the

concentration of the absorbing species, allowing for the determination of concentrations of substances such as drugs, biomolecules, environmental pollutants, dyes, forensic materials, and industrial chemicals. Quantitative analysis involves comparing the absorbance of an unknown sample to that of known standards. The principle behind this method lies in the fact that the absorbance of light at a specific wavelength is directly proportional to the concentration of the absorbing substance. UV-Vis spectrometer is equipped with a monochromator, which allows one to select the desired wavelength, making it possible to target specific components in the sample.

b) Qualitative analysis: UV-Vis spectroscopy is employed for qualitative analysis, enabling the identification of unknown compounds based on their characteristic absorption or transmission spectra. Every substance is composed of molecules, which in turn consist of atoms arranged in a specific configuration. Each molecule exhibits its unique absorption spectra, characterized by position and intensities of absorption peaks. Determining molecular structure through UV-Vis spectrophotometry is more challenging than quantitative or qualitative analysis. The molecular weight of the compound can be measured spectrophotometrically by preparing the suitable derivatives of the compounds. For instance, if we want to determine the molecular weight of amine, it is converted into amine picrate. Then, the known concentration of amine picrate is dissolved in one liter of solvent, and its optical density is measured at 380nm. This application is particularly useful in fields such as pharmaceuticals, veterinary medicine, forensics, and environmental monitoring.

c) Kinetic studies: UV-Vis spectroscopy is widely successfully and effectively used to monitor chemical reactions in real-time by measuring changes in absorbance or transmission as a function of time. Kinetic studies can provide valuable insights into reaction mechanisms, reaction rates, and reaction kinetics, aiding in the development and optimization of chemical processes.

d) Protein and nucleic acid analysis: UV-Vis spectroscopy is utilized in the analysis of proteins and nucleic acids, such as DNA and RNA. Proteins and nucleic acids exhibit characteristic absorbance in the UV region due to the presence of aromatic amino acids and nucleotide bases. UV-Vis spectroscopy is employed to quantify concentrations, assess purity, and determine the structural features of biomolecules.

e) Environmental monitoring: UV-Vis spectroscopy is widely used in environmental monitoring to analyze water quality, air pollution, and soil contaminants. It can detect presence and actual concentration of pollutants such as heavy metals, organic compounds, and pesticides based on their UV-Vis absorption spectra, aiding in environmental risk assessment and regulatory compliance.

f) Material characterization: UV-Vis spectroscopy is used to characterize the optical properties of materials, including semiconductors, nanoparticles, polymers, and dyes. It

provides information about bandgap energy, electronic transitions, and molecular structure, which are essential for material design, development, and quality control.

g) Pharmaceutical analysis: UV-Vis spectroscopy is extensively used in the pharmaceutical industry for drug formulation, quality control, and stability testing. It enables the quantification of active pharmaceutical ingredients (APIs), detection of impurities, and assessment of drug degradation products, ensuring the safety, efficacy, and quality of pharmaceutical products.

Overall, UV-Vis spectroscopy is a versatile analytical technique with widespread applications in chemistry, biology, environmental science, materials science, pharmaceuticals, and many other fields. It contributes to scientific research, industrial processes, and regulatory compliance.

2.1.4 Limitations of UV-Vis spectroscopy

Though UV-Vis spectroscopy is a powerful and widely used analytical technique, it does have some limitations:

a) Limited sensitivity: UV-Vis spectroscopy may lack sensitivity for samples with low concentrations or weak absorbance. Detection limits can be challenging, particularly for trace analysis or when dealing with samples with low molar absorptivity.

b) Interference between solvents and matrix effects: Solvents and matrix effects can interfere with UV-Vis measurements, particularly if they absorb strongly in the UV-Vis range. This can complicate quantitative analysis and lead to inaccuracies in concentration determinations.

c) Limited information about molecular structure: UV-Vis spectroscopy can provide valuable information about electronic transitions and molecular bonding, but it may not offer detailed insights into molecular structure or conformational changes. Additional techniques, such as infrared spectroscopy or nuclear magnetic resonance spectroscopy, may be needed for a comprehensive structural analysis.

d) Limited chemical specificity: UV-Vis spectroscopy provides limited chemical specificity, mainly when dealing with complex mixtures or overlapping absorption bands. It may be unable to distinguish between different chemical species that absorb at similar wavelengths, leading to potential misinterpretation of results.

e) Limited wavelength range: UV-Vis spectroscopy is limited to the UV and visible regions of the electromagnetic spectrum, typically ranging from 190 to 800 nanometers. This restricts its applicability for studying electronic transitions outside this wavelength range, such as in the infrared or far-ultraviolet regions.

f) Sample compatibility: UV-Vis spectroscopy requires samples to be in solution or sufficiently transparent to UV-Vis light. Solid samples or samples with high turbidity may require additional sample preparation steps or alternative techniques for analysis.

g) Lack of structural information: While UV-Vis

spectroscopy can provide information about electronic transitions, it keeps silent about detailed structural information about molecular geometry, bond angles, or stereochemistry. Techniques such as X-ray crystallography or nuclear magnetic resonance spectroscopy are needed for structural elucidation. Despite these limitations, UV-Vis spectroscopy remains a valuable analytical tool with widespread applications in chemistry, biochemistry, environmental science, materials science, and pharmaceuticals. Understanding these limitations is essential for appropriate experimental design and data interpretation.^[22]

2.2 Ultraviolet photoemission spectroscopy

Ultraviolet photoemission spectroscopy (UPS) is an effective and powerful surface analysis technique used to investigate the electronic structure of materials. By irradiating a sample with UV photons and measuring the kinetic energy of emitted photoelectrons, UPS provides valuable information about the energy levels, band structure, and electronic properties of surfaces and thin films.^[23]

The working principle of UPS is based on the photoelectric effect, where electrons in a material absorb photons of sufficient energy, causing them to be ejected from the surface. The energy of the emitted photoelectrons depends on the energy of the incident photons and the binding energy of the electrons in the material. UPS reveals information about the electronic states near the material's Fermi level by analyzing the kinetic energy and intensity of emitted photoelectrons as a function of incident photon energy.^[24-30]

UPS measurements are typically conducted in an ultrahigh vacuum (UHV) environment to minimize sample contamination and surface reactions. The sample is illuminated with monochromatic UV light generated by a UV lamp or a synchrotron light source, which provides tunable photon energies over a wide range. The emitted photoelectrons are then collected and analyzed by an electron energy analyzer, such as a hemispherical analyzer or a cylindrical mirror analyzer, which measures their kinetic energy and angular distribution.^[31]

UPS spectra consist of peaks corresponding to the binding energies of electrons in different electronic states of the material. These peaks reveal information about the density of electronic states, work function, valence band structure, and the Fermi-level position of the material. Researchers can identify electronic transitions, bandgaps, and surface electronic properties by comparing experimental UPS spectra with theoretical calculations or reference materials.^[32]

UPS is widely used in materials science, surface chemistry, and solid-state physics to investigate the electronic structure of surfaces, interfaces, and thin films. It provides insights into surface reactions, interface phenomena, electronic properties of semiconductor devices, and energy levels of organic and inorganic materials. UPS is also valuable for studying surface chemistry, catalysis, and electronic devices, contributing to photovoltaics, sensors, and semiconductor technology

advancements. UPS is a technique that measures the kinetic energy of photoelectrons emitted by molecules that have absorbed ultraviolet photons. This helps to determine the molecular orbital energies in the valence region. UPS is used to study the electronic structure of solids, atoms, and molecules. It can also be used to determine the work function

of a material.

In summary, UPS offers a non-destructive and susceptible method for probing the electronic structure of materials, providing valuable insights into surface and interface properties with atomic-scale resolution illustrated in Table 1.^[33]

Table 1: Distinguish between UPS and X-ray photoelectron spectroscopy.^[34]

Aspect	UPS	X-ray photoelectron spectroscopy (X-rays)
Principle	Measures the kinetic energy of electrons emitted from a material when irradiated with ultraviolet light.	Measures the kinetic energy of electrons emitted from a material when irradiated with X-rays.
Photon Energy Range	Uses ultraviolet photons (10–50 eV), typically from a He= 21.2eV or He= 40.8eV source	Uses higher energy X-rays 1-1.5 Kev typically from Al K α or Mg K α source.
Type of Information	Provides information about the valence electrons and electronic structure of materials.	Provides information about core-level electrons, chemical composition, and oxidation states of elements.
Depth of Analysis	Surface-sensitive technique, probing the top 1–5 nm of the material.	It is also surface-sensitive but probes deeper than UPS (typically 5–10 nm).
Electron Energy Range	Analyzes low-energy valence electrons with binding energies typically in the range of 0–50 eV.	Analyzes high-energy core-level electrons with binding energies typically in the range of 1–1500 eV.
Sensitivity to Chemical Environment	Sensitive to the electronic structure and chemical environment of valence electrons (<i>e.g.</i> , molecular orbitals).	Sensitive to the chemical environment of core electrons, providing detailed information on oxidation states and chemical shifts.
Resolution	Provides high-resolution information about electronic states near the Fermi level.	Provides detailed information about elemental composition and chemical states, but with lower resolution for valence states.
Applications	Used to study the valence band structure, work function, molecular orbitals, and surface states of materials.	Used to identify elements, quantify elemental composition, and analyze chemical bonding and oxidation states.
Elemental Sensitivity	Primarily sensitive to the electronic structure of surface atoms; limited elemental sensitivity.	Highly sensitive to element identification and chemical state analysis (oxidation states, bonding environments).
Work Function Measurement	Highly effective for measuring the work function and electronic states of surfaces.	Not typically used for work function measurements; more focused on core-level electronic structure.
Sample Type	Primarily used for conductive or semi-conductive materials due to the need to emit low-energy electrons.	It can be used for a wide variety of materials, including insulators, conductors, and thin films.
Depth Profiling	Not typically used for depth profiling due to its extreme surface sensitivity.	It can be combined with sputtering techniques for depth profiling, enabling analysis of subsurface layers.
Instrument Setup	Requires an ultraviolet light source (typically a helium discharge lamp) and an electron energy analyzer.	Requires an X-ray source (Al K)
Non-Destructive	Typically non-destructive for most samples but sensitive to surface contamination.	Generally non-destructive, though ion sputtering for depth profiling can damage samples.

2.2.1 Instrumentation of UPS

The instrumentation for UPS typically includes several key components designed to generate, analyze, and detect photoelectrons emitted from a sample surface when illuminated with UV light.^[35-38] The Schematic diagram of UPS is displayed in Fig. 2. Here's an overview of the essential elements of UPS instrumentation:

a) UV light source: The UV light provides monochromatic UV radiation with tunable photon energies. Common UV sources used in UPS gas discharge lamps are helium lamps, deuterium lamps, and synchrotron radiation. However, sometimes other gases are also filled, such as Neon and Argon. The photons emitted by Helium gas have energies of 21.2 and 40.8 eV. Synchrotron radiation offers a broad range of photon energies and high photon flux, enabling high-resolution and high-sensitivity UPS measurements.

b) Sample chamber: The sample chamber houses the sample and provides a vacuum environment for UPS measurements. It typically consists of a vacuum chamber with sample manipulators, heating/cooling stages, and ports for introducing gases or performing surface treatments. The sample chamber is maintained at UHV conditions to minimize sample contamination and surface reactions during measurements. In this chamber, the sample can be heated to a temperature of up to 3000 °C using a resistance oven, the heat of a discharge lamp, or circulating hot liquid. Unique oven design can increase the temperature to 100 °C, and laser vaporization can achieve a temperature of 200-250 °C. Gaseous or vapor samples were used in the classical UPS analysis. However, in modern UPS instrumentations, solid samples can also be analyzed.

c) Sample holder: The sample holder provides a stable platform for mounting and positioning the sample during UPS measurements. It may include mechanisms for sample rotation, tilt, translation, and temperature control to optimize the measurement conditions and acquire spectra from different sample orientations or surface regions.

d) Electron energy analyzer: The electron energy analyzer measures the kinetic energy and angular distribution of photoelectrons emitted from the sample surface. Common types of analyzers used in UPS include hemispherical analyzers, cylindrical mirror analyzers, and electrostatic analyzers. The analyzer selectively filters and detects photoelectrons based on their energy and direction, providing information about their energy distribution and emission angles.

e) Detector system: The detector system captures and records the energy distribution of emitted photoelectrons. It may include electron multipliers or microchannel plates to detect the electron current, channel electron multipliers (CEMs), or electron energy analyzers coupled with position-sensitive detectors. The detector system converts the incoming electrons into electrical signals, which are then processed and analyzed to generate UPS spectra. (1) Electron Multiplier: An electron current of less than 1-14 A can be detected with an

electron multiplier. Channeltrons, a type of electron multiplier, are often used instead of copper beryllium dynodes. Channeltrons are small, inexpensive, and rugged. (2) Microchannel plates: Microchannel plates are another type of detector that can collect electrons within the range of energies simultaneously. (3) Data collection and processing: the computer collects data and processes.

f) Data acquisition and control system: The data acquisition and control system coordinates instrument operation, data acquisition, and data processing. It includes hardware components such as analog-to-digital converters (ADCs), signal amplifiers, and pulse generators, as well as software for instrument control, spectral acquisition, data analysis, and visualization. The data acquisition system uses a gas discharge lamp to produce ultraviolet photons that induce the photoelectric effect. The emitted photoelectrons are then analyzed by energy to produce a spectrum of peaks. The data acquisition system typically includes components: (1) Glass discharge lamp: The glass discharge lamp produces UV photons, usually filled with helium, but other inert gases like argon or neon can also be used; (2) Advantage software: This allows for complete instrumental control, data acquisition, and data processing.

g) Vacuum system: UPS instrumentation requires a high vacuum environment to minimize electron scattering and absorption by gas molecules in the analysis chamber. The vacuum system includes vacuum pumps, valves, and gauges to achieve and maintain the required vacuum level during UPS measurements.

h) Monochromator: In synchrotron-based UPS setups, a monochromator is used to select and tune the photon energy of the synchrotron radiation. The monochromator ensures the production of monochromatic UV radiation with high energy resolution and tunability over a broad range of photon energies.

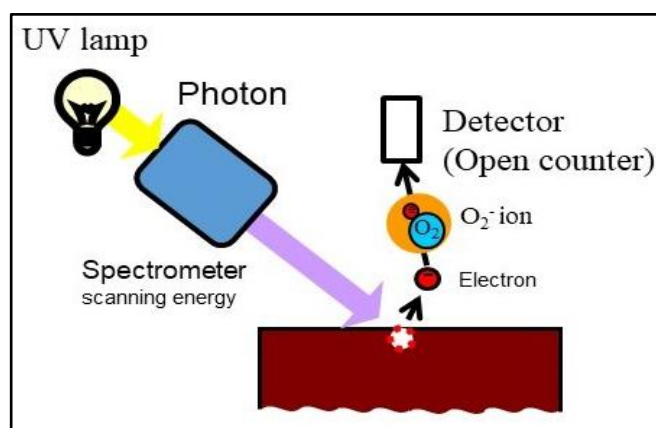


Fig. 2: Schematic diagram of Ultraviolet Photoemission spectroscopy

These components work together to enable the generation, analysis, and detection of photoelectrons emitted from the sample surface when illuminated with UV light, providing valuable insights into the electronic structure, band structure, and surface properties of materials under study.

2.2.2 Principle of UPS

The principle of UPS is based on the photoelectric effect and the interaction between ultraviolet photons and electrons in a material. UPS provides valuable insights into the electronic structure, band structure, and surface properties of materials at the atomic scale. Here's how UPS works:

a) Photoelectric effect: When a material is illuminated with photons of sufficient energy, electrons in the material can absorb the photons and be ejected from the surface. This phenomenon is known as the photoelectric effect and forms the basis of UPS. The energy of the ejected electron corresponds to the secondary electron cut-off. The phenomenon of the photoelectric effect was discovered by Einstein in 1905, and he was awarded the Nobel Prize in Physics for the same.

b) Photon absorption: In UPS, the sample is irradiated with monochromatic UV light generated by a UV lamp or a synchrotron light source. The incident photons have energies corresponding to the UV wavelength, typically in the range of a few electron volts to a few tens of electron volts. UPS is a technique that measures the kinetic energy of photoelectrons emitted by molecules that have absorbed ultraviolet photons. The method uses low-energy ultraviolet light to study materials' electronic properties and valence band structures.

c) Electron emission: When a photon with sufficient energy, *i.e.*, equal to or higher than binding energy, strikes the surface of the material, it can excite an electron from a bound state within the material to a free state. Ultimately, the energy of the emitted photoelectron depends on the difference between the photon energy and the binding energy of the electron in the material. UPS is performed by placing a solid sample in a vacuum chamber and irradiating it with ultraviolet light. A high-energy laser beam is applied to the sample in a vacuum chamber, causing electrons to be emitted. The emitted electrons are then analyzed by energy to produce a spectrum of peaks. The energy of the ejected electrons cut off, and the spectrum of peaks can be used to determine the concentration of specific elements. UPS is used to study the bonding and valence electron levels of atoms and is particularly used for investigating valence and conducting bands.

d) Energy conservation: According to the law of energy conservation, the energy of the emitted photoelectron (kinetic energy) is equal to the difference between the energy of the incident photon and the electron's binding energy in the material

$$E = h\nu \quad (3)$$

where E is the kinetic energy of the emitted photoelectron, ν is the energy of the incident photon (photon energy), and $h = 6.625 \times 10^{-34}$ Js.

e) Electron analysis: The emitted photoelectrons are collected and analyzed by an electron energy analyzer, such as a hemispherical analyzer or a cylindrical mirror analyzer. The analyzer measures the kinetic energy and angular distribution of the emitted photoelectrons.

f) Spectral analysis: By varying the energy of the incident

photons (photon energy) over a range of values, the kinetic energy of the emitted photoelectrons is measured at each photon energy. This results in a spectrum called the UPS spectrum, which consists of peaks corresponding to the binding energies of electrons in different electronic states of the material. The analysis of the photoelectron spectrum allows researchers to determine the distribution of photoelectrons, which in turn helps them understand the properties of the material being studied.

g) Electronic structure: The positions and intensities of peaks obtained in the UPS spectrum provide information about the density of electronic states, valence band structure, Fermi level position, and work function of the material. Thus, by comparing experimental UPS spectra with theoretical calculations or reference materials, researchers can identify electronic transitions, bandgaps, and surface electronic properties.

Overall, UPS offers a powerful method for probing the electronic structure and surface properties of materials with high sensitivity and resolution, providing valuable insights into surface chemistry, catalysis, semiconductor physics, and materials science.

2.2.3 Applications of UPS

UPS is used across various scientific and technological fields due to its ability to provide valuable insights into the electronic structure and surface properties of materials. UPS is a technique that uses ultraviolet light to measure the kinetic energy of photo-electrons emitted from a material's surface. The technique is based on the photo-electric effect and uses low-energy ultra-violet light to study the electronic properties of materials. UPS can be used to study absorbed species and their binding to the surface, as well as their orientation on the surface. A useful result from the characterization of the solid by UPS is the determination of the work function of the material. UPS can be used to:

- Determine work function: The energy required to remove an electron from the surface.
- Determine valence band maximum: The highest occupied energy state below the Fermi level.
- Study electronic structure: UPS is typically used to study the electronic structure of solids.

Some key applications of UPS include:

a) Surface chemistry and catalysis: UPS finds huge applications in chemistry and is used to study the electronic structure of catalyst surfaces and elucidate the mechanisms of surface reactions. It provides information about the energy levels, band structure, and electronic states of catalyst materials, helping researchers optimize catalyst design and understand catalytic processes involved in chemical reactions. UPS can help in understanding the chemical properties of known systems and studying newly synthesized molecular frameworks.

b) Semiconductor physics and device characterization: UPS is used in semiconductor research and device fabrication

to investigate the electronic properties of semiconductor materials and device interfaces. It provides information about the band alignment, energy levels, and surface states of semiconductor surfaces, aiding in the development of advanced semiconductor devices such as transistors, photovoltaic cells, and light-emitting diodes. UPS can help understand how organic semiconductors work in devices by studying the alignment of electrodes and the interfaces between organic layers.

c) Organic and molecular electronics: Apart from the determination of surface structure and catalytic properties, UPS can be used to study the electronic structure of organic semiconductors, molecular films, and organic-inorganic interfaces in organic electronic devices. It provides insights into the energy levels, charge transport mechanisms, and interfacial interactions in organic materials, facilitating the design and optimization of organic electronic devices for applications such as organic light-emitting diodes, organic solar cells, and organic field-effect transistors. UPS can help develop and understand organic electronic devices, materials, and interfaces by accurately measuring the electronic structures of organic semiconductors.

d) Surface science and materials chemistry: UPS is widely used in surface science and materials chemistry to investigate the electronic properties of surfaces, thin films, and nanostructures. It provides information about surface band bending, work function, chemical bonding, and adsorbate-surface interactions, enabling researchers to understand surface phenomena, surface reactions, and surface modification processes for applications in materials science, corrosion protection, and nanotechnology.

e) Thin film growth and interface engineering: This film is an important point of research in modern science and technology. UPS is employed in thin film deposition and interface engineering to monitor the electronic structure and interface properties of thin film systems. It provides insights into the growth kinetics, surface morphology, and electronic properties of thin films, hetero-structures, and interfaces, guiding the design and fabrication of functional materials for microelectronics, optoelectronics, and spintronics applications.

f) Surface modification and functionalization: In the modern period, surface modification is gaining importance. Many times, we need to functionalize the surface of the material by anchoring or loading suitable nanomaterial. UPS is used to study surface modification techniques such as chemical functionalization, surface doping, and surface passivation. It provides information about changes in the electronic structure, surface chemistry, and surface energetics induced by surface treatments, facilitating the development of tailored surface properties for applications in biomaterials, sensors, and coatings.

g) Corrosion science and materials degradation: Corrosion can be described as electrochemical destruction or decay of material. Every year, there is tremendous economic loss due to material corrosion. UPS is employed in corrosion science

and materials degradation studies to investigate the electronic properties of corroded surfaces, passive films, and protective coatings. It provides insights into the corrosion mechanisms, oxidation states, and electronic structure of materials exposed to corrosive environments, contributing to the development of corrosion-resistant materials and coatings for infrastructure, aerospace, and automotive applications.^[39-45]

These applications highlight the versatility and importance of UPS in advancing scientific knowledge, supporting technological innovations, and addressing challenges in various fields of research, development, and manufacturing.

2.2.4 Limitations of UPS

UPS has several limitations:

a) Surface sensitivity: UPS primarily probes the surface region of materials, typically the top few atomic layers. As UPS can provide information only about the topmost layer, even minor surface contamination can affect the accuracy of results, leading to incorrect interpretation of the material's electronic structure. Therefore, even a minute amount of gas adsorbed or contamination from air exposure can significantly alter the UPS spectra, which is a crucial limitation for practical or industrial applications. It may not provide information about the bulk electronic structure or buried interfaces, limiting its applicability to surface-sensitive systems. This can make it difficult to obtain information about the bulk properties of the material, especially if surface contamination or oxidation occurs.

b) Limited chemical information: UPS does not provide detailed chemical information, which limits its applications in chemical analysis.

c) Sample preparation: UPS requires samples to be vacuum-compatible and free from surface contaminants, oxides, or adsorbates that can interfere with the analysis. Sample preparation methods such as cleaning, drying, and handling may introduce artifacts or alter the surface properties, affecting the accuracy and reproducibility of measurements.

d) Sample requirement: UPS requires a sample that can efficiently emit low-energy electrons, which restricts the applications to conductive or semi-conductive materials. Non-conductive or insulating material may charge under UV light, making it difficult to obtain reliable data.

e) Limited depth profiling: UPS lacks depth profiling capabilities and cannot provide information about subsurface layers or buried interfaces without additional techniques. It primarily provides information about the electronic structure of the outermost surface layers, making it less suitable for studying buried interfaces or multilayered structures.

f) High vacuum requirement: UPS requires UHV conditions to prevent electron scattering in air and to maintain surface cleanliness. This limits the in situ analysis of samples exposed to atmospheric conditions.

g) Surface charging: Charging effects can occur during UPS measurements, particularly for insulating or semiconducting materials. The accumulation of charge on the sample surface

can distort the energy levels and band structure, leading to inaccurate measurements and difficulty in interpreting UPS spectra.

h) Work function variability: The work function of a material can be affected by surface adsorbate or contaminants, which affects the measurement of electronic states and complicates data interpretation.

i) Photon energy range: UPS measurements are limited to the range of photon energies provided by the UV light source. While synchrotron radiation offers tunable photon energies over a broad range, conventional UV lamps have limited energy tunability and may not cover the entire energy range of interest for some materials. The low photon energy of UV light restricts the ability to provide elemental composition data or identify different oxidation states as it does not probe deeper core levels.

j) Quantitative analysis: Quantitative analysis with UPS can be challenging due to surface roughness, sample charging effects, and variations in photoionization cross-sections for different materials. Accurate quantification of electronic states, band energies, and density of states may require careful calibration, standards, and consideration of instrumental parameters and sample conditions.

k) Poor sensitivity to light elements: Light elements like hydrogen have weak photoemission, cross-section, and UV-photon energy ranges, making them harder to detect in UPS as compared with other instruments.

l) Instrumentation complexity: UPS instrumentation can be complex and requires specialized equipment, including UV light sources, electron analyzers, vacuum systems, and data acquisition systems. Maintaining and operating UPS instruments requires expertise in surface science, vacuum technology, photonics, and data analysis.

m) Complex data interpretation: overlap of valence states: The valence bands from elements in a material can overlap in energy, making it challenging to distinguish between contributions from different atoms or molecules.

n) Access and availability: Access to UPS facilities may be limited, particularly in regions with fewer research infrastructure resources. The high cost of instrumentation and maintenance may restrict the accessibility of UPS for researchers and industrial users, leading to challenges in obtaining timely and cost-effective measurements.

Despite these limitations, UPS remains a valuable tool for investigating the electronic structure and surface properties of materials, providing valuable insights into surface chemistry, semiconductor physics, and materials science. Addressing these limitations through methodological improvements and complementary techniques continues to be an active area of research in the field of UPS.^[46-48]

2.3 Fluorescence spectroscopy

Fluorescence is the phenomenon of emission of electromagnetic radiation in the visible region from certain materials by absorbing electromagnetic radiation in the

invisible region. Fluorescence spectroscopy, a powerful analytical technique, has evolved significantly since its discovery in the late 19th century.^[49]

The roots of fluorescence can be traced back to Sir George Stokes' observations in 1852, who discovered the phenomenon while studying quinine sulfate. He noted that certain substances absorbed light at a specific wavelength and re-emitted it at a longer wavelength, a process termed "fluorescence" after the mineral fluorite.

The scope of fluorescence spectroscopy expanded rapidly with advancements in technology and understanding. In the early 20th century, scientists like Wood and Wurzburg explored fluorescence's applications in medicine and biology. Later, during the mid-20th century, the development of synthetic fluorophores and sophisticated instrumentation propelled fluorescence spectroscopy into diverse fields including biochemistry, environmental science, materials science, and pharmacology.^[50]

Modern fluorescence spectroscopy encompasses a broad range of techniques and applications. From steady-state fluorescence measurements to time-resolved and single-molecule fluorescence spectroscopy, researchers can investigate molecular structures, dynamics, interactions, and environmental changes with unparalleled sensitivity and specificity. Fluorescence is pivotal in drug discovery, environmental monitoring, molecular biology, and nanotechnology.^[51-54]

Manufacture of modern fluorescence instruments involves cutting-edge technologies. Lasers with precise wavelength control, sensitive detectors, and advanced data analysis software enable researchers to push the boundaries of fluorescence spectroscopy.

In recent years, there has been a surge in miniaturization and automation of fluorescence instrumentation, making it more accessible to a wider range of users. Handheld fluorescence analyzers find applications in field measurements, while integrated systems with high-throughput capabilities streamline drug discovery and clinical diagnostics screening processes.^[55-58]

In summary, fluorescence spectroscopy has come a long way since its serendipitous discovery in the 19th century. Its rich history, coupled with modern advancements in technology and methodology, has cemented its position as a cornerstone in analytical chemistry and interdisciplinary research.^[59]

2.3.1 Instrumentation of Fluorescence spectroscopy

The instrumentation used in fluorescence spectroscopy plays a crucial role in facilitating the measurement and analysis of fluorescence signals from samples.^[59-60] The instrumentation representation of fluorescence spectroscopy is shown in Fig. 3. The key components of fluorescence spectroscopy are described below:

a) Light source: Fluorescence spectroscopy requires a light source to excite the sample molecules. Common sources

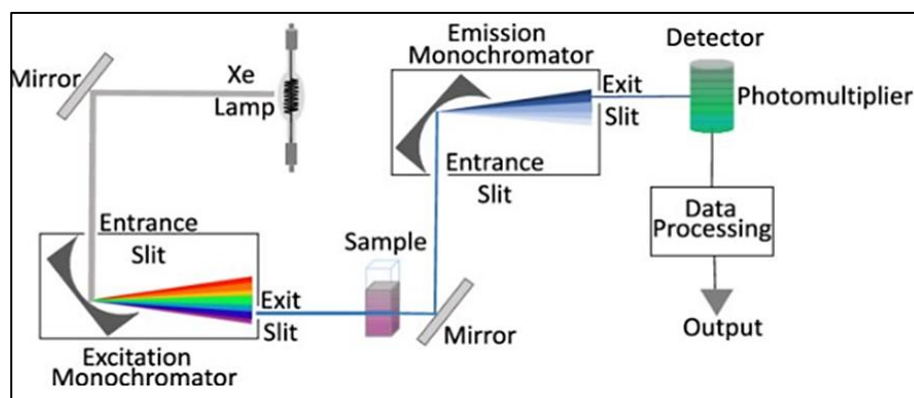


Fig. 3: A pictorial representation of fluorescence spectroscopy. [61]

include xenon arc lamps, mercury lamps, tungsten-halogen lamps, and laser systems. The choice of light source depends on factors such as desired excitation wavelength, intensity, and stability.

b) Monochromator: The monochromator is used to select a specific wavelength of light for sample excitation. A monochromator produces a beam of light with an extremely narrow bandwidth or light of a single color. It is used in optical measuring instruments where tunable monochromatic light is sought. It ensures that only the desired wavelength reaches the sample, thus enabling precise control over excitation conditions. Monochromators can be prism-based or diffraction grating-based. There are three types: single, double, and quadruple.

c) Sample compartment: This is the space where the sample is placed for analysis. It is usually equipped with adjustable holders to accommodate different sample formats, such as cuvettes, microplates, or solid samples.

d) Optical filters: Optical filters are used to selectively transmit fluorescence emission while blocking the excitation light and other stray light. They help improve the signal-to-noise ratio and enhance the sensitivity of fluorescence measurements.

e) Detector: Detectors capture the fluorescence emission from the sample and convert it into an electrical signal. Photomultiplier tubes (PMTs) are commonly used detectors due to their high sensitivity and wide dynamic range. Alternatively, CCDs or PMTs may also be employed, depending on the application.

f) Data acquisition system: The data acquisition system digitizes and records the detector output, allowing for the analysis and interpretation of fluorescence spectra. It may include ADCs, data processing units, and software for data visualization and analysis.

g) Spectrometer: In some fluorescence spectroscopy setups, a spectrometer is used to disperse the emitted fluorescence light into its spectral components. This allows for the measurement of fluorescence spectra, which provide valuable information about the molecular properties of the sample.

h) Control and software: Instrumentation is typically controlled by dedicated software that allows users to set

experimental parameters, acquire data, and perform data analysis. Modern software packages often include advanced features such as spectral deconvolution, kinetics analysis, and multivariate analysis.

These components work together seamlessly to enable the accurate and precise measurement of fluorescence signals, making fluorescence spectroscopy a versatile and powerful analytical technique in various scientific fields.

2.3.2 Principle of fluorescence spectroscopy

Fluorescence spectroscopy is based on the principle of fluorescence, a phenomenon exhibited by specific molecules called fluorophores. Fluorescence spectroscopy uses a beam of light that excites the electrons in molecules of certain compounds and causes them to emit light. Fluorescence is a member of the ubiquitous luminescence family of processes in which susceptible molecules emit light from electronically excited states created by either a physical, mechanical, or chemical mechanism. The generation of luminescence through the excitation of a molecule by ultraviolet or visible light photons is a phenomenon termed photoluminescence, which is formally divided into two categories, fluorescence, and phosphorescence, depending upon the electronic configuration of the excited state and the emission pathway. Fluorescence is the property of some atoms and molecules to absorb light at a particular wavelength and to subsequently emit light of a longer wavelength after a brief interval, termed the fluorescence lifetime. The fluorescence is governed by three important events, all of which occur on timescales that are separated by several orders of magnitude. Excitation of a susceptible molecule by an incoming photon happens in femtoseconds while vibrational relaxation of excited state electrons to the lowest energy level is much slower and can be measured in picoseconds. The final process, emission, is the emission of a longer wavelength photon and the return of the molecule to the ground state, which occurs in a relatively long time period of nanoseconds. Although the entire molecular fluorescence lifetime, from excitation to emission, is measured in only a billionths of a second, the phenomenon is a stunning manifestation of the interaction between light and matter that forms the basis for the expansive fields of steady-state and

time-resolved fluorescence spectroscopy and microscopy. The term fluorescence was coined on the basis of blue-white fluorescent mineral fluorite. Stokes also discovered the wavelength shift to longer values in emission spectra that bears his name.^[62]

Fluorescent probes are materials that can undergo electronic transitions that ultimately result in fluorescence. The fluorophore is a fluorescent probe or fluorochrome-conjugated to a larger macromolecule through absorption or covalent bonds. In general, fluorophores are divided into two broad classes, termed intrinsic and extrinsic. Intrinsic fluorophores, such as aromatic amino acids, neurotransmitters, porphyrins, and green fluorescent proteins, are those that occur naturally. Extrinsic fluorophores are synthetic dyes or modified biochemicals that are added to a specimen to produce fluorescence with specific spectral properties. The principle can be summarized in several key steps:^[63]

a) Absorption of light: When a fluorophore molecule is exposed to light of a specific wavelength (known as the excitation wavelength), it absorbs the energy from the incident photons. This absorption promotes an electron from the ground state to an excited state.

b) Relaxation to excited state: Upon absorption of light, the fluorophore molecule transitions from its ground state (lower energy) to an excited state (higher energy). This excited state is typically short-lived, lasting only for a fraction of a nanosecond to a few microseconds.

c) Emission of light: After a brief period in the excited state, the fluorophore molecule returns to its ground state by releasing the excess amount of energy as light. The emitted light, known as fluorescence emission, is of a longer wavelength (lower energy) than the absorbed excitation light. This phenomenon is governed by Stokes' law, which states that the fluorescence emission takes place at a longer wavelength than the absorbed light.

d) Stokes shift: The difference in wavelength between the excitation and emission light is called the Stokes shift. It is a characteristic property of each fluorophore and provides valuable information about the molecule's structure, environment, and interactions.

e) Fluorescence intensity: The intensity of fluorescence emission is proportional to the concentration of fluorophores in the sample and their quantum yield, which is the efficiency of fluorescence emission relative to other non-radiative relaxation pathways.

f) Fluorescence spectrum: By scanning the emission wavelength while keeping the excitation wavelength constant (or vice versa), a fluorescence spectrum can be obtained. This spectrum represents the intensity of fluorescence emission at different wavelengths and provides valuable insights into the properties of the fluorophores and their environment.

Overall, fluorescence spectroscopy exploits the unique electronic transitions and energy transfer processes of fluorophore molecules to investigate various molecular phenomena, including structure, dynamics, interactions, and

environmental changes. It is widely used in biochemistry, veterinary medicine, environmental science and engineering, biophysics, chemistry, materials science, medicine, and other scientific disciplines for both qualitative and quantitative analysis.

2.3.3 Applications of fluorescence spectroscopy

Fluorescence spectroscopy finds diverse applications across numerous scientific fields due to its sensitivity, selectivity, and versatility. Some prominent applications include:

a) Biomedical research: Fluorescence spectroscopy is extensively utilized in biomedical research for studying biomolecules, cells, and tissues. It is employed in protein structure analysis, DNA sequencing, detecting biomarkers, monitoring enzyme kinetics, and studying cellular processes such as membrane potential changes and intracellular ion concentrations.

b) Clinical diagnostics: In clinical diagnostics, fluorescence spectroscopy plays a crucial role in disease diagnosis and monitoring. It is used to detect biomolecules indicative of diseases such as cancer, diabetes, infectious diseases, and cardiovascular disorders. Fluorescent probes and markers enable sensitive and specific detection of target analytes in biological samples like blood, urine, and tissue biopsies.

c) Pharmaceutical analysis: Fluorescence spectroscopy is widely applied in pharmaceutical analysis for drug discovery, formulation development, and quality control. It facilitates the characterization of drug compounds, including their purity, stability, and interactions with other molecules. Fluorescence assays are also employed in high-throughput screening of potential drug candidates.

d) Environmental monitoring: Fluorescence spectroscopy finds a wide range of applications in environmental science for monitoring pollutants, contaminants, and natural organic matter in water, soil, and air. Therefore, it can be used to assess environmental quality, track sources of pollution, and study ecological processes such as biodegradation and photochemical reactions.^[64-70]

e) Food and beverage industry: In the food and beverage industry, fluorescence spectroscopy is employed for quality control, authentication, and safety assessment. It can detect contaminants, adulterants, and spoilage markers in food products. In addition to this, fluorescence techniques are used to monitor food processing, assess nutritional content, and study flavor compounds.

f) Material science: Fluorescence spectroscopy is valuable in material science for analyzing the properties and behavior of materials, including polymers, nanoparticles, and semiconductor materials. It is used to investigate material composition, structure, surface properties, and defects. Fluorescence-based sensors are also employed to detect analytes in materials and monitor manufacturing processes.

g) Forensic analysis: In forensic science, fluorescence spectroscopy aids in the analysis of forensic evidence, such as fibers, paints, inks, and biological fluids. It helps identify and

characterize trace evidence, differentiate between materials, and link evidence to specific sources or individuals.

h) Environmental science and engineering: Fluorescence spectroscopy is widely used in environmental science and engineering to study natural organic matter, pollutants, and contaminants in water, soil, and air. Pollution of natural resources is a major worldwide problem that directly or indirectly affects a large number of flora and fauna. Fluorescence spectroscopy helps to assess water quality, monitor pollution, and understand environmental processes such as biodegradation and photochemical reactions.^[71-75]

2.3.4 Limitations of fluorescence spectroscopy

While fluorescence spectroscopy is a powerful analytical technique with broad applicability, it also has certain limitations:

a) Overlapping emission spectra: Fluorescent molecules often have emission spectra that overlap with those of other molecules in complex samples. This can lead to difficulties in distinguishing between different fluorophores and interpreting fluorescence signals accurately.

b) Quantum yield variation: The fluorescence quantum yield, which represents the efficiency of fluorescence emission, can vary significantly among different fluorophores and environmental conditions. Variations in quantum yield can affect the accuracy and reproducibility of quantitative measurements.

c) Inner filter effect: In samples with high absorbance at the excitation wavelength, the incident light can be partially absorbed before reaching the fluorophores, leading to a decrease in fluorescence intensity. This phenomenon, known as the inner filter effect, can distort fluorescence measurements and complicate data analysis.

d) Photo-bleaching: Prolonged exposure to excitation light can cause photo-bleaching, where fluorophores lose their fluorescence activity over time due to irreversible chemical reactions or physical degradation. Photo-bleaching limits the duration of fluorescence measurements and can affect the stability of fluorescence signals.

e) Background autofluorescence: The phenomenon of autofluorescence, which arises from endogenous fluorophores, is observed in certain biological materials such as flavins, collagen, and lipofuscin. Background autofluorescence can interfere with detecting specific fluorescent probes and reduce the signal-to-noise ratio, particularly in complex biological matrices.

f) Limited depth penetration: Fluorescence signals from samples are typically detected from the surface or at a shallow depth within the sample. This limitation restricts the application of fluorescence spectroscopy to surface-sensitive measurements and may not be suitable for analyzing samples with significant light scattering or absorption properties.

g) Temperature sensitivity: Fluorescence intensity and emission spectra can be sensitive to changes in temperature, affecting the stability and reproducibility of measurements.

Temperature control is often required to minimize temperature-induced artifacts and ensure the accuracy of fluorescence data.

h) Sample preparation requirements: Fluorescence spectroscopy often requires specific sample preparation procedures to optimize the signal-to-noise ratio and minimize artifacts. Sample purification, labeling with fluorescent probes, and removal of interfering substances may be necessary, which can increase the complexity and time required for analysis.^[76-78]

2.4 Raman spectroscopy

Raman spectroscopy, a pivotal analytical technique, traces its roots to the groundbreaking discovery by Sir Raman and Krishnan in 1928. He was awarded the Nobel Prize in Physics for the discovery of the Raman Effect. He is the only Scientist from India who has been awarded the Nobel Prize. While studying the scattering of light, they observed that a small fraction experienced a shift in frequency, later termed the “Raman effect”. This phenomenon revolutionized molecular analysis, offering insights into the vibrational and rotational modes of molecules.^[79-83]

Over the years, Raman spectroscopy has found wide-ranging applications. It elucidates molecular structures and interactions in chemistry, aiding pharmaceutical research, materials science, and environmental monitoring. Its capability to characterize materials extends to diverse fields, including semiconductor analysis, art conservation, and geology. Biomedical applications leverage Raman's ability to probe biological tissues and fluids, facilitating disease diagnosis and drug development.^[84,85]

Modern Raman spectroscopy owes much of its efficacy to technological advancements. Laser technology, particularly solid-state lasers, delivers precise and intense excitation light, enhancing sensitivity and signal-to-noise ratios. Optical components, such as filters and gratings, optimize light collection and dispersion, ensuring accurate spectral analysis. Detectors like CCDs enable sensitive detection and rapid data acquisition.

Modern Raman spectroscopy owes much of its efficacy to technological advancements. Laser technology, particularly solid-state lasers, delivers precise and intense excitation light, enhancing sensitivity and signal-to-noise ratios. Optical components, such as filters and gratings, optimize light collection and dispersion, ensuring accurate spectral analysis. Instrumentation has also evolved, with sophisticated spectrometers featuring automated sample stages and intuitive software for data analysis. Portable Raman spectrometers have emerged, facilitating on-site and field applications across industries. Quality control measures guarantee instrument performance, meeting the demands of scientific research, industrial processes, and regulatory standards.^[86,87]

Thus, Raman spectroscopy's historical journey from its serendipitous discovery to modern applications epitomizes scientific innovation. Its scope encompasses diverse

disciplines driven by technological progress and interdisciplinary collaboration. As manufacturing techniques advance, Raman spectroscopy continues to shape our understanding of the molecular world and address contemporary challenges across science and industry.

2.4.1 Instrumentation of Raman spectroscopy

The instrumentation of Raman spectroscopy comprises several key components that work together to generate, collect, and analyze Raman scattering signals. [88-90] The schematic diagram of Raman spectroscopy is presented in Fig. 4. Here is an overview of the essential components of Raman Spectroscopy:

a) Laser source: Raman spectroscopy typically employs a monochromatic laser as the excitation source. Common lasers used in Raman Spectroscopy include diode lasers, solid-state lasers (such as diode-pumped and solid-state lasers), and semiconductor lasers. The choice of laser depends on factors like excitation wavelength, power, and stability.

b) Sample illumination: The laser beam is directed onto the sample through a microscope objective or focusing lens. This ensures precise control over the laser spot size and position on the sample, enabling localized measurements and spatial mapping.

c) Optical filters: Optical filters are effectively used to block the Rayleigh scattered light (light that undergoes elastic scattering at the same frequency as the incident laser) and transmit the Raman scattered light alone. Filters help to improve the signal-to-noise ratio by removing unwanted background noise from the spectrum.

d) Dispersion element: A dispersive element, such as a diffraction grating or prism, disperses the Raman scattered light into its component wavelengths. This allows the detection of Raman shifts corresponding to vibrational modes of molecules.

e) Detector: Raman scattered light is detected by a sensitive detector, typically a CCD camera or PMT. These detectors convert optical signals into electronic signals for data acquisition and analysis.

f) Spectral analyzer: A spectral analyzer, such as a spectrometer or spectrograph, is used to separate and analyze the Raman scattered light according to its wavelengths. The spectrometer may employ a diffraction grating or prism to disperse the light and capture the Raman spectra.

g) Data acquisition system: The electronic signals from the detector are processed and recorded by a data acquisition system. This system digitizes the signals and stores them for further analysis, often in the form of Raman spectra.

h) Software: Dedicated software is used for data analysis, including spectral processing, peak identification, and quantification of Raman signals. Advanced software packages may also offer features like spectral deconvolution, baseline correction, and multivariate analysis.

i) Optional accessories: Additional accessories, such as sample stages with temperature control, polarization optics,

and confocal microscopy attachments, can be incorporated into the Raman spectroscopy setup to enhance measurement capabilities and address specific experimental requirements.

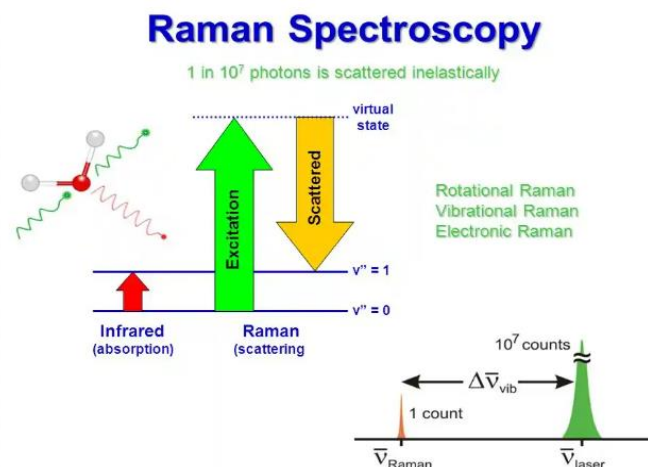


Fig. 4: Schematic diagram of Raman spectroscopy. [91]

Overall, the instrumentation of Raman spectroscopy combines precision optics, lasers, detectors, and advanced data analysis software to enable the accurate and sensitive detection of Raman scattering signals from samples, making it a versatile technique for molecular analysis across various scientific disciplines.

2.4.2 Principle of Raman spectroscopy

Raman spectroscopy is based on the principle of inelastic light scattering, known as the Raman effect, discovered by Sir Raman and Krishnan in 1928. The principle can be summarized in several key steps:

a) Incident light interaction: A monochromatic light source, typically a laser, irradiates the sample. The photons from the incident light interact with the sample molecules, leading to elastic and inelastic scattering processes.

b) Elastic scattering (Rayleigh scattering): The majority of incident photons undergo elastic scattering, known as Rayleigh scattering, where the scattered light maintains the same energy (frequency) as the incident light. This scattered light contributes to the background and is much more intense than Raman's scattered light.

c) Inelastic scattering (Raman scattering): A small fraction of incident photons undergo inelastic scattering, known as Raman scattering. During this process, the scattered photons gain or lose energy due to interactions with the vibrational and rotational modes of the sample molecules.

d) Energy transfer: In Raman scattering, the energy of the scattered photon is shifted relative to the incident photon by an amount corresponding to the energy of a specific vibrational or rotational transition within the molecule. This shift in energy is known as the Raman shift and is characteristic of the molecular structure of the sample.

e) Raman spectra: The Raman scattered light is collected and dispersed by a spectrometer, separating it into component

wavelengths. The resulting spectrum, known as a Raman spectrum, represents the intensity of scattered light as a function of Raman shift. Each peak in the spectrum corresponds to a specific vibrational or rotational mode within the molecule.

f) Information content: The Raman spectrum provides valuable information about the molecular structure, chemical composition, and bonding environment of the sample. It can identify functional groups, detect molecular conformations, and differentiate between different chemical species.

Thus, Raman spectroscopy exploits the Raman effect to probe the vibrational and rotational modes of molecules, offering a non-destructive, label-free, and highly sensitive analytical technique for molecular analysis in various scientific fields, including chemistry, materials science, biology, and pharmaceuticals.

2.4.3 Applications of Raman spectroscopy

Raman spectroscopy, some of the applications can be enlisted as below:

a) Chemical analysis: Raman spectroscopy is extensively used for chemical analysis, including qualitative and quantitative determination of molecular composition, identification of functional groups, and characterization of chemical reactions. It is employed in fields such as pharmaceuticals, polymers, organic chemistry, and forensic science.

b) Material characterization: Raman spectroscopy is an important analytical tool and provides information regarding structure, composition and properties of the material under study. This spectroscopic technique can provide information about crystalline phases, detect polymorphs and contaminants assess material purity and homogeneity and investigate phase transition and defects. Applications include semiconductor analysis, mineralogy, ceramics, and thin-film characterization.

c) Biomedical and veterinary sciences: Raman spectroscopy is used to study the biological tissues, cells, and biomolecules. It can provide information about bio-molecular structure, identify disease markers, assess cell viability and metabolic activity, and monitor drug interactions and distribution in tissues. Also can be employed in detection of neoplastic tissues, tissue engineering, drug discovery, and microbiology.

d) Pharmaceutical analysis: Raman spectroscopy plays a crucial role in the pharmaceutical industry for design and development of drug, formulation analysis, and quality control. It can identify active pharmaceutical ingredients (APIs), detect polymorphs and impurities, assess drug stability and degradation, and monitor drug-polymer interactions in formulations.

e) Environmental monitoring: We observe environmental pollution. The environmental pollution highly affects the human life and livestock. Therefore, environmental monitoring is an important parameter and Raman spectroscopy is employed in environmental science for measurement of pollutants, contaminants, and natural

resources. It can identify organic and inorganic pollutants in water, soil, and air, assess environmental degradation and remediation processes, and study microbial communities and biogeochemical cycles.

f) Art and cultural heritage conservation: Raman spectroscopy is used for non-destructive analysis and authentication of artworks, historical artifacts, and cultural heritage objects. It can identify pigments, dyes, varnishes, and degradation products, assess material aging and deterioration, and guide conservation treatments and restoration efforts.

g) Food and beverage analysis: Raman spectroscopy finds applications in the food and beverage industry for quality control, authentication, and safety assessment. It can identify ingredients, detect contaminants and adulterants, assess nutritional content, and monitor food processing and storage conditions.^[92,93]

2.4.4 Limitations of Raman spectroscopy

Raman spectroscopy is a powerful analytical technique with numerous applications, and it also has several limitations:

a) Fluorescence interference: Fluorescence from the sample or impurities can interfere with Raman scattering signals, particularly when the fluorescence intensity is comparable to or higher than the Raman signal. Fluorescence background can obscure Raman peaks, reducing the signal-to-noise ratio and complicating spectral analysis.

b) Sample heating: The high-intensity laser used in Raman spectroscopy can heat the sample, especially when analyzing sensitive or thermally unstable samples. Sample heating may cause thermal decomposition, phase transitions, or changes in molecular conformation, potentially altering Raman spectra and leading to inaccurate results.

c) Limited depth penetration: Raman scattering is a surface-sensitive technique, with the depth of penetration typically limited to a few micrometers. This limits its applicability for analyzing bulk materials or samples with significant light scattering or absorption properties, as the Raman signal may not adequately represent the entire sample volume.

d) Sample preparation: Raman spectroscopy often requires minimal sample preparation; however, certain samples may require specific preparation techniques to optimize signal quality and reproducibility. Sample preparation steps such as drying, grinding, and mounting can introduce artifacts or alter sample properties, potentially affecting Raman spectra.

e) Low sensitivity for trace analysis: Raman spectroscopy has lower sensitivity compared to some other analytical techniques, particularly for trace analysis or detection of low-concentration analytes. The Raman effect is inherently weak, resulting in relatively low signal intensities, which may limit its applicability for trace-level detection or quantification.

f) Instrumentation costs: High-quality Raman spectroscopy instrumentation can be costly, particularly for systems with advanced features such as high-resolution spectrometers, sensitive detectors, and multiple laser excitation options. The initial investment and maintenance costs may be prohibitive

for some research labs or industrial settings.

g) Complex spectral interpretation: Raman spectra can be complex and challenging to interpret, especially for samples with overlapping peaks, multiphase materials, or mixtures of compounds. Spectral interpretation may require expertise in vibrational spectroscopy and molecular structure analysis, making it time-consuming and subjective.

Despite these limitations, Raman spectroscopy remains a valuable analytical tool with widespread applications in diverse scientific fields, and ongoing advancements in instrumentation and data analysis techniques continue to address many of these challenges.

3. Electron spectroscopy

Electron spectroscopy encompasses a range of techniques that utilize the interaction between electrons and matter to investigate the electronic structure, composition, and properties of materials. One of the fundamental principles underlying electron spectroscopy is the measurement of the energy distribution of electrons emitted or scattered from a sample in response to various excitation processes. This interaction provides valuable information about the electronic states, bonding characteristics, and surface properties of the material under investigation.^[94-96]

In electron spectroscopy, electrons are typically generated and accelerated towards the sample using an electron source such as a heated filament or a field emission cathode. The energy of the incident electrons can be controlled and varied to probe different depths within the sample, allowing for surface-sensitive or bulk-sensitive analysis depending on the technique employed.

One of the most widely used electron spectroscopy techniques is X-ray photoelectron spectroscopy (XPS), also known as electron spectroscopy for chemical analysis (ESCA). In XPS, monochromatic X-rays are used to eject core electrons from atoms within the sample. The kinetic energy and intensity of the emitted photoelectrons are then measured to determine the elemental composition, chemical bonding, and oxidation states of the surface atoms. By analyzing the binding energy of the photoelectrons, XPS can provide information about the chemical environment and electronic structure of the material's surface layers.

Another important electron spectroscopy technique is Auger electron spectroscopy (AES), which detects the characteristic Auger electrons emitted following the relaxation of core holes created by an incident electron beam. AES is particularly sensitive to the composition and elemental distribution of atoms near the sample surface, making it a valuable tool for surface analysis and depth profiling.

Transmission electron energy loss spectroscopy (TEM-EELS) is another powerful electron spectroscopy technique that measures the energy loss of electrons passing through a thin sample in a transmission electron microscope (TEM). By analyzing the energy loss spectrum, TEM-EELS can provide information about the electronic structure, bonding, and

elemental composition of materials with high spatial resolution.

Overall, electron spectroscopy techniques offer valuable insights into the electronic properties and composition of materials, making them indispensable tools for scientific research and technological applications in fields such as materials science, nanotechnology, surface chemistry, and catalysis.^[97]

3.1 X-ray photoelectron spectroscopy

X-ray photoelectron spectroscopy (XPS) was officially introduced as a technique for chemical analysis in the 1960s. XPS capitalizes on the photoelectric effect discovered by Hertz in 1887, which was later developed by Siegbahn and awarded the Nobel Prize in 1981. It allowed researchers to study the elemental composition, chemical states, and electronic structures of materials with high sensitivity and surface specificity. It offers valuable insights into the surface chemistry and structure of solids, liquids, and thin films with high sensitivity and resolution.

At its core, XPS relies on the photoelectric effect, first described by Albert Einstein in 1905, where photons (X-rays) incident on a material surface eject photoelectrons with energies characteristic of the atomic species present and their chemical environment. This emitted photoelectron energy is measured to determine elemental composition and chemical states within the top few nanometers of the sample surface.

The basic setup of an XPS instrument consists of an X-ray source, typically employing a monochromatic X-ray beam, which irradiates the sample. The emitted photoelectrons are then collected and analyzed by an electron energy analyzer, which separates electrons based on their kinetic energy. A detector records the electron energy distribution, producing a spectrum that reveals the binding energies of photoelectrons, which are specific to the elements and chemical states present in the sample.

XPS is highly versatile and finds applications across various scientific fields and industries. In materials science, it is used to study surface composition, chemical reactions, and thin film growth. XPS provides insights into molecular structure, surface functionalization, and catalysis in chemistry. Semiconductor research is employed to characterize semiconductor surfaces, interfaces, and electronic devices. Additionally, XPS is utilized in environmental science for analyzing pollutants, biomedical research for studying biomaterials and biological surfaces, and forensic science for trace analysis and materials identification.

Modern XPS instruments offer advanced capabilities, including high spatial resolution, chemical state mapping, and depth profiling, enabling detailed characterization of complex materials and interfaces. Furthermore, sophisticated data analysis software allows for accurate peak fitting, quantitative analysis, and interpretation of XPS spectra.

Thus, X-ray Photoelectron Spectroscopy is a valuable analytical tool that provides fundamental insights into the

composition, structure, and properties of materials at the nanometer scale, driving advancements in research, development, and technology across diverse scientific disciplines and industrial sectors.

3.1.1 Instrumentation of XPS

The XPS instrument comprises several key components that work together to generate, analyze, and interpret XPS spectra. Here is an overview of the essential components of X-ray photoelectron Spectroscopy, and the pictorial representation is shown in Fig. 5.

a) X-ray source: The X-ray source generates monochromatic X-rays to irradiate the sample surface. Common X-ray sources include Mg K α (1253.6 eV), Al K α (1486.6 eV), and Cu K α (8048 eV). These X-rays are produced using X-ray tubes or synchrotron radiation sources.

b) Sample chamber: The sample chamber holds the specimen under investigation and provides a vacuum environment to minimize air-induced contamination and allow the interaction of the X-ray beam with the sample surface.

c) Detector: The detector records the energy and intensity of the emitted photoelectrons as a function of their kinetic energy. Modern XPS detectors include multi-channel electron detection systems, such as delay-line detectors or microchannel plate detectors, which offer high sensitivity and spatial resolution.

d) Energy analyzer: The energy analyzer selects the desired energy range of the emitted photoelectrons and rejects unwanted background signals, such as secondary electrons or Auger electrons. It ensures that only photoelectrons with energies corresponding to the elements of interest are detected.

e) Data acquisition system: Modern XPS is equipped with the data acquisition system digitizes and records the signals from the detector, producing XPS spectra that represent the energy distribution of emitted photoelectrons. It may include signal amplifiers, ADCs, and data processing units.

f) Vacuum system: XPS requires maintenance of the vacuum. A vacuum system maintains a high vacuum environment inside the sample chamber to minimize collisions between photoelectrons and air molecules, ensuring accurate XPS

measurements. It typically includes vacuum pumps, vacuum gauges, and vacuum seals.

g) Software: Dedicated software is used for instrument control, data acquisition, and spectral analysis. Different software are developed and used. It allows users to set experimental parameters, acquire XPS spectra, perform peak fitting and quantification, and visualize and interpret the results.

Thus, we notice that the instrumentation of XPS combines precise X-ray sources, electron analyzers, detectors, and advanced data analysis software to enable the accurate and sensitive analysis of the elemental composition, chemical states, and surface properties of materials, making it a versatile tool for surface science, materials characterization, and chemical analysis.

3.1.2 Principle of XPS

XPS operates on the principle of the photoelectric effect and the subsequent analysis of emitted photoelectrons. The principle can be summarized in several key steps:

a) X-ray irradiation: The sample under investigation is irradiated with monochromatic X-rays, typically generated by an X-ray source such as a magnesium (Mg K α), aluminum (Al K α), or copper (Cu K α) X-ray tube. These X-rays have sufficient energy to penetrate the surface layers of the sample.

b) Photoelectric effect: When the X-rays interact with the sample, they can excite inner-shell electrons (typically core electrons) of atoms within the sample. If the energy of the incident X-ray photons exceeds the binding energy of an inner-shell electron, the electron can be ejected from the atom. This process is known as the photoelectric effect.

c) Emission of photoelectrons: The ejected inner-shell electron, now referred to as a photoelectron, exits the atom with kinetic energy equal to the difference between the energy of the incident X-ray photon and the binding energy of the electron. The kinetic energy of the emitted photoelectron is characteristic of the element from which it originated and is specific to the electron's initial binding state within the atom.

d) Energy analysis: The emitted photoelectrons are collected and energy-analyzed using an electron energy analyzer. The

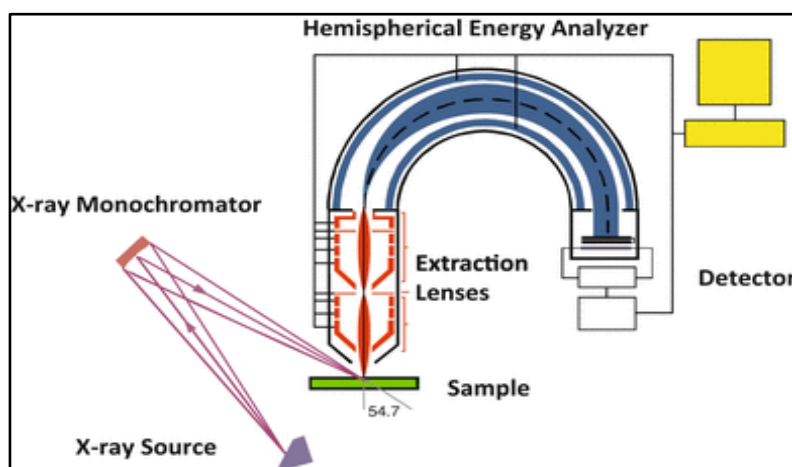


Fig. 5: A pictorial representation of X-ray photoelectron spectroscopy.

analyzer selects photoelectrons with specific kinetic energies and separates them according to their energy distribution. This process allows for the determination of the binding energy of the photoelectrons, which is related to the chemical environment and electronic structure of the atoms from which they originated.

e) Spectral interpretation: The energy-analyzed photoelectron signals are recorded, typically as a function of binding energy, to generate an XPS spectrum. Peaks in the spectrum correspond to the kinetic energies of emitted photoelectrons, providing information about the elemental composition, chemical states, and surface chemistry of the sample.

f) Data analysis: XPS spectra are analyzed using specialized software to identify and quantify the elements present in the sample, determine their chemical states, and extract information about chemical bonding and electronic structure. Peak fitting algorithms, reference databases, and spectral deconvolution techniques are commonly employed for data analysis and interpretation.

Overall, XPS exploits the photoelectric effect to probe the elemental composition, chemical states, and surface properties of materials with high sensitivity and specificity, making it a valuable tool for surface analysis, materials characterization, and chemical analysis in various scientific and industrial applications.

3.1.3 Applications of XPS

XPS finds diverse applications across numerous scientific fields, such as chemical sciences, physics, veterinary science, health, agriculture, pharmaceutical science, environmental sciences and engineering, *etc.* It is also widely used in industrial sectors, research, and development and can provide detailed information about the chemical composition, electronic structure, and surface properties of materials. Some prominent applications include:

a) Surface analysis: XPS is widely used for surface analysis of solid materials, providing valuable insights into surface composition, elemental distribution, chemical states, and surface contamination. It is employed in fields such as catalysis, corrosion science, adhesion studies, and surface modification.

b) Materials characterization: XPS is an important technique widely and effectively used for materials characterization, enabling researchers to analyze the chemical composition and electronic structure of materials with high sensitivity and resolution. It is employed in materials science, nanotechnology, metallurgy, and semiconductor research to study thin films, coatings, nanoparticles, and composite materials.

c) Catalysis and catalytic materials: XPS is utilized in catalysis research to investigate the surface chemistry and catalytic properties of catalyst materials. It provides information about catalyst composition, active sites, oxidation states, and surface reactions, aiding in the design and

optimization of catalysts for various industrial processes, including energy conversion, environmental remediation, and chemical synthesis.

d) Semiconductor industry: XPS plays a critical role in the semiconductor industry for characterizing semiconductor materials, device surfaces, and interfaces. It is used to analyze thin film deposition, interface engineering, doping profiles, oxide layers, and surface passivation in semiconductor devices such as integrated circuits, solar cells, and microelectronic components.

e) Polymer science: XPS is employed in polymer science for studying polymer surfaces, interfaces, and functionalization. It provides information about polymer composition, chemical bonding, cross-linking, and surface modifications, facilitating the development of advanced materials for applications in coatings, adhesives, biomaterials, and electronics.

f) Biomedical research: XPS finds applications in biomedical research for analyzing biomaterials, implants, and biological surfaces. It is used to study the surface properties of biomaterials, characterize tissue-engineered constructs, assess implant integration and biocompatibility, and investigate interactions between biomolecules and surfaces.

g) Environmental science and engineering: XPS is widely used in environmental science and engineering to study environmental samples, pollutants, and contaminants. It provides information about surface chemistry, oxidation states, and speciation of environmental pollutants, aiding in environmental monitoring, pollution control, and remediation efforts.

h) Coatings and thin films: XPS is employed in coatings and thin film industries for quality control, composition analysis, and surface characterization. It helps optimize coating formulations, assess film thickness, detect defects, and study the adhesion and durability of coatings in applications such as automotive coatings, protective coatings, and electronic thin films.

Overall, XPS is a versatile analytical technique with widespread applications in scientific research, materials development, quality control, and industrial manufacturing, contributing to advancements in various technological fields and addressing complex scientific and engineering challenges.

3.1.4 Limitations of XPS

XPS is a highly valuable technique for surface analysis and materials characterization, it also has several limitations:

a) Surface sensitivity: XPS is primarily a surface-sensitive technique, probing only the top few nanometers to micrometers of the sample surface. Consequently, it may not provide information about the bulk composition or buried interfaces of thick or layered materials.

b) Sample preparation requirements: XPS requires clean and flat sample surfaces for accurate analysis. Sample preparation techniques such as polishing, sputtering, and cleaning are often necessary to remove surface contaminants, oxides, and roughness, which can affect XPS measurements.

c) Limited depth profiling: While depth profiling can be performed using XPS, it is typically limited to shallow depths due to the interaction volume of X-ray photons with the sample. Depth profiling techniques such as ion sputtering or angle-resolved XPS may be employed, but they have practical limitations in terms of depth resolution and analysis depth.

d) Surface charging: XPS measurements can be affected by surface charging phenomena, particularly for insulating or semiconducting samples. Surface charging can distort the kinetic energy of emitted photoelectrons, leading to shifts in peak positions and spectral distortions, which may complicate data interpretation.

e) Quantitative analysis challenges: Quantitative analysis in XPS can be challenging due to factors such as differential sensitivity factors, matrix effects, and variations in electron escape depths. Accurate quantification of elemental concentrations and chemical states requires careful calibration, reference standards, and consideration of instrumental parameters.

f) Limited elemental sensitivity: XPS has limited sensitivity for detecting light elements (*e.g.*, hydrogen, helium) and elements with low atomic number (*e.g.*, carbon, nitrogen) due to their low X-ray photoionization cross-sections. Consequently, XPS may not be suitable for analyzing samples containing trace amounts of these elements.

g) Chemical state information: While XPS provides valuable information about elemental composition and oxidation states, it may not always distinguish between chemically similar species or subtle chemical variations. Interpretation of XPS spectra requires expertise in spectral deconvolution, peak fitting, and chemical bonding analysis.

h) Cost and accessibility: High-quality XPS instruments can be expensive to purchase, operate, and maintain. Additionally, XPS facilities may not be readily accessible to all researchers, limiting its widespread adoption and utilization in certain academic and industrial settings.

Despite these limitations, XPS remains a powerful and widely used technique for surface analysis, materials characterization, and chemical analysis, offering valuable insights into the composition, structure, and properties of materials at the nanometer scale. Ongoing advancements in instrumentation, data analysis techniques, and sample preparation methods continue to address many of these challenges, expanding the capabilities and applications of XPS in scientific research and industrial applications.

3.2 Electron energy loss spectroscopy

Electron energy loss spectroscopy (EELS) is a powerful analytical technique used to investigate the electronic and structural properties of materials at the nanometer scale. It operates on the principle of measurement of energy loss of electrons once they interact with the sample material. [98- 101] This spectroscopic technique can provide useful information related with elemental composition, chemical bonding and electronic excitation. Fig. 6 addresses a pictorial

representation of electron energy loss spectroscopy.

In EELS, a high-energy electron beam is directed onto a thin sample, such as a nanomaterial, semiconductor, or biological specimen. As the electrons pass through the sample, they undergo interactions with the sample's atoms and electrons, resulting in energy losses due to processes such as ionization, plasmon excitation, and inner-shell excitation.

The energy loss spectrum obtained from EELS contains information about the energy distribution of electrons after interacting with the sample. By analyzing the shape, intensity, and position of peaks in the energy loss spectrum, researchers can extract valuable information about the sample's composition, chemical environment, and electronic structure.

EELS offers several advantages over other analytical techniques. It provides high spatial resolution, allowing researchers to probe nanoscale features and interfaces within materials. It also offers elemental and chemical specificity, enabling the identification of specific elements and chemical bonding configurations within complex samples.

EELS finds diverse applications in various scientific fields, such as veterinary science, environmental science and engineering, agriculture, physics, chemistry, materials science, nanotechnology, semiconductor physics, and biology. It is used to study a wide range of materials, including catalysts, nanoparticles, polymers, biomolecules, and electronic devices.

Overall, EELS is a valuable tool for investigating the electronic and structural properties of materials at the nanometer scale, offering detailed insights into their composition, bonding, and functionality. Its versatility and high sensitivity make it an indispensable technique for advancing scientific understanding and technological innovation in diverse research areas.

3.2.1 Instrumentation of EELS

The instrumentation of EELS involves several key components that work together to generate, analyze, and interpret electron energy loss spectra. Here's an overview of the essential components present in EELS:

a) Electron source: EELS instruments utilize an electron source capable of producing a high-energy electron beam. This is commonly achieved using a field emission gun (FEG) or a thermal electron source, such as a tungsten filament. The electron source emits electrons with energies typically ranging from a few hundred electron volts (eVs) to several kilo-electron volts (keVs).

b) Electron optics: Electron optics, including electron lenses and apertures, are used to control and focus the electron beam onto the sample. These components ensure that the electron beam is well-collimated and properly aligned for accurate and precise measurements.

c) Sample stage: A sample stage holds the specimen under investigation and allows for precise positioning and manipulation during the experiment. It may feature motorized stages for automated sample positioning and rotation, as well as heating or cooling capabilities to control sample

temperature during analysis.

d) Energy filter: The heart of the EELS instrument is the energy filter, which selects and analyzes the energy loss of electrons that pass through the sample. The energy filter consists of an electron energy analyzer, typically a magnetic prism or a monochromator, which disperses electrons based on their energy and selects a specific energy range for analysis.

e) Detector system: The detector system records the energy loss spectrum of electrons after they interact with the sample. Common detectors include scintillation detectors, semiconductor detectors, and electron multipliers. The choice of detector depends on factors such as sensitivity, energy resolution, and signal-to-noise ratio requirements.

f) Data acquisition system: Modern EELS is facilitated with an efficient data acquisition system digitizes and processes the signals from the detector, producing an electron energy loss spectrum. It may include signal amplifiers, ADCs, and data processing units to handle the large amount of data generated during the experiment.

g) Vacuum system: Like earlier discussed instruments, EELS also requires evacuated condition. Once vacuum is created the instrument can operate efficiently by minimizing electron scattering and interaction with air molecules, which gives accurate and reliable measurement. The vacuum system typically includes vacuum pumps, vacuum gauges, and vacuum seals to maintain a stable vacuum environment within the instrument.

h) Control and analysis software: Dedicated software controls the instrument operation, acquires data, and performs spectral analysis. It allows users to set experimental parameters, acquire EELS spectra, and analyze spectral features such as peak positions, intensities, and shapes. Advanced software packages may also offer tools for spectral deconvolution, background subtraction, and quantitative analysis.

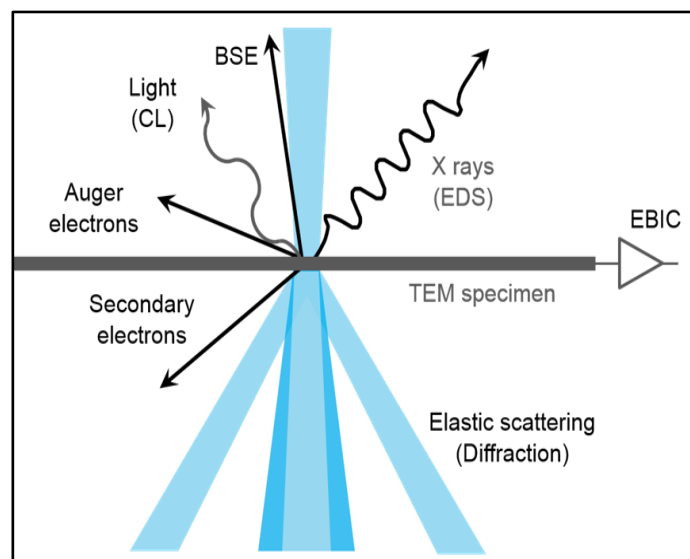


Fig. 6: A pictorial representation of Electron energy loss spectroscopy.

Overall, the instrumentation of EELS combines precision electron sources, optics, energy filters, detectors, and data analysis software to enable the accurate and sensitive analysis of electron energy loss spectra, providing valuable insights into the electronic and structural properties of materials at the nanometer scale.

3.2.2 Principle of EELS

EELS operates on the principle of measuring the energy loss of electrons as they interact with a sample, providing valuable information about the sample's electronic structure, chemical composition, and bonding characteristics. The principle can be summarized in several key steps:

a) High-energy electron beam: In EELS, a high-energy electron beam is directed onto a thin sample, such as a nanomaterial, semiconductor, or biological specimen. The energy of the incident electrons typically ranges from a few hundred eVs to several keVs.

b) Interaction with sample: As the high-energy electrons pass through the sample, they undergo interactions with the sample's atoms and electrons. These interactions can lead to various energy loss mechanisms, including:

c) Ionization: Electrons may lose energy by ionizing atoms within the sample, causing the ejection of inner-shell electrons.

d) Plasmon excitation: Electrons may excite collective oscillations of conduction electrons, known as plasmons, within the sample material.

e) Inner-shell excitation: Electrons may excite core electrons to higher energy levels, resulting in inner-shell ionization and Auger electron emission.

f) Energy loss spectrum: The energy loss spectrum obtained from EELS represents the energy distribution of electrons after interacting with the sample. It is typically recorded as a function of energy loss, measured in eVs.

g) Analysis of energy loss spectrum: By analyzing the energy loss spectrum, researchers can extract valuable information about the sample's electronic structure, chemical composition, and bonding characteristics. Peaks in the spectrum correspond to specific energy loss mechanisms, providing insights into the types of electron transitions and excitations occurring within the sample.

h) Quantitative analysis: Quantitative analysis of EELS spectra involves determining the relative intensities and positions of energy loss peaks, as well as comparing them to reference spectra and theoretical calculations. This allows for the identification of elemental constituents, chemical bonding states, and electronic excitations within the sample.

Overall, EELS provides a powerful means of probing the electronic and structural properties of materials at the nanometer scale, offering detailed insights into their composition, bonding, and electronic excitations. It is widely used in materials science, nanotechnology, semiconductor physics, and biological research for investigating a wide range of materials and phenomena.

3.2.3 Applications of EELS

EELS is a versatile analytical technique with numerous applications across various scientific disciplines. Some prominent applications include:

a) Materials science and nanotechnology: The electronic and structural properties of the materials at nanoscale can be investigated with EELS. It is employed to study nanoparticles, nanocomposites, thin films, and nanostructured materials, providing insights into their composition, chemical bonding, and electronic excitations. EELS is particularly valuable for characterizing advanced materials with tailored properties for applications in electronics, photonics, catalysis, and energy storage.

b) Semiconductor physics: modern electronics is based upon semiconductors. It becomes very important to study electronic structure; bandgap, and EELS give good insight into it. It provides information about band alignments, dopant distributions, interface states, and electronic defects, aiding in the optimization of semiconductor devices such as transistors, LEDs, photovoltaics, and sensors.

c) Catalysis and surface chemistry: Catalysts are the materials that can alter the rate of chemical reactions. EELS finds wide applications in catalysis research for studying the surface chemistry, catalytic mechanisms, and active sites of catalyst materials. It is used to investigate catalyst-support interactions, adsorbate species, reaction intermediates, and surface reconstruction phenomena, facilitating the design and development of efficient catalysts for chemical synthesis, environmental remediation, and energy conversion processes.

d) Biomaterials and biological imaging: Biomedical research is a developing sector. In biomedical research, EELS finds applications to study biomaterials, biological specimens, and cellular structures at the nanoscale. The chemical composition, elemental distribution, and structural organization of biological materials, including proteins, nucleic acids, membranes, and organelles, can be determined using EELS. Today, imaging techniques are widely used in medicine. Imaging is possible using various advanced techniques such as compact topographic scan and magnetic resonance imaging (MRI)-Spectroscopy. The imaging technique is a non-invasive technique and can be employed to determine abnormalities or malfunctions of the internal organs. Also, fractures and cracks in bone can be determined using imaging techniques. EELS is also used for biological imaging applications, such as electron energy loss imaging (EELI), to visualize cellular ultrastructure and molecular interactions.

e) Plasmonics and photonics: EELS plays a crucial role in plasmonics and photonics research for investigating the optical properties, plasmonic resonances, and light-matter interactions of nanostructured materials and plasmonic devices. It enables the characterization of plasmonic excitations, surface plasmon polaritons (SPPs), and localized surface plasmons (LSPs), guiding the design of plasmonic sensors, nanophotonic devices, and metamaterials with tailored optical properties and functionalities.

f) Environmental science and geosciences: EELS is utilized in environmental science and geosciences to study mineral compositions, geological specimens, and environmental samples. It provides information about mineral phases, crystal structures, elemental abundances, and chemical weathering processes, aiding in the identification and characterization of minerals, ores, and geological formations, as well as the assessment of environmental pollutants and contaminants.

Overall, EELS offers a powerful and versatile approach for investigating the electronic and structural properties of materials across diverse scientific disciplines, driving advancements in materials research, device fabrication, biomedical imaging, environmental monitoring, and fundamental scientific understanding.

3.2.4 Limitations of EELS

While EELS is a powerful technique for investigating the electronic and structural properties of materials, it also has several limitations:

a) Low spatial resolution: Achieving high spatial resolution in EELS can be challenging, particularly for thick or heterogeneous samples. The spatial resolution is limited by the electron beam diameter and the interactions of electrons with the sample, which may result in broadening of features and reduced spatial detail.

b) Sample thickness: EELS is primarily suited for analyzing thin samples or regions with low electron scattering, as thicker samples can lead to increased inelastic scattering and multiple scattering events, which can distort the energy loss spectrum and complicate data interpretation. Thicker samples may require thinning techniques such as ion milling or focused ion beam (FIB) milling to reduce scattering effects.

c) Surface sensitivity: EELS is inherently a surface-sensitive technique, probing only the top few nanometers to micrometers of the sample surface. Consequently, it may not provide information about bulk properties or buried interfaces within thick or layered materials.

d) Instrumentation complexity: EELS instruments are complex and require specialized expertise for operation and data analysis. The setup typically involves sophisticated electron optics, energy filters, detectors, and data acquisition systems, as well as stringent vacuum conditions to minimize electron scattering and interactions.

e) Beam damage and electron radiation effects: High-energy electron beams used in EELS can induce sample damage, such as electron beam-induced deposition, radiation damage, and specimen charging effects, particularly for sensitive or beam-sensitive materials. Minimizing beam damage requires careful optimization of beam parameters and sample handling procedures.

f) Quantitative analysis challenges: Quantitative analysis in EELS can be challenging due to factors such as detector efficiency, sample thickness variations, background signals, and uncertainties in energy calibration. Accurate quantification of elemental concentrations and electronic

excitations requires careful calibration, reference standards, and consideration of instrumental parameters.

g) Signal-to-noise ratio: EELS spectra may suffer from poor signal-to-noise ratio, particularly for low-energy loss features or weak signals, which can affect the detection and identification of spectral features. Signal averaging techniques and background subtraction methods may be employed to improve the signal-to-noise ratio, but these techniques may increase acquisition times and data processing complexity.

Despite these limitations, EELS remains a valuable technique for investigating the electronic and structural properties of materials at the nanometer scale, offering detailed insights into their composition, bonding, and electronic excitations. Ongoing advancements in instrumentation, data analysis techniques, and sample preparation methods continue to address many of these challenges, expanding the capabilities and applications of EELS in scientific research and industrial applications.

4. Magnetic resonance spectroscopy

Magnetic resonance spectroscopy (MRS) is a powerful analytical technique used to investigate the chemical structure, composition, and dynamics of molecules and materials by measuring the interactions of atomic nuclei with external magnetic fields and electromagnetic radiation. The principle of MRS is grounded in the fundamental properties of NMR, which describes how atomic nuclei with non-zero spin generate magnetic moments when placed in a magnetic field and how these magnetic moments can interact with RF radiation.

In MRS, a sample containing nuclei with non-zero spins, such as hydrogen (^1H), carbon (^{13}C), phosphorus (^{31}P), or fluorine (^{19}F), is placed in a strong and homogeneous magnetic field generated by a superconducting magnet. This magnetic field causes the nuclei to align with or against the field, resulting in a slight energy difference between the two alignments, known as the Zeeman splitting.

When the sample is irradiated with RF pulses at the resonant frequency corresponding to the Zeeman splitting, nuclei transition between energy levels, absorbing or emitting energy in the form of RF radiation. This phenomenon is known as NMR. The resonance condition, determined by the Larmor equation, depends on the strength of the magnetic field and the gyromagnetic ratio of the nucleus.

MRS exploits the unique chemical environments and interactions experienced by nuclei within a molecule or material, which influence their resonance frequencies and relaxation properties. By applying different RF pulse sequences and varying experimental parameters, such as pulse duration, repetition time, and magnetic field strength, MRS can selectively excite and manipulate nuclei of interest, leading to the acquisition of spectral data containing valuable information about the sample's chemical composition, molecular structure, and dynamic processes.

The acquired spectra, typically displayed as plots of signal

intensity versus frequency or chemical shift, reveal characteristic peaks corresponding to specific nuclei and chemical environments within the sample. By analyzing the positions, intensities, and line shapes of these peaks, MRS enables researchers to identify molecular species, quantify concentrations, elucidate chemical structures, and study molecular interactions in a wide range of applications, including chemistry, biochemistry, veterinary medicine, pharmacology, agriculture, medicine, materials science, and environmental science and engineering.

4.1 Nuclear magnetic resonance spectroscopy

Nuclear magnetic resonance (NMR) spectroscopy is a powerful analytical technique used to study the structure, composition, and dynamics of particularly organic molecules at the atomic level. It exploits the magnetic properties of atomic nuclei and their interactions with external magnetic fields to provide detailed information about molecular structure, chemical bonding, and molecular motion.

At its core, NMR spectroscopy relies on the principle of nuclear spin—the intrinsic angular momentum possessed by atomic nuclei due to the presence of unpaired protons or neutrons. When placed in an external magnetic field, these nuclei align themselves with or against the field, resulting in two energy states with slightly different energies. This energy difference between the two states corresponds to a characteristic resonant frequency known as the Larmor frequency, which is specific to each nucleus and its local chemical environment.

To probe these resonant frequencies, NMR spectroscopy employs RF radiation, typically in the radiofrequency region of the electromagnetic spectrum. By applying RF pulses at the Larmor frequency of the nuclei of interest, the nuclei can be selectively excited from one energy state to another. When the RF pulse is turned off, the excited nuclei relax back to their lower energy state, emitting energy in the form of RF radiation, which is detected and analyzed.

The information obtained from NMR spectroscopy is encoded in the form of a spectrum, which represents the intensity of the emitted RF radiation as a function of frequency. Fig. 7 shows a schematic diagram of NMR spectroscopy. Each peak in the spectrum corresponds to a specific resonance frequency associated with different nuclei within the molecule. The chemical shifts, intensities, and multiplicity of these peaks provide valuable insights into the chemical environment, connectivity, and spatial arrangement of atoms within the molecule.

NMR spectroscopy offers several advantages, including non-destructive analysis, high sensitivity, and the ability to study samples in solution, solid state, or even living organisms. It is widely used in various scientific disciplines, including chemistry, veterinary science, physics, biochemistry, pharmaceuticals, materials science, medical diagnostics, environmental science, and engineering for applications such as structure elucidation, compound identification, reaction

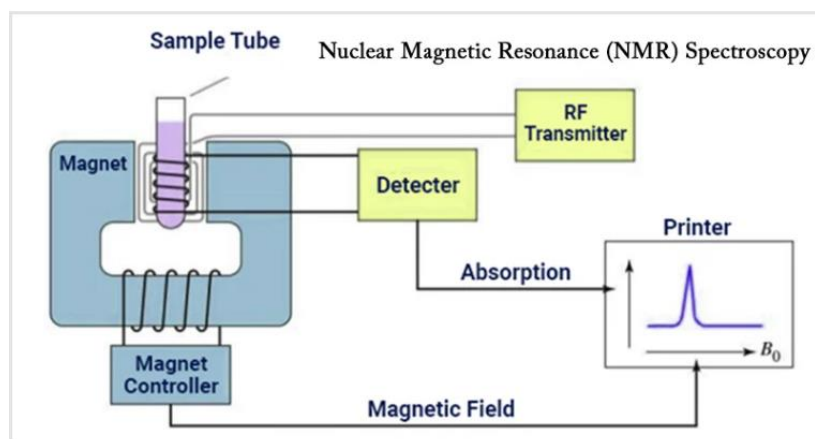


Fig. 7: A pictorial representation of Nuclear magnetic resonance spectroscopy.

monitoring, and protein folding studies.

Overall, NMR spectroscopy is a versatile and indispensable tool for chemical analysis and molecular characterization, providing valuable information about the properties and behavior of molecules in diverse research and industrial settings.

4.1.1 Instrumentation of NMR spectroscopy

Here's an overview of its key components:

a) Magnet: The primary component of an NMR spectrometer is a powerful magnet, typically a superconducting magnet, which generates a strong and uniform magnetic field. The strength of the magnetic field is measured in units of Tesla (T) or Gauss (G), with higher field strengths providing better resolution and sensitivity.

b) Radiofrequency system: The RF system generates radiofrequency pulses used to manipulate the nuclear spins in the sample and to detect the resulting NMR signals. It consists of RF transmitters, amplifiers, and antennas (coils), which generate the magnetic fields necessary to excite and detect nuclear spin transitions.

c) Sample holder: The sample holder, or probe, holds the NMR sample in a precisely defined position within the magnet's magnetic field. The sample may be in the form of a solution, solid, or gas, depending on the type of NMR experiment being performed. The probe typically contains RF coils for transmitting and receiving RF signals, as well as temperature control systems to maintain sample temperature.

d) Gradient coils: Gradient coils are used to generate spatially varying magnetic field gradients, which are essential for spatial encoding in multidimensional NMR experiments. By applying gradient pulses during signal acquisition, spatial information can be encoded along different dimensions, allowing for the spatial localization of NMR signals.

e) Receiver system: The receiver system amplifies and detects the NMR signals emitted by the sample following RF excitation. It consists of RF receivers, preamplifiers, and signal-processing electronics, which amplify and digitize the NMR signals for further processing and analysis.

f) Computer control and data acquisition system: Modern

NMR spectrometers are controlled by computer software, which enables users to set experimental parameters, acquire NMR spectra, and perform data processing and analysis. The computer also controls the timing of RF pulses, gradient pulses, and data acquisition, ensuring precise and accurate measurements.

g) Shimming systems: Shimming systems are used to optimize the homogeneity of the magnetic field within the NMR sample. This is achieved by adjusting the currents in shim coils located around the main magnet to minimize magnetic field variations and distortions, which can lead to the broadening of NMR signals and reduced spectral resolution.

h) Sample changer and automation: Some NMR spectrometers are provided with sample changers and automation systems, which allow for the automated handling of multiple samples. This enables high-throughput NMR experiments and sample screening, particularly in applications such as metabolomics, drug discovery, pharmaceutical science, veterinary medicine, environmental science, engineering quality control, etc.

Overall, the instrumentation of NMR spectroscopy comprises a combination of powerful magnets, RF systems, gradient coils, detectors, and computer control systems, which work together to generate, manipulate, and detect NMR signals from the sample. These sophisticated instruments enable the characterization of molecular structure, dynamics, and interactions in a wide range of chemical and biological systems.

4.1.2 Principle of NMR spectroscopy

The principle can be summarized as follows:

a) Magnetic properties of nuclei: Atomic nuclei possess a property known as nuclear spin, which arises from the presence of protons and neutrons. Nuclei with an odd number of protons or neutrons have a nonzero nuclear spin and behave like tiny magnets with a magnetic moment.

b) Alignment in magnetic field: When placed in an external magnetic field, the nuclear magnetic moments of the nuclei align either parallel (low-energy state) or antiparallel (high-energy state) with the direction of the magnetic field. The

energy difference between these two states is proportional to the strength of the magnetic field and is determined by the nuclei's gyromagnetic ratio.

c) Resonance condition: If electromagnetic radiation with the appropriate frequency is applied to the sample, it can induce transitions between the low-energy and high-energy nuclear spin states. This frequency, known as the Larmor frequency, is determined by the strength of the magnetic field and the gyromagnetic ratio of the nuclei according to the equation:

$$\omega = \gamma B_0 \quad (1)$$

where ω is the angular frequency, γ is the gyromagnetic ratio, and B_0 is the magnetic field strength.

d) Absorption of radiation: When the Larmor frequency matches the frequency of the applied radiation, a resonance condition occurs, and the nuclei absorb energy from the radiation, transitioning from the lower-energy state to the higher-energy state. This absorption of radiation results in the excitation of nuclear spins.

e) Relaxation processes: After excitation, the nuclei return to their original state through relaxation processes known as spin-lattice relaxation (T1) and spin-spin relaxation (T2). During these processes, the excess energy is released as NMR signals, which can be detected and analyzed to provide information about the chemical environment and molecular structure of the sample.

f) NMR signals: The NMR signals are detected as variations in the absorption or emission of electromagnetic radiation by the sample. These signals, typically recorded as a function of frequency or chemical shift, represent the resonance frequencies of different nuclear species in the sample and provide information about their chemical composition and molecular environment.

Overall, NMR spectroscopy exploits the magnetic properties of atomic nuclei to probe the chemical structure, composition, and dynamics of molecules in a sample. By measuring the resonance frequencies and intensities of NMR signals, valuable information about molecular structure, conformation, and interactions can be obtained, making NMR spectroscopy a powerful tool in chemistry, biochemistry, materials science, and many other scientific disciplines.

4.1.3 Applications of NMR spectroscopy

Some prominent applications of NMR spectroscopy include:

a) Chemical analysis and structure elucidation: NMR spectroscopy is widely used for the determination of the structure of organic compounds and organometallic compounds and the study of the physical, chemical, and biological properties of matter also medical and veterinary doctors are widely using magnetic resonance imaging, a multidimensional NMR imaging technique for diagnosis of disease or internal injury. This analytical technique provides valuable information regarding molecular connectivity, stereochemistry and various functional groups present. Thus, NMR spectroscopy is a popular analytical tool based on synthetic organic chemistry, natural product chemistry, and

drug discovery.

b) Protein and biomolecular structure determination: NMR spectroscopy finds extensive applications in structural biology and biochemistry for structural elucidation, particularly of giant bio-molecular complexes, proteins, nucleic acids, *etc.* It provides insights into protein folding, dynamics, and interactions, facilitating the study of biomolecular mechanisms, drug binding, and disease pathways.

c) Metabolomics and metabolic profiling: Veterinary pathology is an important branch of veterinary science that deals with the study of causes of disease. Since earlier periods, attempts have been made to elucidate the cause of disease, and with the progress of time, the methods for the detection of disease have been replaced. In modern scenarios, various advanced techniques are used in the diagnosis of diseases.

d) Medical imaging and diagnosis: NMR spectroscopy is utilized in MRI for non-invasive medical imaging and diagnosis of anatomical structures and pathological conditions in the body. Magnetic Resonance Imaging technique can be used for high-resolution imaging of soft tissues, organs, and physiological processes. This technique can be used to detect fractures, the presence of neoplastic tissues, *etc.* However, it is worth mentioning that this is not a confirmatory test for the diagnosis of a tumor. We would like to mention here glioma is a deadly brain tumor. It is a primary tumor. Hence, every good physician should try to co-relate the clinical symptoms and should not merely depend upon MRI imaging.

e) Material characterization and solid-state NMR: NMR spectroscopy is employed in materials science for characterizing the chemical composition, molecular structure, and physical properties of materials such as polymers, catalysts, nanoparticles, and surfaces. Solid-state NMR techniques provide insights into crystalline and amorphous materials, interfaces, and nanoscale structures, aiding in materials design, development, and quality control.

f) Environmental analysis and pollution monitoring: In the modern period, environmental pollution is increasing day by day. Attempts are going on globally to protect the environment and earth. Environmental pollution can be estimated using various techniques. One of the important techniques for analyzing environmental pollution is NMR spectroscopy. It provides information about the composition, transformation, and fate of environmental substances, supporting environmental monitoring, risk assessment, and remediation efforts.

g) Food and beverage analysis: NMR spectroscopy is applied in the food and beverage industry for analyzing the composition, quality, and authenticity of food products, ingredients, and beverages. It enables the detection of nutritional components, additives, contaminants, and adulterants, contributing to food safety, quality control, and product development.

h) Pharmacology and drug development: NMR spectroscopy is used in pharmacology and drug development

to study drug-receptor interactions, drug metabolism, and drug formulation. It provides insights into drug binding sites, affinity, and specificity, facilitating the design and optimization of therapeutic agents and pharmaceutical formulations.

Overall, NMR spectroscopy offers a versatile and powerful analytical tool for addressing a wide range of scientific and technological challenges in chemistry, biology, medicine, veterinary science, agriculture, materials science, environmental science, and food science, driving advancements in research, innovation, and industry.

4.1.4 Limitations of NMR spectroscopy

Some of the limitations of NMR spectroscopy are mentioned below.

a) Resolution: Despite advancements in NMR instrumentation, achieving high-resolution spectra can be challenging for complex samples or samples with overlapping signals. Spectral resolution may be limited by factors such as line broadening due to molecular motion, signal-to-noise ratio, and spectral congestion.

b) Sensitivity: NMR spectroscopy typically requires a relatively large amount of sample, often in the milligram to microgram range, depending on the sensitivity of the NMR instrument and the properties of the sample being analyzed. This requirement can be challenging for samples with limited availability or low concentration.

c) Signal overlap: When complex molecule is analyzed signal overlap may take place. In case of signal overlap, it becomes difficult to analyze the compounds. This can complicate data interpretation and analysis, particularly in metabolomics, natural product chemistry, and environmental samples.

d) Instrument cost and maintenance: NMR instrument is a complex instrument and is expensive. It is expensive to purchase, maintain and even requires skilled man power. Apart from these the cost of consumable, such as deuterated solvents and NMR tubes, can also be significant.

e) Sample preparation: The sample cannot be directly used as obtained. For the analysis we need to carefully prepare the sample. The sample preparation requires skilled hands and needs to maintain sample purity, stability and solubility. For biological samples, such as proteins and nucleic acids, isotopic labeling may be necessary to enhance sensitivity and spectral resolution, adding complexity and cost to experiments.

f) Magnetic field homogeneity: NMR spectroscopy relies on a uniform magnetic field for accurate and reliable measurements. Magnetic field inhomogeneities, arising from imperfections in the magnet or sample positioning, can lead to spectral distortions, line broadening, and reduced sensitivity.

g) Limited information depth: Although NMR Spectroscopy is a powerful analytical tool, it is effective only up to the upper strata of the material. The internal structure of the material cannot be determined using NMR techniques. Consequently, it may not provide the required information about the bulk of the material under study. It is to be noted that

if the surface composition and bulk composition are different, then perhaps NMR cannot give detailed information about the bulk composition of the material.

h) Time consumption: NMR experiments can be time-consuming, particularly for multidimensional experiments or samples requiring long acquisition times to achieve a sufficient signal-to-noise ratio. This can limit the throughput of NMR analyses, especially in high-throughput screening applications.

Despite these limitations, NMR spectroscopy remains a valuable and widely used technique for structural elucidation, chemical analysis, and molecular characterization in chemistry, biology, materials science, and many other scientific disciplines. Ongoing advancements in NMR instrumentation, methodologies, and data analysis techniques continue to address many of these challenges, expanding the capabilities and applications of NMR spectroscopy in research, industry, and academia.

5. Mass spectroscopy

Mass spectrometry (MS) is an analytical technique used to identify and quantify molecules based on their mass-to-charge ratio (m/z). The principle of mass spectrometry relies on the separation, detection, and analysis of ions generated from a sample through the interaction with high-energy particles or electromagnetic radiation. The process involves several key steps such as ionization, ion separation, detection, and analysis. The difference between magnetic and mass spectroscopy is illustrated in Table 2. Overall, the principle of mass spectrometry enables the accurate and sensitive analysis of a wide range of compounds, including small molecules, proteins, peptides, nucleic acids, and polymers. Mass spectrometry finds applications in various fields such as chemistry, biochemistry, pharmaceuticals, veterinary medicine, nanoscience and technology, material sciences, agriculture sciences, agro-chemistry and pest management, environmental science, forensics, and proteomics, providing valuable insights into the composition, structure, and function of molecules in complex samples.

5.1 Time-of-flight mass spectrometry

Time-of-Flight Mass Spectrometry (TOF-MS) is a powerful analytical technique used for the rapid and accurate determination of the mass-to-charge ratio (m/z) of ions in a sample. It operates on the principle of measuring the time taken by ions to travel a known distance under the influence of an electric field, allowing for the determination of their mass-to-charge ratio.

In TOF-MS, the sample is ionized, typically using techniques such as electron impact ionization, electrospray ionization (ESI), or matrix-assisted laser desorption/ionization (MALDI). The resulting ions are accelerated into the flight tube, a vacuum chamber maintained at low pressure, by applying a high-voltage electric field. Fig. 8 indicates the schematic diagram of time-of-flight mass spectroscopy. As the

Table 2: Differentiate between Magnetic Spectroscopy and Mass Spectroscopy.

Aspect	Magnetic spectroscopy	Mass spectroscopy
Principle	Measures the interaction of magnetic fields with magnetic moments or spin states of electrons and nuclei.	Measures the mass-to-charge ratio (m/z) of ions to identify and quantify molecules.
Interaction Type	Involves magnetic field interactions with electron/nuclear spins or magnetic properties.	Involves ionizing molecules and separating ions based on their mass-to-charge ratio in electric/magnetic fields.
Energy Range	Involves low-energy transitions, typically in the radiofrequency or microwave range.	Ions are accelerated to high kinetic energies; m/z ratios span from low to very high, depending on the ion.
Techniques	Nuclear Magnetic Resonance, Electron Spin Resonance (ESR/EPR), Mössbauer Spectroscopy.	Time-of-flight (TOF), Quadrupole Mass Spectrometry, Matrix-Assisted Laser Desorption Ionization (MALDI), Electrospray Ionization (ESI).
Type of Information	Provides information on magnetic properties, spin states, electronic environments, and hyperfine interactions.	Provides molecular mass, structural information, and isotopic composition of compounds.
Sample Types	Applicable to solids, liquids, and biological samples containing magnetic nuclei or unpaired electrons.	Applicable to gases, liquids, solids, biomolecules, and complex mixtures.
Sample Ionization	No ionization of the sample is required.	Ionization is essential (e.g., by electron impact, MALDI, or ESI) to create ions for analysis.
Data Output	Spectra showing resonance frequencies, spin transitions, and magnetic interactions.	Spectra displaying peaks that correspond to mass-to-charge (m/z) ratios of ions.
Sensitivity	Sensitive to magnetic environments and unpaired electron or nuclear spin states.	Sensitive to molecular mass and structure; capable of detecting trace quantities of molecules.
Quantitative Analysis	Generally qualitative, although NMR can provide quantitative information about molecular environments.	Highly quantitative, allowing precise determination of molecular weights and concentrations.
Magnetic Sensitivity	Sensitive to magnetic properties, spin states, and hyperfine interactions.	Not sensitive to magnetic properties. Focuses on the mass and charge of ions.
Applications	Used for studying molecular structures, magnetic properties, and local atomic environments.	Used for molecular identification, structural elucidation, quantification of biomolecules, and proteomics.
Sample Preparation	Requires magnetic field alignment; may need specific solvents or conditions (e.g., deuterated solvents for NMR).	Requires ionization, may involve sample fragmentation, and often requires sample purification.
Destructive/Non-destructive	Typically, non-destructive (NMR, ESR), though some techniques (Mössbauer) may slightly alter samples.	Often destructive due to ionization and fragmentation of molecules.
Resolution	Resolves fine details of electron/nuclear spin environments.	Resolves fine details of molecular mass and isotopic composition with high mass resolution.

ions travel through the flight tube, they experience a constant accelerating force and gain kinetic energy proportional to their charge-to-mass ratio. Heavier ions, with lower charge-to-mass ratios, take longer to reach the detector compared to lighter ions, with higher charge-to-mass ratios.

The time taken by ions to traverse the flight tube is directly proportional to their mass-to-charge ratio, according to the equation:

$$T = kmz \quad (5)$$

where T is the time of flight, k is a proportionality constant, m is the mass of the ion, and z is the charge of the ion.

By measuring the time of flight of ions to the detector, TOF-MS allows for the determination of their mass-to-charge ratios. The resulting mass spectrum displays peaks representing ions of different mass-to-charge ratios present in the sample.

TOF-MS offers several advantages over other mass spectrometry techniques, including high sensitivity, high mass

resolution, wide mass range, and fast data acquisition rates. It is particularly well-suited for analyzing complex mixtures, high-throughput screening, and characterizing biomolecules such as proteins, peptides, nucleic acids, and metabolites.

Overall, TOF-MS is a versatile and widely used analytical technique with applications spanning chemistry, biochemistry, veterinary sciences, agriculture, agro-chemistry, pharmaceuticals, environmental science, and materials science, providing valuable insights into the composition, structure, and behavior of molecules in diverse sample types.

5.1.1 Instrumentation of TOF-MS

The instrumentation of TOF-MS involves several key components that work together to ionize, accelerate, and detect ions based on their mass-to-charge ratio (m/z). The components involved in TOF-MS are discussed as follows:

a) Ion source: The ion source is responsible for generating ions from the sample molecules. Standard ionization techniques used in TOF-MS include electron impact ionization, ESI, MALDI, and laser ablation. The choice of ion source depends on the nature of the sample and the desired ionization method.

b) Ion acceleration region: In TOF-MS, ions are accelerated into the flight tube or drift region by applying a high-voltage electric field. This region typically consists of a series of electrodes or grids that create a uniform electric field to accelerate the ions toward the detector.

c) Flight tube or drift region: The flight tube or drift region is a vacuum chamber maintained at low pressure where ions travel under the influence of the electric field. The flight tube is typically several meters long and may contain additional focusing elements to improve mass resolution and ion transmission.

d) Ion detector: At the end of the flight tube, ions are detected by an ion detector, which converts the arrival time of ions into electrical signals. Common types of ion detectors used in TOF-MS include electron multipliers, microchannel plates, and Faraday cups. The choice of ion detector depends on factors such as sensitivity, dynamic range, and response time.

e) Data acquisition system: The data acquisition system digitizes and processes the electrical signals from the ion detector, recording the arrival times and intensities of ions as a function of time. This system typically includes ADCs, signal processing electronics, and data acquisition software for real-time data analysis and visualization.

f) Time-to-digital converter: In TOF-MS, the time of flight of ions is measured with high precision using the TDC. The TDC converts the arrival times of ions into digital signals with nanosecond or picosecond resolution, allowing for accurate determination of mass-to-charge ratios.

g) Control electronics and software: Like other modern characterization tools, TOF-MS is also equipped with electronics and software, which enable users to set experimental parameters, control instrument operation, and acquire mass spectra. Advanced software packages may also

offer tools for data processing, peak deconvolution, and spectral interpretation.

Time of Flight Mass Spectrometry

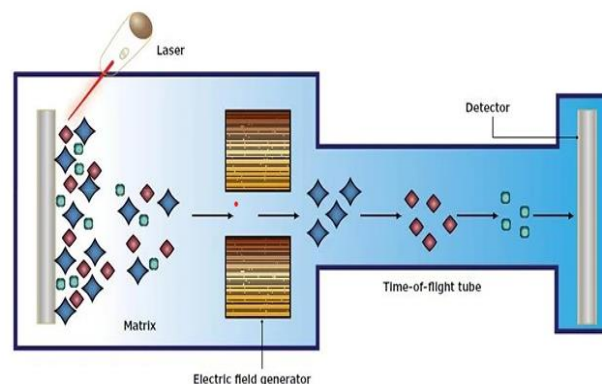


Fig. 8: A pictorial representation of time-of-flight mass spectroscopy.

Overall, the instrumentation of TOF-MS combines ion sources, ion acceleration regions, flight tubes or drift regions, chemical sensors that act as ion detectors, data acquisition systems, and control electronics to enable the rapid and accurate analysis of ions based on their mass-to-charge ratios. These components work together to provide high sensitivity, high mass resolution, and wide mass range capabilities for a wide range of applications in chemistry, biology, environmental science, and materials science.

5.1.2 Principle of TOF-MS

The TOF-MS operates on the principle of measuring the time taken by ions to traverse a known distance under the influence of an electric field, allowing for the determination of their mass-to-charge ratio (m/z).

The principle can be summarized in several key steps:

a) Ionization: The sample is ionized using various ionization techniques such as electron impact ionization, ESI, MALDI, or laser ablation. This process generates ions from the sample molecules, which may be positively or negatively charged depending on the ionization method.

b) Acceleration: Once the ions are generated, it is required to accelerate them. The acceleration of ions is possible in the flight tube or drift region by applying the appropriate amount of high voltage. This electric field imparts kinetic energy to the ions, causing them to accelerate towards the detector.

c) Flight: As the ions travel through the flight tube or drift region, they experience a constant accelerating force and gain kinetic energy proportional to their charge-to-mass ratio (m/z). The ions do not travel at uniform speed. Heavier ions, with lower m/z ratios, take longer to traverse the flight tube compared to lighter ions, with higher m/z ratios.

d) Time measurement: The time taken by ions to travel from the ion source to the detector is measured with high precision using a TDC. This device converts the arrival times of ions into digital signals with nanosecond or picosecond resolution.

e) Detection and analysis: The ions are detected at the end of the flight tube or drift region by an ion detector, which converts their arrival times into electrical signals. These signals are then processed and analyzed to generate the mass spectrum, providing information about the mass distribution and composition of the sample.

Overall, TOF-MS provides a rapid, sensitive, and accurate method for analyzing ions based on their mass-to-charge ratios, making it a valuable tool in a wide range of scientific and industrial applications, including chemistry, biology, pharmaceuticals, environmental science, and materials science.

5.1.3 Applications of TOF-MS

Some prominent applications of TOF-MS include:

a) Proteomics and protein analysis: Biological systems have many supramolecular structures, including proteins, carbohydrates, nucleic acids, *etc.* TOF-MS can be effectively used in proteomics for the identification, characterization, and quantification of proteins and peptides. It enables the analysis of complex protein mixtures, post-translational modifications, protein-protein interactions, and protein expression profiling, aiding in biomarker discovery, disease diagnosis, and drug development.

b) Metabolomics and metabolic profiling: Metabolism is the fundamental characteristic of living beings. There are certain types of disorders arising due to metabolic dysfunctions. Therefore, the determination of metabolic pathways is of prime importance in the treatment of such conditions. The advanced characterization tool TOF-MS is employed in metabolomics for analyzing the metabolites present in biological samples such as blood, urine, and tissues. It allows for the comprehensive profiling of metabolites, metabolic pathways, and metabolic fluxes, providing insights into cellular metabolism, physiological processes, and disease states. Thus, it becomes possible to monitor the disease and plan effectively the treatment.

c) Pharmaceutical analysis and drug discovery: For non-surgical treatment of illness medicines are important. Pharmaceutical science is the domain concerned with the manufacture of medicinal drugs. TOF-MS is utilized in pharmaceutical analysis for the identification, quantification, and characterization of drug compounds, metabolites, and impurities. The metabolism of the drug is another important factor. In order to determine the metabolism of the drug, we determine the half-life period. TOF-MS enables high-throughput screening of compound libraries, drug metabolism studies, pharmacokinetic analysis, and quality control of pharmaceutical products.

d) Environmental monitoring and toxicology: TOF-MS is used in environmental science and engineering to analyze pollutants, contaminants, and natural organic matter in environmental samples such as water, soil, and air. It allows for the detection and quantification of environmental pollutants, pesticides, toxins, and emerging contaminants,

supporting environmental monitoring, risk assessment, and regulatory compliance.

e) Food and beverage analysis: TOF-MS is applied in the food and beverage industry for analyzing the composition, quality, and safety of food products, ingredients, and beverages. It enables the detection of food contaminants, additives, allergens, and adulterants, as well as the profiling of flavor compounds and nutritional components, contributing to food safety, quality assurance, and product development.

f) Clinical diagnostics and biomarker discovery: With the onset and advancement of the disease, various conditions change from normal physiological conditions. Thus, the biomarkers can be used for the effective detection of a diseased condition of an individual. TOF-MS is employed in clinical diagnostics to detect biomarkers associated with various diseases and medical conditions. It allows for identifying and quantifying disease biomarkers in biological samples such as blood, urine, and cerebrospinal fluid, aiding in early disease detection, prognosis, and personalized medicine.

g) Materials analysis and characterization: TOF-MS is used in materials science to analyze the composition, structure, and properties of materials such as polymers, nanoparticles, and thin films. It enables the detection of elemental composition, surface chemistry, molecular weight distribution, and degradation products, supporting materials characterization, quality control, and product development.

h) Forensic analysis and homeland security: TOF-MS is employed in forensic science for analyzing forensic evidence such as drugs, explosives, and trace materials. It allows for the detection of illicit substances, identification of chemical signatures, and forensic profiling of samples, assisting in criminal investigations, forensic casework, and border security.

Thus, TOF-MS offers a versatile and powerful analytical tool for addressing a wide range of scientific, industrial, and societal challenges, providing valuable insights into the composition, structure, and behavior of molecules in diverse sample types.

5.1.4 Limitations of TOF-MS

The limitation of TOF-MS can be encountered as below:

a) Resolution: TOF-MS typically offers lower mass resolution compared to other mass analyzers such as quadrupole and ion trap mass spectrometers. This limitation can affect the ability to resolve ions with similar mass-to-charge ratios, leading to spectral overlap and potential misinterpretation of results.

b) Dynamic range: TOF-MS may have limitations in its dynamic range, particularly for samples with a wide range of ion abundances. High-abundance ions can saturate the detector, leading to signal suppression or loss of sensitivity for lower-abundance ions. This can affect the accuracy and reliability of quantitative measurements.

c) Mass accuracy: High mass accuracy in TOF-MS can be challenging, particularly for large biomolecules or high-mass

ions. Factors such as ionization efficiency, flight tube length, and detector calibration can introduce errors in mass measurements, affecting the accuracy of molecular identification and quantification.

d) Sample introduction: The ionization techniques used in TOF-MS may require careful sample preparation and optimization to achieve efficient ionization and ion transmission. Variability in sample composition, matrix effects, and ion suppression can affect the reproducibility and reliability of results, particularly for complex samples.

e) Data processing and analysis: Like other modern scientific instruments, TOF-MS is equipped with electronic circuits and soft-wares. This can work on spectral processing, peak deconvolution, and molecular identifications. Incorrect parameter settings or data processing methods can lead to artifacts, false positives, or misinterpretation of results.

f) Fragmentation: TOF-MS may produce significant fragmentation of analyte ions, particularly for large biomolecules such as proteins and peptides. Fragmentation can complicate spectral interpretation and reduce the ability to obtain intact molecular ions for structural analysis.

g) Sensitivity to ion velocity: TOF-MS is sensitive to variations in ion velocity, which can arise from differences in ionization conditions, flight tube length, and ion mobility effects. Variations in ion velocity can lead to the broadening of peaks and reduced mass resolution, particularly for ions with different charge states or kinetic energies.

h) Sample throughput: TOF-MS may have limitations in sample throughput compared to other mass spectrometry techniques, particularly for high-resolution analyses or complex multidimensional experiments. Longer data acquisition times and data processing requirements can limit the number of samples that can be analyzed within a given time frame.

Despite these limitations, TOF-MS remains a valuable analytical technique with widespread applications in chemistry, biology, pharmaceuticals, environmental science, and materials science. Ongoing advancements in instrumentation, data analysis methods, and sample preparation techniques continue to address many of these challenges, expanding the capabilities and applications of TOF-MS in scientific research and industrial settings.

5.2 Matrix-assisted laser desorption/ionization mass spectrometry

Matrix-assisted laser desorption/ionization mass spectrometry (MALDI-MS) is a powerful analytical technique used for the rapid and sensitive analysis of biomolecules, polymers, and other organic compounds. It enables the characterization and identification of molecules based on their mass-to-charge ratio (m/z) and is widely used in various scientific disciplines, including proteomics, metabolomics, lipidomics, and clinical diagnostics.

In MALDI-MS, the sample molecules are co-crystallized or mixed with a matrix compound on a solid support, such as

a stainless steel target plate. The matrix serves multiple purposes, including absorbing laser energy, promoting desorption/ionization of analyte molecules, and enhancing ionization efficiency. Common matrix compounds used in MALDI-MS include α -cyano-4-hydroxycinnamic acid (CHCA), 2,5-dihydroxybenzoic acid (DHB), and sinapinic acid.

Overall, MALDI-MS is a versatile and powerful analytical technique that enables the rapid and sensitive analysis of a wide range of molecules, making it an indispensable tool in modern analytical chemistry and life sciences. Fig. 9 represents the MALDI-mass spectroscopy.

5.2.1 Instrumentation of MALDI-MS

MALDI-MS is a powerful analytical technique used to analyze biomolecules such as peptides, proteins, nucleic acids, and carbohydrates. The instrumentation of MALDI-MS typically consists of several key components:

a) Laser system: A pulsed laser system is used to irradiate the sample, causing desorption and ionization of analyte molecules from the sample matrix. The laser wavelength is typically in the UV or near-infrared (NIR) range.

b) Sample target: The sample to be analyzed is applied onto a target plate. This target plate is typically made of a conductive material such as stainless steel or indium tin oxide (ITO) coated glass. The sample is mixed with a matrix compound, which helps to facilitate desorption and ionization of analyte molecules.

c) Matrix solution: The matrix solution is prepared by dissolving a suitable organic compound in a solvent. Common matrices used in MALDI-MS include α -cyano-4-hydroxycinnamic acid, sinapinic acid, and 2,5-dihydroxybenzoic acid. The choice of matrix depends on the type of analyte and the desired ionization mechanism.

d) Ion optics: After desorption and ionization, the ions are accelerated and focused using a series of ion optics components such as ion lenses and ion mirrors. These components help to improve the efficiency of ion transmission from the ion source to the mass analyzer.

e) Mass analyzer: The mass analyzer separates the ions based on their mass-to-charge ratio (m/z). Common types of mass analyzers used in MALDI-MS include TOF, ion trap, and Fourier transform ion cyclotron resonance (FT-ICR) analyzers. TOF analyzers are most commonly used in MALDI-MS due to their high sensitivity and high mass resolution.

f) Detector: The detector detects the ions that reach the mass analyzer and generates an electrical signal proportional to their abundance and mass-to-charge ratio. PMTs or microchannel plate detectors are commonly used as detectors in MALDI-MS instruments.

g) Data acquisition and analysis software: Data acquisition software controls the instrument operation and collects mass spectra from the detector. Analysis software is used to process and interpret the mass spectra, identify analyte peaks, and perform quantitation and characterization of analytes.

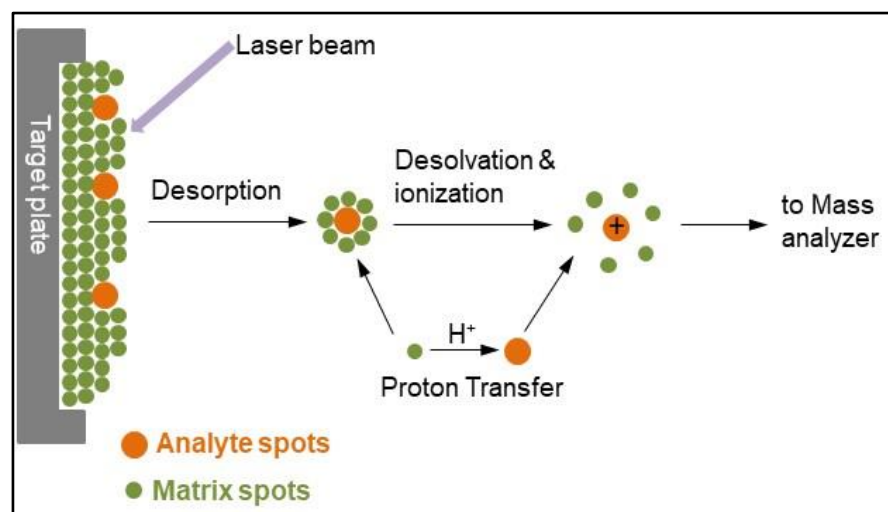


Fig. 9: A pictorial representation of MALDI-mass spectroscopy.

Overall, MALDI-MS instrumentation combines laser technology, ion optics, mass analyzers, and sophisticated data analysis software to enable high-throughput and sensitive analysis of biomolecules with minimal sample preparation.

5.2.2 Principle of MALDI-MS

MALDI-MS is a technique used for the analysis of biomolecules, such as peptides, proteins, nucleic acids, and carbohydrates. The principle of MALDI-MS involves several key steps:

a) Sample preparation: The sample cannot be directly analyzed using MALDI-MS. The sample requires preparation; here, we mix the sample with a matrix compound. The matrix compound may work for different purposes, including assisting in the desorption and ionization of the analyte to be analyzed mixed with a matrix compound. The matrix serves multiple purposes, including assisting in the desorption and ionization of analyte molecules, protecting them from degradation during the process, and facilitating the formation of ions suitable for mass analysis. The common matrix used in the process may be benzoic acid and cinnamic acids. The inorganic material such as graphite can also be used as the matrix.

b) Desorption and ionization: The sample-matrix mixture is applied to a target plate and allowed to dry, forming a thin film. A pulsed laser beam is then directed onto the sample, causing rapid heating and vaporization of the matrix molecules. This process desorbs and ionizes the analyte molecules present in the sample, generating gas-phase ions.

c) Ionization mechanism: The exact mechanism of ionization in MALDI-MS is complex and can vary depending on the analyte and matrix used. However, it generally involves processes such as protonation, deprotonation, and adduct formation. The matrix absorbs the laser energy and transfers it to the analyte molecules, leading to their ionization.

d) Ion acceleration and detection: The generated ions are accelerated into the mass analyzer by an electric field. In TOF mass analyzers, which are commonly used in MALDI-MS,

ions are accelerated to high velocities and separated based on their mass-to-charge ratio (m/z). Lighter ions reach the detector before heavier ions, allowing for their separation and detection.

e) Mass analysis and detection: The ions reaching the detector generate an electrical signal proportional to their abundance and mass-to-charge ratio. This signal is then processed to produce a mass spectrum, which represents the distribution of ions as a function of their m/z values. The mass spectrum provides information about the mass and relative abundance of the analyte molecules present in the sample.

Overall, the principle of MALDI-MS involves the use of a matrix compound to facilitate the desorption and ionization of analyte molecules upon laser irradiation, followed by mass analysis and detection of the generated ions. This technique enables the sensitive and selective analysis of biomolecules with high throughput and minimal sample preparation.

5.2.3 Applications of MALDI-MS

MALDI-MS finds applications across various fields due to its ability to analyze a wide range of biomolecules with high sensitivity and accuracy. Some of the critical applications include:

a) Proteomics: Proteomics is the complete evaluation of the function and structure of proteins to understand an organism's nature. MALDI-MS is extensively used in proteomics to analyze proteins and peptides. This technique can be used to identify, characterize, and quantify the protein molecules in complex mixtures, such as cell lysates, tissues, and body fluids. MALDI-MS can be coupled with techniques like liquid chromatography (LC) to enhance proteome coverage and identify post-translational modifications. MALDI-MS can be connected with techniques like LC to enhance proteome coverage and identify post-translational modifications.

b) Clinical diagnostics: Correct diagnosis of the disease is the key factor in success of treatment. There is continuous evolution in the process of clinical diagnosis. In the earlier day's doctors or then practitioners used to detect the disease

based on their own skills. However, in modern world the disease is generally confirmed with the help of advanced instruments. Few decades back various tests such as hematological tests were performed based upon biochemical tests. With the advancement in electronics, many of the tests are replaced with electronic instruments. Today we are aware that blood glucose assay is possible in home also by unskilled personnel's also. Similarly, MALDI-MS has emerged as a valuable tool in clinical diagnostics for the analysis of biomarkers associated with diseases such as cancer, cardiovascular disorders, and infectious diseases. It enables the rapid and sensitive detection of disease-specific molecules in patient samples, offering potential for early diagnosis, prognosis, and monitoring of therapeutic responses.

c) Pharmaceutical analysis: MALDI-MS is used in pharmaceutical research and development for drug discovery, pharmacokinetic studies, and quality control. It facilitates the analysis of small molecules, peptides, and proteins, aiding in the identification of drug candidates, determination of drug metabolism pathways, and assessment of drug purity and stability.

d) Glycomics: The tool of glycomics can be used for several clinical applications such as disease diagnosis, detection of cancer, personalized medicine, *etc.* MALDI-MS is employed in glycomics for the analysis of carbohydrates and glycoproteins. It enables the profiling of glycan structures and quantification of glycan abundances in biological samples. MALDI-MS can provide valuable insights into the role of glycans in various physiological and pathological processes, including cell signaling, immune response, and disease progression.

e) Metabolomics: Metabolomics is the large-scale study of small molecules commonly known as metabolites within cells, biofluids, tissues, or organisms. MALDI-MS is used in metabolomics for the comprehensive analysis of small-molecule metabolites present in biological samples. It enables the profiling of metabolite fingerprints and the identification of metabolite biomarkers associated with metabolic pathways, diseases, and environmental exposures. MALDI-MS can contribute to the understanding of metabolic regulation and disease mechanisms.

f) Imaging mass spectrometry: Imaging is an essential phenomenon in modern veterinary sciences and medicine. If MALDI-MS is equipped with imaging techniques, the technique's effectiveness will be further enhanced. Imaging mass spectrometry enables the visualization and mapping of molecular distributions, allowing researchers to study molecular changes in diseased tissues, drug distribution in pharmacokinetic studies, and biomarker localization in biological samples.

g) Environmental science and engineering: MALDI-MS is applied in environmental analysis to detect and quantify pollutants, toxins, and environmental contaminants in soil, water, and air samples. It offers high sensitivity and selectivity

for analyzing organic and inorganic compounds, contributing to environmental monitoring and risk assessment efforts.^[102-106]

Overall, MALDI-MS has diverse applications spanning basic research, clinical diagnostics, pharmaceutical analysis, and environmental monitoring, making it a versatile tool in analytical chemistry and life sciences.

5.2.4 Limitations of MALDI-MS

The limitations of MALDI-MS are encountered as below:

a) Sample preparation: MALDI-MS often requires extensive sample preparation, including matrix application and purification steps, which can be time-consuming and labor-intensive. Sample preparation can also introduce variability and affect the reproducibility of results.

b) Matrix interference: The choice of matrix can significantly impact the quality of MALDI-MS spectra. Matrix interference, such as matrix adducts and background signals, can complicate data interpretation and affect sensitivity and specificity.

c) Analyte size: The detection efficiency of MALDI-MS depends on the size of the analyte. It is observed that MALDI-MS is more suitable for the analysis of small to medium-sized molecules (up to 100 kDa) compared to larger biomolecules such as intact proteins (>100 kDa). Analyzing larger molecules with MALDI-MS may require specialized techniques or additional sample preparation steps.

d) Ionization efficiency: The ionization efficiency of MALDI-MS can vary depending on the analyte and matrix, leading to differences in signal intensity and detection limits for different compounds. Some analytes may exhibit poor ionization efficiency, limiting their detection sensitivity.

e) Mass accuracy and resolution: While MALDI-MS offers high throughput and sensitivity, it may have limitations in mass accuracy and resolution compared to other mass spectrometry techniques such as ESI-MS or Fourier transform ion cyclotron resonance (FT-ICR) MS.

f) Matrix suppression: In complex samples, matrix suppression effects can occur, where the ionization of analyte molecules is suppressed by coexisting matrix compounds or sample contaminants. This can lead to reduced sensitivity and compromised detection of low-abundance analytes.

g) Matrix clustering: The sensitivity and effectiveness of the instrument depend upon the concentration of the analyte also. At higher concentrations, there is the possibility of the formation of clusters or aggregates, and these clusters or aggregates may interfere with the ionization and can lead to spectral artifacts. However, controlling matrix concentration and optimizing sample preparation conditions can help mitigate this issue.

h) Limited mass range: MALDI-MS may have limitations in the mass range it can effectively analyze. While MALDI-TOF instruments typically have high mass accuracy and resolution, they may not be suitable for analyzing very high or very low mass ions.

i) Surface effects: MALDI-MS is sensitive to surface effects, and variations in sample topography or surface roughness can affect ionization efficiency and spectral quality. To minimize surface effects, homogeneous sample application and careful target preparation is essential.

Despite these limitations, MALDI-MS remains a valuable technique for analyzing biomolecules in various fields, and ongoing advancements in instrumentation, sample preparation, and data analysis continue to address many of these challenges.

6. Thermal spectroscopy

Thermal spectroscopy encompasses a range of techniques that utilize the interaction between matter and electromagnetic radiation, particularly in the infrared region of the electromagnetic spectrum, to study the thermal properties of materials. The principle of thermal spectroscopy revolves around the measurement and analysis of how materials absorb, emit, or transmit infrared radiation as a function of temperature. This interaction provides valuable information about the molecular vibrations, chemical bonds, and thermal behavior of the sample under investigation.

The principle of thermal spectroscopy can be elucidated through several key components such as infrared radiation, sample interaction, spectral measurement, temperature dependence, and data analysis. Overall, the principle of thermal spectroscopy enables the characterization and analysis of materials based on their thermal properties and infrared spectra. Thermal spectroscopy finds applications in various fields, such as materials science, chemistry, polymer science, pharmaceuticals, and food science, providing valuable insights into the composition, structure, and thermal behavior of a wide range of materials.^[107-113]

6.1 Thermogravimetric analysis

Thermogravimetric analysis (TGA) is a thermal analysis technique used to study changes in the mass of a sample as a function of temperature or time under controlled atmospheric conditions. This method provides valuable insights into the composition, thermal stability, and decomposition kinetics of materials.^[114-117]

In a typical TGA experimentation process, the sample is placed in a crucible or pan made of inert materials such as alumina or platinum. The crucible, suspended from a highly sensitive balance, is subjected to a programmed temperature ramp within a furnace. With the change in temperature of the system, the sample undergoes physical and chemical changes such as evaporation, decomposition, oxidation, or reduction. These changes cause a corresponding change in the sample's mass, which is continuously monitored by the balance.

The temperature ramp can be either isothermal or non-isothermal, depending on the experimental requirements. Additionally, TGA experiments are conducted under controlled atmospheres, such as inert gases (e.g., nitrogen or argon) or reactive gases (e.g., oxygen) supplied by a gas flow system. The resulting thermogravimetric curve provides a

graphical representation of the mass loss or gain of the sample as a function of temperature or time.^[118,119]

6.2 Instrumentation to TGA

Thermogravimetric analysis spectroscopy is described in Fig. 10. The critical components of the TGA instrument are described below:

a) Sample holder: The sample holder is usually a crucible or pan made of inert materials like alumina or platinum. It holds the sample during the analysis and is suspended from a compassionate balance.

b) Furnace: The sample holder, with the sample, is placed inside a furnace. The furnace provides controlled heating according to the experimental parameters. It can ramp the temperature at a constant rate (non-isothermal) or maintain a constant temperature (isothermal) for specific time intervals.

c) Balance: A highly sensitive balance continuously measures the mass of the sample throughout the experiment. It detects small changes in mass, often in the sub-microgram range, and provides accurate mass measurements.

d) Temperature control system: The temperature control system regulates and maintains the desired temperature profile inside the furnace. It ensures precise temperature control throughout the analysis, which is critical for obtaining reliable data.

e) Gas flow system: TGA experiments are conducted under controlled atmospheric conditions. A gas flow system supplies the desired atmosphere, typically inert gases like nitrogen or argon or reactive gases like oxygen. The flow rate and composition of the gas can be adjusted as per experimental requirements.

f) Data acquisition and analysis software: Specialized software controls the instrument operation and collects data from the balance and temperature control system. It processes the data to generate thermogravimetric curves, which plot mass changes against temperature or time. Additionally, the software may offer tools for data analysis and interpretation.

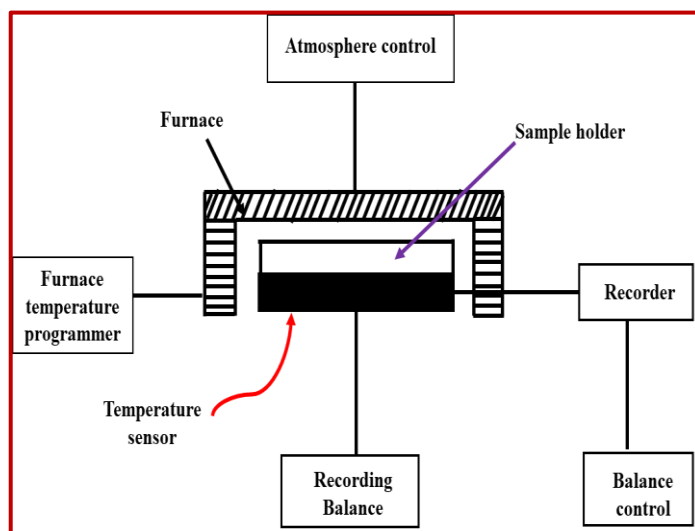


Fig. 10: A pictorial representation of Thermo-gravimetric analysis spectroscopy.

These components work together seamlessly to enable accurate and reproducible TGA measurements. They provide insights into the thermal properties, decomposition kinetics, and compositional changes of materials under varying temperatures and atmospheric conditions. TGA instruments are essential tools in materials science, polymers, pharmaceuticals, and other fields for characterizing a wide range of materials and understanding their behavior under thermal stress.

6.3 Principal of TGA

The principle of TGA revolves around measuring the change in mass of a sample as a function of temperature (or time) while subjecting it to controlled heating in a defined atmosphere. This technique provides valuable information about the thermal stability, composition, and decomposition kinetics of materials. The basic steps involved in TGA are as follows:

a) Sample preparation: A small amount of the sample (usually a few milligrams to grams) is placed in a crucible or pan, which is typically made of inert materials such as alumina or platinum. The sample is then loaded into the TGA instrument.

b) Heating profile: The sample is subjected to a programmed temperature profile within a furnace. The temperature can be increased at a constant rate (non-isothermal) or maintained at a constant level (isothermal) for specific time intervals. The chosen heating profile depends on the experimental requirements and the thermal behavior of the sample.

c) Mass measurement: A highly sensitive balance continuously measures the mass of the sample throughout the experiment. Any change in mass, whether due to evaporation, decomposition, oxidation, reduction, or other processes, is recorded by the balance.

d) Data analysis: The mass data collected by the balance is processed to generate a thermogravimetric curve, which plots the mass of the sample as a function of temperature (or time). This curve provides insights into the thermal stability and decomposition behavior of the sample. Key parameters extracted from the curve include onset temperature, peak temperature, rate of mass loss, and total mass change.

e) Controlled atmosphere: TGA experiments are often conducted under controlled atmospheric conditions. The atmosphere can be inert (*e.g.*, nitrogen or argon) or reactive (*e.g.*, oxygen) depending on the sample and the phenomena under investigation. A gas flow system supplies the desired atmosphere to the furnace chamber.

By analyzing the thermogravimetric curve, researchers can determine various characteristics of the sample, such as its decomposition temperature, degradation kinetics, moisture content, and composition. TGA is widely used in materials science, polymers, pharmaceuticals, and other fields for characterizing the thermal behavior of a wide range of materials and understanding their thermal stability and decomposition mechanisms.

6.4 Applications of TGA

Some critical applications of TGA include:

a) Polymer characterization: Polymers are widely used in various domains of science and technology, such as veterinary science and medicine. The polymeric materials can be used for effective drug delivery. Therefore, it is essential to have correct knowledge regarding the structure, properties, and thermal stability of polymeric molecules. TGA is an efficient characterization tool used to determine the thermal stability, decomposition temperature, degradation kinetics, and composition of polymeric materials. It helps in assessing factors such as polymer purity, the presence of additives, and polymer aging.

b) Quality control in materials manufacturing: TGA is employed for quality control purposes in industries such as plastics, rubber, composites, and textiles. It allows manufacturers to monitor the thermal properties of raw materials, intermediate products, and finished goods, ensuring consistency and reliability in product performance.

c) Pharmaceuticals: TGA is utilized in pharmaceutical research and development to assess the purity, stability, and compatibility of drug substances and formulations. It helps identify degradation products, determine shelf-life, and optimize formulation processes.

d) Environmental science and engineering: TGA is applied in environmental science to analyze organic and inorganic contaminants, pollutants, and waste materials. It provides insights into the thermal decomposition behavior of ecological samples, aiding in pollution control, waste management, and remediation efforts.

e) Catalyst characterization: TGA is used to study the thermal stability and catalytic properties of catalyst materials in various chemical processes. It helps evaluate catalyst activity, selectivity, and lifetime, contributing to the development and optimization of catalytic reactions.

f) Food science: Food ingredients require thermal treatment, which we observe in our daily lives. Thus, it becomes important to determine the thermal stability, composition, and quality of food ingredients and products. TGA can be effectively used for this and helps to study dehydration, decomposition, and oxidation of food materials.

g) Minerals and ceramics: TGA is utilized to characterize minerals, ceramics, and refractory materials to understand their thermal behavior, phase transitions, and sintering properties. It aids in optimizing processing parameters and developing advanced materials.

h) Fuel analysis: TGA is employed in the analysis of fuels, including fossil fuels, biomass, and alternative energy sources, to study their combustion behavior, calorific value, and thermal degradation characteristics. It contributes to the development of efficient and sustainable energy systems.

Overall, TGA is a versatile technique that plays a crucial role in materials science, chemistry, pharmaceuticals, environmental science, and various other fields, providing valuable information for research, development, and quality

control purposes.

6.5 Limitations of TGA

The limitations of TGA are encountered as below:

a) Sample size and homogeneity: The size and structure of the sample play a crucial role in TGA analysis. TGA instrumentation requires small sample sizes, which may not always be uniformly present inside the bulk structure. Also, the homogeneous nature of the material to be studied is important. Heterogeneous samples can lead to variability in measurements.

b) Temperature range: TGA instruments have a limited temperature range, and they may not be suitable for studying materials that undergo extreme temperature changes or have high melting points. Some materials may decompose or evaporate outside the instrument's temperature range.

c) Atmospheric effects: The choice of the atmosphere (*e.g.*, inert, oxidative, or reducing) in TGA experiments can significantly influence the thermal behavior of materials. Contaminants in the atmosphere or reactive gases may interact with the sample, leading to inaccurate results.

d) Calibration: Accurate calibration of the TGA instrument is essential for obtaining reliable data. Variations in the calibration procedure or instrument performance can affect the accuracy and reproducibility of measurements.

e) Sensitivity to environmental conditions: TGA measurements can be sensitive to environmental conditions such as humidity and air flow. Changes in these parameters can affect the mass readings and introduce errors in the analysis.

f) Interpretation of results: Interpreting TGA data requires careful consideration of various factors, including sample composition, heating rate, atmosphere, and experimental conditions. Incorrect interpretation or misidentification of thermal events can lead to erroneous conclusions.

g) Limited kinetic information: While TGA provides information about the onset temperature and rate of mass change, it may not always yield detailed kinetic parameters such as activation energy or reaction mechanisms. Additional techniques, such as differential scanning calorimetry or kinetic modeling, may be required for a comprehensive analysis.

h) Instrumental limitations: TGA instruments may have limitations in terms of sensitivity, resolution, and detection limits. Some materials may exhibit subtle thermal changes that are beyond the detection capabilities of standard TGA setups. Despite these limitations, TGA remains a valuable tool for studying the thermal properties, decomposition behavior, and composition of materials in various fields. Careful experimental design, calibration, and data interpretation are essential for mitigating potential challenges and obtaining meaningful results.

7. Vibrational spectroscopy

Vibrational spectroscopy encompasses techniques that utilize the interaction between matter and electromagnetic radiation

to probe molecular vibrations within a sample. The principle of vibrational spectroscopy is grounded in the fundamental concept that molecules exhibit characteristic vibrational modes that absorb or emit energy at specific frequencies corresponding to these vibrational motions. The principle of vibrational spectroscopy can be elucidated through several key components such as infra-red radiation, sample interaction, spectral measurement, Raman scattering, selection rules, and data analysis.

Table 3 lists the distinction between Optical Spectroscopy and Vibrational Spectroscopy. Overall, the principle of vibrational spectroscopy enables the characterization and analysis of molecules based on their vibrational modes and infrared spectra. Vibrational spectroscopy finds applications in various fields such as chemistry, biochemistry, materials science, pharmaceuticals, and environmental science, providing valuable insights into the composition, structure, and properties of a wide range of materials and compounds.

7.1 Fourier transform infrared spectroscopy

Fourier transform infrared spectroscopy (FT-IR) is a versatile analytical technique used to qualitatively and quantitatively analyze the chemical composition of materials based on their interaction with infrared radiation. FT-IR mainly provides information regarding the various functional groups present in the molecule. The data obtained from FT-IR can be combined with the data obtained from another analytical technique, giving a picture of organic molecules. Thus, FT-IR is a versatile and essential tool in the determination of the structure of organic molecules.^[120-122]

The basic principle of FT-IR involves passing a beam of IR radiation through a sample and measuring the intensity of transmitted or reflected light at different wavelengths. The sample interacts with the IR radiation, causing selective absorption or scattering of specific frequencies corresponding to the vibrational modes of its constituent molecules. These vibrational modes result from the stretching, bending, and twisting motions of chemical bonds within the molecules.

In an FT-IR spectrometer, the IR radiation is generated by an IR source such as a heated filament or a laser. The IR beam is directed through an interferometer, which modulates the frequency of the radiation using a moving mirror. The modulated IR beam is then passed through the sample and onto a detector, where the intensity of the transmitted or reflected light is measured as a function of wavelength. The interferogram generated by the detector contains information about the sample's IR absorption spectrum, which represents the unique fingerprint of the sample's molecular composition. The interferogram is subjected to Fourier transformation, a mathematical process that converts the time-domain data into a frequency-domain spectrum. The resulting FT-IR spectrum displays absorption peaks at specific wavenumbers (reciprocal of wavelength), corresponding to the vibrational frequencies of the sample's chemical bonds. Fig. 11 shows the schematic diagram of Fourier transform infrared spectroscopy. FT-IR

Table 3. Differentiate between Optical Spectroscopy and Vibrational Spectroscopy.

Aspect	Optical Spectroscopy	Vibrational Spectroscopy
Principle	Measures the interaction of light UV-visible, or near-infrared) with matter, typically focusing on electronic transitions.	Measures vibrational transitions in molecules, typically in the infrared or Raman spectra.
Type of Interaction	Involves absorption, emission, or scattering of light related to electronic energy levels in atoms or molecules.	Involves absorption or scattering of light due to vibrational transitions within molecules.
Energy Range	Higher energy transitions (UV-Vis: 1.5–6 eV; NIR: 0.5–1.5 eV).	Lower energy transitions (IR: 0.01–0.5 eV). Raman covers similar energy transitions but with a scattering process.
Information Provided	Provides information about electronic states, band gaps, and chromophores in materials.	Provides detailed information on molecular vibrations, bond strengths, and molecular structure.
Spectral Region	UV-Vis (200–800 nm) and near-infrared (800–2500 nm).	Infrared (IR: 2.5–25 μm) and Raman (typically near visible or IR).
Sample Types	Suitable for gases, liquids, and solids, especially molecules with electronic transitions.	Suitable for molecules, polymers, crystals, biological samples, and materials with distinct vibrational modes.
Sensitivity to Molecular Structure	Sensitive to electronic configurations and environments of atoms within molecules.	Sensitive to functional groups, molecular bonds, and interactions between atoms.
Common Techniques	UV-Vis Spectroscopy, Fluorescence Spectroscopy, Absorption/Emission Spectroscopy.	Infrared Spectroscopy, Raman Spectroscopy.
Instrumentation	Requires a light source (<i>e.g.</i> , UV-Vis lamp), monochromator, and detector.	Requires an IR source or laser (for Raman), and a detector such as a photodiode or CCD camera.
Data Output	Spectra showing absorption/emission as a function of wavelength or frequency.	Spectra showing vibrational peaks as a function of wavenumber (cm^{-1}).
Chemical Sensitivity	Can detect chromophores and determine electronic transitions in molecules and materials.	Sensitive to molecular vibrations, providing information on bond types and molecular symmetry.
Applications	Used for studying electronic transitions, material band gaps, and the presence of specific chromophores.	Used for identifying functional groups, studying molecular structures, and analyzing chemical bonds.
Sample Preparation	Typically minimal, it requires transparent or thin samples for UV-Vis studies.	May require solid, liquid, or gas phase samples, often needing specific conditions for Raman or IR studies.
Non-Destructive	Typically non-destructive, especially in absorption spectroscopy.	It can be non-destructive, particularly in Raman spectroscopy. IR can cause some heating.

spectra are highly informative and can identify functional groups, determine molecular structures, analyze mixtures, and monitor chemical reactions.

FT-IR is widely applied in various fields, including chemistry, materials science, pharmaceuticals, environmental science, forensic analysis, and polymer science. Its non-destructive nature, high sensitivity, and rapid data acquisition make FT-IR indispensable for characterizing a wide range of organic and inorganic materials.

7.1.1 Instrumentation to FT-IR

FT-IR instruments typically consist of several key components designed to generate and analyze infrared radiation and measure its interaction with a sample. The key components of FT-IR are furnished as below:

a) Infrared source: Like other spectroscopic techniques FT-IR also requires some source to generate electromagnetic radiations. Here in case of FT-IR Spectroscopy we require longer wavelength infra-red source of radiations. FT-IR instruments utilize various sources to generate infrared

radiation across a broad spectral range. The source is always some form of filament which is maintained at red or white heat by electric current. Common sources include nichrome wire filaments, silicon carbide globars, and ceramic heaters. Also Nernst filament, consisting of a spindle rare earth oxides about 1 inch long 0.1 inch in diameter, and globar filament, a rod of carborundum, somewhat thicker and longer than the Nernst. Some advanced FT-IR instruments may also incorporate quantum cascade lasers (QCLs) or other tunable lasers for specific applications.

b) Interferometer: The heart of an FT-IR spectrometer is the interferometer, which modulates the infrared radiation and enables Fourier transformation of the resulting interferogram. The most common type of interferometer used in FT-IR instruments is the Michelson interferometer, consisting of a beam splitter, fixed mirror, and moving mirror. Other interferometer configurations such as the Rocking Arm and Fourier transform spectrometers may also be employed.

c) Sample compartment: The sample compartment houses the sample being analyzed. It is designed to accommodate various sample forms, including solids, liquids, gases, and thin films. Some FT-IR instruments offer accessories such as sample holders, transmission cells, and attenuated total reflectance (ATR) accessories to facilitate sample handling and analysis. The sample is held between the plates of polished mineral salts. Pure liquids are studied between the thickness of 0.01 mm while solutions are studied in the thickness ranging from 0.1–10 mm thickness, depending upon the dilution. Gas samples up to pressure 1 atm or greater are contained in glass cells either 5 or 10 cm long, closed their ends with a rock salt window. A special long-path cell in which radiation is repeatedly reflected up and down the cell may be used for gases at low pressure. The detection of solid samples is difficult as they reflect and scatter the incident radiations and low transmittance is observed. If the suitable solvent for the dissolution of solid material is not available, it is examined by grinding it very finely in paraffin oil and thus forming a suspension or mull. This can then be held between salt plates in the same way as pure liquid or solution. Another way to analyze the solid sample is by grinding the solid sample with potassium bromide. Under very high pressure, the material will flow slightly, and the mixture can be pressured in dye to make a transparent disc. Now, the compressed mixture of solid material and potassium bromide may be placed directly in the infra-red beam in a suitable holder.

d) Detector: The detector captures the modulated infrared radiation transmitted or reflected by the sample. Common detectors used in FT-IR instruments include thermal detectors (e.g., pyroelectric detectors, thermopiles) and photon detectors (e.g., photodiodes, PTMs) depending upon photoconductivity. Each detector type offers specific advantages in terms of sensitivity, dynamic range, and spectral response. In both, the greater the effect (temperature or rise in conductivity) at a given frequency, the greater the transmittance of the sample at that frequency. The detector based upon the heating effect is

found in the Golay cell, which is pneumatic in operation. The radiation is incident on tiny cells containing air, and the temperature changes are measured in terms of pressure changes within the cell, which can be recorded directly as the transmittance. The detector based upon the heating effect is a sensitive thermocouple or bolometer. Pyroelectric detectors such as deuterated triglycine sulfate are commonly used in FT-IR. They are thermal detectors and are sensitive across the entire infrared range. The other type of detector is based on photoconductivity. The phenomenon of conductivity is supposed to be based on the movement of free electrons through the lattice. On the basis of electrical conductivity, the materials can be classified into three types: (1) Insulators: Here, no loosely bound free electrons are available that can conduct the charges. Therefore, insulators do not show electrical conductivity. (2) Semiconductor: As the name indicates, semiconductors have a certain number of free electrons capable of charge conduction. A pure semiconductor is known as an intrinsic semiconductor. The number of electrons present in conduction bands can be tuned by doping. The doped semiconductor is known as an extrinsic semiconductor. The semiconductors can be doped with trivalent or pentavalent impurities. The semiconductors doped with trivalent impurities are known as p-type semiconductors, and the semiconductors doped with pentavalent impurities are known as n-type impurities. The intrinsic and extrinsic types of semiconductors show intermediate conductivity between insulators and conductors. (3) Conductors: These are the materials that have a large number of free electrons capable of conduction of charge. In terms of band theory, conductor valence bands and conduction bands overlap. Photoconductors are a particular type of semiconductor in which energy required comes from incident radiations, and some materials like lead sulfide are sensitive enough to infrared radiations. Therefore, it can be used for the manufacturing of detectors. There are some other types of photodetectors, such as Indium Antimonide (InSb), Mercury, and Cadmium Telluride. These detectors operate at liquid Nitrogen temperature. The low temperature reduces noise and has a faster response time and sensitivity. When extremely high sensitivity is required, doped photoconductors such as gold or copper-doped germanium detectors can be used. But this works at liquid helium temperature.

e) Data acquisition system: A data acquisition system digitizes the detector output and processes the interferogram to generate an FT-IR spectrum. This system typically includes analog-to-digital converters, signal processing electronics, and digital signal processors. Advanced FT-IR instruments may incorporate real-time signal processing capabilities for rapid data acquisition and analysis.

f) Computer and software: FT-IR instruments are typically controlled and operated using dedicated software installed on a computer. The software allows users to define experimental parameters, acquire spectral data, perform data processing and analysis, and generate reports. It may also include libraries of

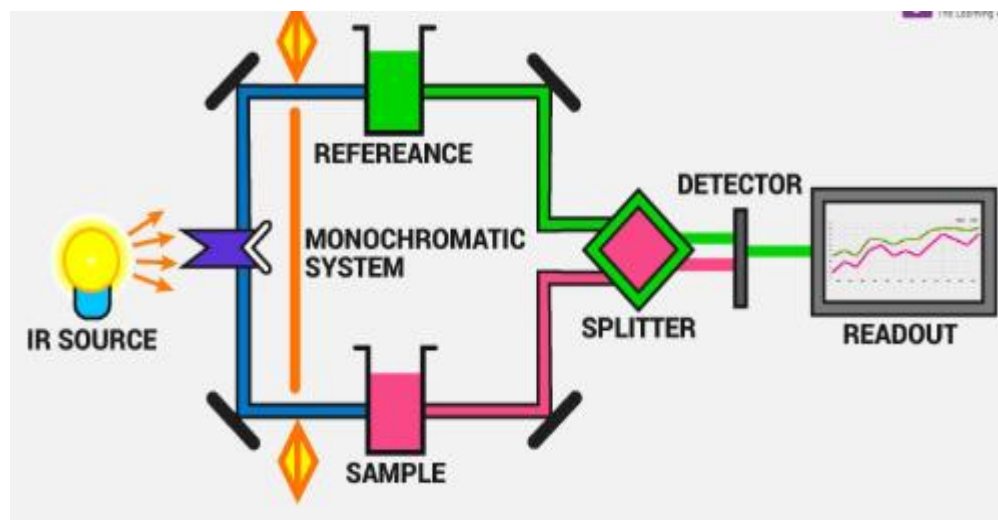


Fig. 11: A pictorial representation of Fourier transform infra-red spectroscopy.

spectral databases for sample identification and comparison.

These components work together seamlessly to enable the acquisition of high-quality infrared spectra and the analysis of molecular composition, functional groups, and chemical bonds in a wide range of samples. FT-IR instruments are essential tools in analytical chemistry, materials science, pharmaceuticals, and other fields for characterizing and identifying organic and inorganic compounds.

7.1.2 Principle of FT-IR

The FT-IR is based on the interaction of infrared radiation with matter and the measurement of the resulting absorption, emission, or reflection of light. FT-IR spectroscopy provides valuable information about the chemical composition, molecular structure, and bonding interactions present in a sample. The basic principle of FT-IR involves the following steps:

a) Generation of infrared radiation: An infrared radiation source, such as a heated filament, globar, or laser, emits infrared radiation across a wide spectral range. The radiation consists of photons with energies corresponding to vibrational and rotational transitions within molecules.

b) Interferometry: The infrared radiation is directed through an interferometer, typically a Michelson interferometer. The interferometer splits the incoming radiation into two beams using a beamsplitter. One beam travels to a fixed mirror, while the other travels to a moving mirror. The moving mirror periodically changes its position, causing the path difference between the beams to vary over time.

c) Interferogram formation: The two beams are recombined after reflection from the mirrors. The interference between the two beams results in an interferogram, a time-domain signal that encodes information about the intensity of the infrared radiation as a function of the optical path difference between the beams.

d) Fourier transformation: The interferogram is subjected to Fourier transformation, a mathematical process that converts the time-domain data into a frequency-domain spectrum. This

transformation reveals the constituent frequencies present in the interferogram, which correspond to the various infrared absorption bands of the sample.

e) Spectral analysis: The resulting FT-IR spectrum represents the absorption of infrared radiation by the sample as a function of wavenumber (reciprocal of wavelength). Each peak in the spectrum corresponds to a specific vibrational mode or combination of vibrational modes within the sample's molecules.

f) Data interpretation: FT-IR spectra are analyzed to identify functional groups, determine molecular structures, and characterize chemical bonds in the sample. Spectral libraries and reference databases are often used to compare and identify unknown samples.

By analyzing the FT-IR spectrum, researchers can obtain valuable information about the molecular composition and structure of materials, making FT-IR spectroscopy a powerful tool in analytical chemistry, materials science, pharmaceuticals, and other fields.

7.1.3 Applications of FT-IR

Some important applications of the FT-IR are mentioned below:

a) Chemical analysis: FT-IR is extensively used for qualitative and quantitative analysis of organic and inorganic compounds. It is employed in the identification of functional groups, characterization of molecular structures, and determination of chemical compositions in a wide range of samples, including liquids, solids, and gases.

b) Polymer science: FT-IR is indispensable in polymer science for studying the composition, structure, and properties of polymers and polymer-based materials. It facilitates the analysis of polymer blends, copolymers, additives, fillers, and degradation products. FT-IR is used in polymer synthesis, quality control, and product development in industries such as plastics, rubber, coatings, and adhesives.

c) Pharmaceuticals: FT-IR is employed in the pharmaceutical industry for quality control, formulation development, and

analysis of drug substances and products. It enables the identification of drug polymorphs, detection of impurities, monitoring of drug stability, and characterization of drug-excipient interactions.

d) Environmental analysis: FT-IR is used in environmental science for the analysis of air pollutants, water contaminants, soil components, and waste materials. It helps in identifying organic and inorganic pollutants, monitoring pollution levels, and assessing environmental impacts.

e) Forensic science: FT-IR plays a crucial role in forensic analysis for the identification of trace evidence, unknown substances, and forensic samples. It assists forensic scientists in analyzing fibers, paints, drugs, explosives, and biological samples. FT-IR is utilized in criminal investigations, forensic laboratories, and crime scene analysis.

f) Materials characterization: Material science is a fast-growing branch of science and technology. Today, extensive research is being done in the domain of material science and nanotechnology. The particular age FT-IR is employed in materials science to characterize the composition, structure, and properties of various materials. It is used to analyze ceramics, metals, composites, semiconductors, nanoparticles, and thin films. FT-IR helps in understanding material behavior, assessing material quality, and optimizing material properties for specific applications.

g) Biomedical research: The biomolecules contain various types of functional groups. Therefore, FT-IR can be effectively used in biomedical research for studying biological tissues, cells, and supramolecular biological compounds such as proteins, carbohydrates, lipids, and other biomolecules. This technique can be used in the diagnosis of diseases, biomarkers, drug discovery, and understanding of biological processes. FT-IR spectroscopic techniques such as attenuated total reflectance (ATR) are used for non-destructive analysis of biological samples.

h) Food and beverage analysis: Organic materials are generally used as food for the animal kingdom. These organic materials possess various functional groups. FT-IR can be effectively used in food and beverage sectors for analysis of contaminants, additives, and various ingredients found in food materials, monitoring food safety and quality standards.

Overall, FT-IR spectroscopy is a versatile analytical technique with broad applications in chemistry, materials science, pharmaceuticals, veterinary medicine, environmental science, forensics, biomedical research, and food analysis, among others. Its non-destructive nature, rapid data acquisition, and high sensitivity make it an indispensable tool for research, development, quality control, and regulatory compliance in various industries.

7.1.4 Limitations of FT-IR

a) Sample preparation: FT-IR often requires samples to be prepared as thin films, pellets, or solutions to obtain reliable spectra. During pellets preparation we use potassium bromide with material to be analyzed. The material is mulled with

Potassium Bromide in mortar and pestle and then pellets are prepared. Sample preparation can introduce artifacts or alter the sample's properties, potentially affecting the accuracy and interpretation of results.

b) Surface sensitivity: FT-IR is primarily a surface-sensitive technique, which means it mainly probes the outer layers of a sample. This limitation can be problematic for heterogeneous samples with depth variations or surface treatments.

c) Water absorption: Water vapor strongly absorbs infrared radiation in certain spectral regions, particularly in the mid-infrared range. This absorption can interfere with the analysis of samples containing moisture or when measuring samples in humid environments.

d) Limited sensitivity for dilute samples: FT-IR may lack sensitivity for dilute samples or trace components, particularly in complex matrices. The presence of strong absorbers can mask the signals of weaker components, making their detection challenging.

e) Instrumental drift and noise: FT-IR instruments may suffer from instrumental drift and noise, which can affect the accuracy and reproducibility of measurements. Drift can occur due to temperature, humidity, or instrument stability changes, while noise can arise from electronic interference or detector instabilities.

f) Overlapping bands: FT-IR spectra often exhibit overlapping absorption bands, especially in complex samples containing multiple components. Overlapping bands can complicate spectral interpretation and make distinguishing between different functional groups or chemical species challenging.

g) Limited quantitative accuracy: While FT-IR can provide qualitative information about sample composition and structure, its quantitative accuracy may be limited, especially for mixtures or samples with overlapping peaks. Quantitative analysis often requires calibration with standards and careful consideration of sample thickness and path length factors.

h) Chemical interference: FT-IR spectra can be affected by chemical interferences, such as sample interactions with the sample holder or container. Contaminants or impurities present in the sample or sample environment can also interfere with spectral interpretation.

Despite these limitations, FT-IR spectroscopy remains a valuable tool for chemical analysis, material characterization, and structural elucidation in various fields. Understanding these limitations is essential for obtaining accurate and reliable results in FT-IR experiments.

7.2 Surface-enhanced Raman spectroscopy

Surface-enhanced Raman Spectroscopy (SERS) is a powerful analytical technique that enhances the Raman scattering signal from molecules adsorbed on or near specially prepared surfaces, leading to significantly increased sensitivity and detection capabilities. When a beam of light is passed through a transparent substance, a small amount of the radiation energy is scattered, the scattering persisting even if all dust particles

or other extraneous matter are rigorously excluded from the substance. Suppose monochromatic radiation, or radiation of a very narrow frequency band, is used. In that case, the scattered energy will consist almost entirely of incident frequency radiation, known as Rayleigh scattering. Also, some discrete frequencies above and below the incident beam will be scattered. These scatterings with frequency different than incident frequency are known as Raman scattering.

The quantum theory of radiation can explain the phenomenon of Raman scattering. This treats radiation of frequency ν as consisting of a stream of particles having energy $h\nu$ where h is Planck's constant. Photons can be imagined to undergo oscillations with molecules, and if the collision is perfectly elastic, they will be deflected unchanged. A detector placed to collect energy at right angles to the incident beam will thus receive photons of energy $h\nu$, *i.e.*, radiation of frequency ν . Elastic scattering can be likened to a ball bearing striking a rigid table. The ball bearing bounces off the table without any loss of energy.

Overall, the principle of Surface-Enhanced Raman Spectroscopy exploits the unique electromagnetic properties of metal nanoparticles to amplify the Raman scattering signal from molecules, enabling sensitive detection and analysis of trace analytes in various fields such as chemistry, biology, environmental science, and materials science.^[123-125]

7.2.1 Instrumentation of SERS

SERS instrumentation involves specialized components to enhance Raman signals from molecules adsorbed on or near roughened metal surfaces, and the schematic diagram is presented in Fig. 12. Here's an overview of the critical components:

a) Raman spectrometer: The core of SERS instrumentation is a Raman spectrometer, which is equipped with a laser source, a monochromator or dispersive element, and a detector. The laser source typically emits light in the visible or near-infrared range, commonly at wavelengths such as 532 nm or 785 nm. The monochromator or dispersive element selects specific Raman scattering wavelengths, and the detector measures the intensity of Raman signals.

b) Metal substrate: SERS requires a metal substrate with a roughened or nanostructured surface to enhance Raman signals. Commonly used metals include gold, silver, and copper. The metal substrate provides localized surface plasmon resonance (LSPR) effects, amplifying the surface's electromagnetic field and enhancing Raman scattering from nearby molecules.

c) Sample deposition: Samples are typically deposited onto the metal substrate using drop-casting, spin-coating, or vapor deposition techniques. The sample may consist of molecules of interest dissolved or dispersed in a solvent. The sample preparation method and concentration can influence the SERS signal intensity and reproducibility.

d) Microscope or optical setup: SERS measurements are often performed using a microscope or optical setup to focus the laser onto the sample and collect Raman signals. The microscope objective provides spatial resolution and controls the spot size of the laser beam. SERS measurements can also be conducted using fiber-optic probes or integrated into microfluidic devices for in situ analysis.

e) Filters and optics: Optical filters and lenses are used to eliminate background fluorescence and scatter from the sample and enhance the collection of Raman signals. Filters with specific wavelength ranges can isolate the desired Raman scattering wavelengths and reject unwanted spectral contributions.

f) Signal processing and analysis: Software tools are used for data acquisition, signal processing, and spectral analysis. Spectral processing may include baseline correction, noise reduction, and peak fitting algorithms to extract meaningful information from the SERS spectra. Spectral databases and chemometric methods may also be employed for sample identification and quantitative analysis.

Overall, SERS instrumentation combines elements of Raman spectroscopy, surface chemistry, and optical microscopy to enable highly sensitive and selective detection of molecules at low concentrations, making it a valuable tool for chemical analysis, biosensing, and materials characterization.

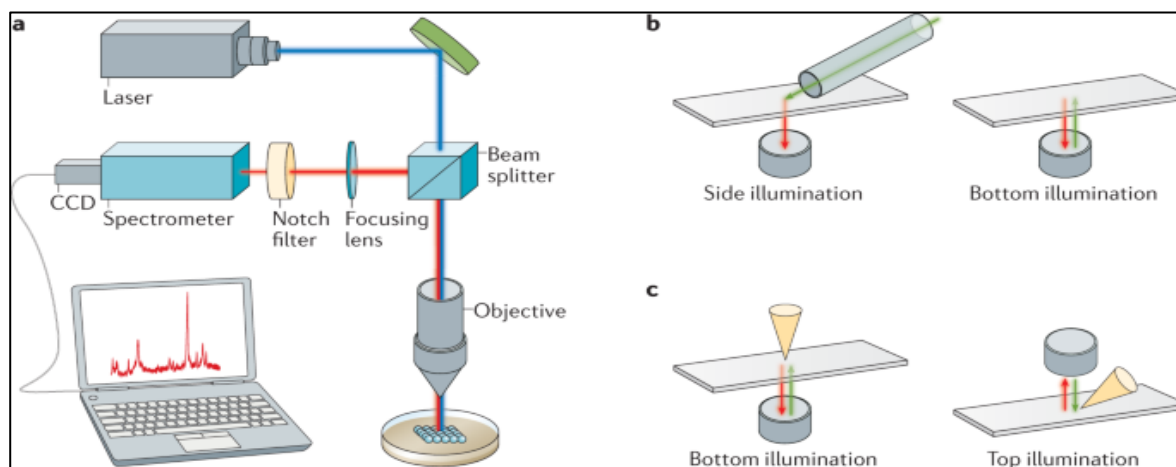


Fig. 12: A pictorial representation of the surface enhances Raman spectroscopy.

7.2.2 Principle of SERS

SERS operates on the principle of enhancing the Raman scattering signals of molecules adsorbed on or near roughened metal surfaces. The enhancement is achieved through two main mechanisms: electromagnetic enhancement and chemical enhancement.

a) Electromagnetic enhancement: When a molecule is in close proximity to a roughened metal surface, such as gold, silver, or copper, the interaction between the incident light and the metal surface generates localized surface plasmon resonance (LSPR) effects. LSPR results from the collective oscillation of conduction electrons in the metal nanostructures when excited by incident light. The enhanced electromagnetic field generated near the metal surface significantly enhances the Raman scattering intensity of molecules located within the "hot spots" of the electromagnetic field. This enhancement is proportional to the fourth power of the local electromagnetic field intensity, leading to extremely high enhancement factors, typically ranging from 10^4 to 10^8 times compared to conventional Raman scattering.

b) Chemical enhancement: Chemical enhancement arises from charge transfer or chemical interactions between the molecules and the metal surface. This mechanism can further amplify the Raman signals, especially for molecules that strongly interact with the metal surface through chemical bonds or charge transfer processes. Chemical enhancement typically contributes to a moderate enhancement of Raman signals, but it can significantly enhance the detection sensitivity for certain classes of molecules.

The combined effect of electromagnetic and chemical enhancement in SERS results in highly amplified Raman scattering signals, enabling the detection of trace amounts of molecules with high sensitivity and specificity. The enhancement is strongly dependent on the morphology, size, and composition of the metal nanostructures, as well as the excitation wavelength of the incident light and the molecular properties of the analyte.

Overall, SERS offers unique advantages for chemical analysis, biosensing, and materials characterization, making it a powerful technique for a wide range of applications in chemistry, biology, medicine, and materials science.

7.2.3 Applications of SERS

Some key applications of SERS include:

a) Chemical analysis: SERS is widely used for the detection and identification of chemical species in complex mixtures. It enables the sensitive detection of trace amounts of molecules, making it valuable for analytical chemistry applications such as forensic analysis, environmental monitoring, and food safety testing.

b) Biosensing: SERS is employed in biosensing applications for the detection of biomolecules such as proteins, DNA, RNA, and small molecules. It offers label-free detection with high sensitivity, enabling the rapid and selective detection of biomarkers for disease diagnosis, drug screening, and medical

diagnostics.

c) Biomedical imaging: SERS imaging techniques allow for high-resolution imaging of biological samples with molecular specificity. SERS nanoparticles functionalized with targeting ligands can selectively bind to specific biomarkers in tissues or cells, enabling the visualization of disease biomarkers, cellular processes, and drug distribution in vivo.

d) Surface science and catalysis: SERS is used in surface science and catalysis research to study the adsorption and reaction kinetics of molecules on catalytic surfaces. It provides insights into surface reactions, intermediate species, and catalytic mechanisms, facilitating the design and optimization of catalysts for chemical processes.

e) Nanotechnology and materials science: The properties of material at nanoscale depends upon its size and shape. Also the material can be functionalized by different materials. SERS is utilized in nanotechnology and materials science for the characterization and analysis of nanomaterials, nanoparticles, and nanostructures. It enables the investigation of surface plasmon resonance effects, surface chemistry, and molecular interactions at the nanoscale.

f) Environmental monitoring: SERS is applied in environmental monitoring for the detection and quantification of pollutants, contaminants, and toxic substances in air, water, and soil. It offers rapid, sensitive, and selective detection capabilities, making it suitable for on-site environmental analysis and monitoring.

g) Food and beverage analysis: Food and beverage is becoming important from industrial point of view also. In earlier day food was a domestic concern, but with the situation is changed. In modern world food and beverage suffers from contamination, adulteration, spoilage and so on. These industries are under the control of Food and Drug Administration. The experts require various instruments for analysis of food and beverage. SERS is used in the food and beverage industry for the analysis of food ingredients, additives, contaminants, and adulterants. It enables the detection of foodborne pathogens, allergens, pesticides, and toxins, contributing to food safety and quality control.

h) Art conservation and cultural Heritage: SERS is employed in art conservation and cultural heritage studies for the analysis of pigments, dyes, and coatings on historical artifacts and artworks. It helps in identifying materials, understanding degradation mechanisms, and designing conservation strategies.

Overall, SERS offers a powerful and versatile platform for chemical analysis, biosensing, imaging, and materials characterization across diverse applications in science, technology, and industry. Its sensitivity, specificity, and ability to provide molecular-level information make it a valuable tool for research, development, and quality control in various fields.

7.2.4 Limitations of SERS

a) Reproducibility: SERS signals can vary significantly between different experiments and substrates due to variations

in the preparation of metal substrates, sample deposition methods, and experimental conditions. Achieving consistent and reproducible results can be challenging, particularly for quantitative analysis.

b) Substrate uniformity: The enhancement of Raman signals in SERS is highly dependent on the morphology and uniformity of the metal nanostructures on the substrate surface. Variations in substrate morphology can lead to spatial variations in enhancement, affecting the reliability and interpretation of SERS measurements.

c) Signal enhancement mechanism: The exact mechanism behind SERS enhancement is still not fully understood, and it can vary depending on the specific experimental conditions and sample characteristics. This variability complicates the interpretation of SERS spectra and the optimization of experimental parameters.

d) Matrix effects: SERS signals can be influenced by the surrounding environment, including the chemical composition of the substrate, solvent effects, and interactions with other molecules present in the sample matrix. These matrix effects can complicate spectral interpretation and affect the accuracy of quantitative analysis.

e) Limited range of substrates: While noble metals such as gold and silver are commonly used as SERS substrates due to their strong plasmonic properties, they are relatively expensive and may not be suitable for all applications. Developing alternative substrates with comparable enhancement properties remains an ongoing challenge in the field of SERS.

f) Sample compatibility: SERS may not be suitable for all types of samples, particularly those with low Raman scattering cross-sections or strong fluorescence background. Certain sample preparation methods, such as drying or immobilization on solid substrates, may alter the sample's properties and affect the SERS measurements.

g) Quantitative analysis: Quantitative analysis using SERS can be challenging due to factors such as substrate heterogeneity, matrix effects, and variations in enhancement factors. Developing reliable calibration methods and standards for quantitative SERS analysis remains an area of active research.

h) Instrumentation complexity: SERS instrumentation often requires specialized equipment, including laser sources, high-quality optics, and precise control of experimental parameters. The complexity of SERS instrumentation can limit its accessibility and practicality for routine analysis in some laboratories.

Despite these limitations, SERS continues to be a valuable tool for chemical analysis, biosensing, materials characterization, and other applications, offering high sensitivity and molecular specificity for a wide range of samples. Addressing these limitations through further research and technological advancements will help unlock the full potential of SERS in various fields.

8. Optical emission spectroscopy

Optical emission spectroscopy (OES) is an analytical technique used to analyze the elemental composition of materials by measuring the characteristic wavelengths of light emitted from excited atoms or ions. The principle of OES is based on the emission of light when atoms or ions transition from higher energy states to lower energy states.

In OES, a sample is subjected to a high-energy excitation source, such as an electric arc, plasma discharge, or laser, which energizes the atoms or ions within the sample. As these excited species return to their ground state, they emit photons of specific wavelengths corresponding to the energy differences between the excited and ground states. Each element emits light at distinct wavelengths, allowing for the identification and quantification of elements present in the sample.

The emitted light is collected and dispersed by an optical system, typically consisting of lenses, mirrors, and diffraction gratings, which separates the emitted wavelengths according to their energies. The dispersed light is then detected by a photodetector, such as a PMT or CCD, which measures the intensity of light at different wavelengths.

The resulting emission spectrum is a graphical representation of the intensity of emitted light as a function of wavelength. Each peak in the spectrum corresponds to a specific element present in the sample, with the peak intensity proportional to the concentration of the element. By comparing the emission spectrum of the sample to known reference spectra, the elemental composition of the sample can be determined.

OES offers several advantages, including rapid analysis, high sensitivity, and the ability to analyze a wide range of elements simultaneously. It is widely used in various industries, including metallurgy, environmental monitoring, pharmaceuticals, and semiconductor manufacturing, for applications such as quality control, material characterization, and process monitoring.

Overall, OES is a valuable analytical technique for elemental analysis, providing rapid and accurate determination of the elemental composition of materials in diverse research and industrial applications.

8.1 Inductively coupled plasma optical emission spectroscopy

Inductively coupled plasma optical emission spectroscopy (ICP-OES) is a powerful analytical technique used for elemental analysis in various samples. It combines the high-temperature capabilities of inductively coupled plasma with the sensitive detection and quantification of elements using optical emission spectroscopy. As the excited atoms return to their ground state, they emit characteristic wavelengths of light corresponding to their elemental composition. These emissions are collected and dispersed by an optical system, typically consisting of lenses, mirrors, and diffraction gratings. The dispersed light is then detected by a photodetector, such

as a PMT or CCD, which measures the intensity of light at different wavelengths.

The resulting emission spectrum is a graphical representation of the intensity of emitted light as a function of wavelength. Each peak in the spectrum corresponds to a specific element present in the sample, with the peak intensity proportional to the concentration of the element. By comparing the emission spectrum of the sample to known reference spectra, the elemental composition and concentration of the sample can be determined. ICP-OES offers several advantages, including high sensitivity, wide elemental coverage, and the ability to analyze a wide range of sample types, including liquids, solids, and gases. It is widely used in various industries, including environmental analysis, pharmaceuticals, food and beverage, and metallurgy, for applications such as quality control, environmental monitoring, and trace elemental analysis.

In summary, ICP-OES is a versatile and powerful analytical technique for elemental analysis, offering rapid, sensitive, and accurate determination of elemental composition in diverse samples.

8.1.1 Instrumentations of ICP-OES

The instrumentation of ICP-OES consists of several key components that work together to generate, analyze, and detect the elemental emissions from a sample and the diagram is as shown in Fig.13. Here's an overview of the main components:

a) Inductively coupled plasma source: The heart of ICP-OES instrumentation is the inductively coupled plasma (ICP) source. This consists of a RF generator that produces a high-frequency alternating current, which is then used to induce a plasma discharge in a flowing inert gas, typically argon. The plasma is sustained by the electromagnetic field generated by a set of induction coils surrounding the torch.

b) Sample introduction system: Samples for analysis are introduced into the ICP source typically via a nebulizer or a spray chamber, where they are aerosolized and introduced into the plasma as a fine mist of droplets. The nebulizer converts

the liquid sample into an aerosol, while the spray chamber removes larger droplets and ensures a consistent sample flow rate into the plasma.

c) Optical system: The optical system in ICP-OES instruments is responsible for collecting, dispersing, and detecting the emitted light from the plasma. It typically consists of a set of lenses, mirrors, and diffraction gratings that collect and disperse the emitted light according to its wavelength. The dispersed light is then directed onto a detector for measurement.

d) Detector: The detector in ICP-OES instruments is usually a photodetector, such as a PMT or a CCD array. It measures the intensity of the emitted light at different wavelengths across the spectrum. The detector converts the optical signal into an electrical signal that can be processed and analyzed by the instrument's electronics.

e) Data acquisition and analysis software: Modern ICP-OES instruments are equipped with sophisticated data acquisition and analysis software. This software controls instrument parameters, records spectral data, performs baseline correction, and calculates elemental concentrations based on calibration curves and known standards. It also allows for visualization and interpretation of spectral data.

f) Cooling system: The ICP-OES instrument requires a cooling system to maintain stable operating conditions and prevent overheating of critical components, such as the plasma torch and the optical components. This typically involves a combination of water-cooled components and temperature control systems.

8.1.2 Principle of ICP-OES

The principle of ICP-OES involves the generation of a high-temperature plasma, the excitation of atoms or ions within the plasma, and the subsequent analysis of the emitted light to determine the elemental composition of a sample.

a) Inductively coupled plasma source: The process begins with the generation of a high-temperature plasma in an ICP torch. This is achieved by passing an inert gas, typically argon, through an RF coil. The RF coil induces a high-frequency

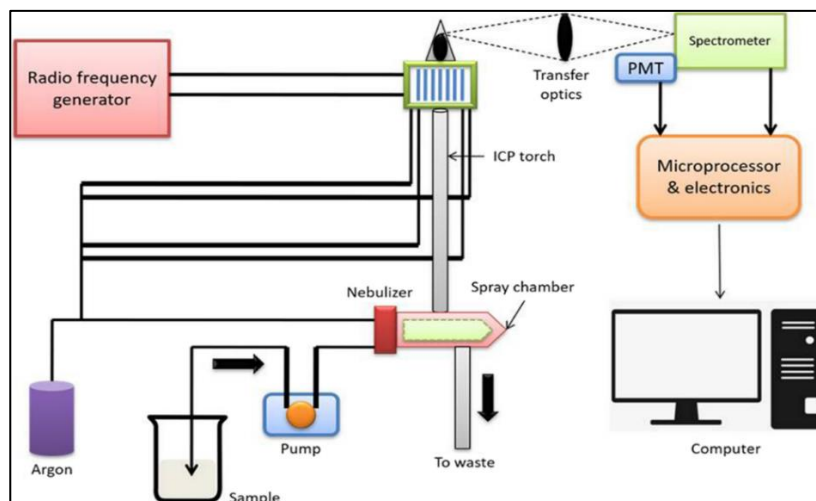


Fig. 13: A pictorial representation of inductively coupled plasma-optical emission spectroscopy.

alternating current in the gas, creating a rapidly oscillating electromagnetic field. This field ionizes the gas, generating a high-temperature plasma flame with temperatures exceeding 10,000 degrees Celsius.

b) Sample introduction and atomization: The sample to be analyzed is introduced into the plasma flame via a nebulizer or spray chamber. The sample is typically in liquid form and is aerosolized into fine droplets by the nebulizer. As the droplets pass through the high-temperature plasma, they are rapidly heated and atomized, converting the sample into a cloud of atoms or ions.

c) Excitation and emission of light: The high-temperature plasma excites the atoms or ions in the sample to higher energy levels. As these excited species return to their ground state, they emit photons of characteristic wavelengths corresponding to the energy differences between the excited and ground states. These emissions form the basis of the elemental fingerprint of the sample.

d) Optical detection: The emitted light is collected and directed into an optical system, which disperses the light into its component wavelengths using diffraction gratings or prisms. The dispersed light is then directed onto a detector, such as a PMT or CCD, which measures the intensity of light at different wavelengths across the spectrum.

e) Data analysis: The resulting emission spectrum, which consists of peaks corresponding to the characteristic wavelengths of the emitted light, is analyzed to determine the elemental composition of the sample. Each peak in the spectrum corresponds to a specific element present in the sample, with the peak intensity proportional to the concentration of the element.

Overall, the principle of ICP-OES relies on the generation of a high-temperature plasma, the excitation of atoms or ions within the plasma, and the subsequent analysis of the emitted light to provide rapid, sensitive, and accurate elemental analysis capabilities for a wide range of samples.

8.1.3 Applications of ICP-OES

ICP-OES is a versatile analytical technique widely used in various industries and research fields for elemental analysis. Some key applications include:

a) Environmental analysis: ICP-OES is used for the analysis of environmental samples such as water, soil, and air. It enables the determination of trace elements, heavy metals, and pollutants, aiding in environmental monitoring, pollution control, and regulatory compliance.

b) Geological exploration: ICP-OES is employed in geological exploration for the analysis of rock, soil, and sediment samples. It helps identify and quantify valuable minerals, trace elements, and rare earth elements, facilitating mineral exploration, resource assessment, and mining operations.

c) Metallurgical analysis: ICP-OES is used in metallurgical industries for quality control and analysis of metal alloys, ores, and industrial materials. It provides information about the

elemental composition, impurities, and trace elements present in metals, ensuring product quality, process optimization, and compliance with industry standards.

d) Pharmaceutical analysis: ICP-OES is employed in the pharmaceutical industry for the analysis of drug formulations, raw materials, and pharmaceutical products. It enables the quantification of elemental impurities, such as heavy metals, in pharmaceuticals, ensuring product safety, purity, and regulatory compliance.

e) Food and beverage analysis: ICP-OES is used for the analysis of food and beverage samples to determine the elemental composition and nutritional content. It helps identify contaminants, such as heavy metals and pesticides, in food products, ensuring food safety and quality control.

f) Clinical and biological analysis: ICP-OES is utilized in clinical and biological research for the analysis of biological samples, tissues, and fluids. It enables the quantification of trace elements and metals in biological specimens, contributing to studies on nutrition, toxicology, and disease diagnosis.

g) Material characterization: ICP-OES is employed for the characterization of materials in materials science and engineering. It provides information about the elemental composition, purity, and homogeneity of materials, supporting research and development in areas such as semiconductor manufacturing, nanotechnology, and advanced materials.

Overall, ICP-OES is a powerful analytical technique with widespread applications in environmental monitoring, industrial analysis, quality control, and scientific research, providing rapid, sensitive, and accurate elemental analysis capabilities for a diverse range of samples and industries.

8.1.4 Limitations of ICP-OES

ICP-OES is highly sensitive and used for elemental analysis and has some limitations:

a) Matrix effects: Interference from matrix components in complex samples can affect the accuracy and precision of quantitative measurements. Matrix effects can lead to spectral overlap, background noise, and spectral interferences, particularly in samples with high concentrations of salts, acids, or organic compounds.

b) Sample preparation: ICP-OES typically requires samples to be in solution form for analysis. Sample preparation can be time-consuming and labor-intensive, involving digestion, dilution, filtration, and other pre-treatment steps to ensure sample homogeneity and compatibility with the instrument.

c) Limited sensitivity for low concentrations: While ICP-OES is highly sensitive, it may lack sensitivity for trace elemental analysis at very low concentrations. Detection limits can vary depending on the element, sample matrix, and instrumental parameters, limiting its applicability for ultra-trace analysis.

d) Limited elemental coverage: While ICP-OES offers simultaneous multi-element analysis, it may have limitations in elemental coverage compared to other techniques, such as

ICP-MS. Certain elements, such as noble gases and isotopes with low atomic mass, may be challenging to detect or quantify using ICP-OES.

e) Cost of instrumentation and maintenance: ICP-OES instruments can be expensive to purchase, operate, and maintain. They require specialized equipment, consumables, and trained personnel for operation, calibration, and maintenance, increasing the overall cost of analysis.

f) Interference from polyatomic species: Interference from polyatomic species, such as molecular ions and oxide or hydroxide ions, can complicate spectral interpretation and quantification. Background correction techniques, such as internal standardization or spectral deconvolution, may be required to mitigate these interferences.

g) Limited spatial resolution: ICP-OES provides bulk elemental analysis of samples and lacks spatial resolution. It cannot provide information about elemental distribution or localization within heterogeneous samples, limiting its applicability for spatially resolved analysis.

Despite these limitations, ICP-OES remains a valuable analytical technique for routine and high-throughput elemental analysis in various industries, research fields, and regulatory applications, providing rapid, sensitive, and accurate determination of elemental composition in diverse samples.

8.2 Ferromagnetic resonance spectroscopy

Ferromagnetic resonance (FMR) spectroscopy is a technique used to study the dynamic magnetic properties of ferromagnetic materials. It relies on the resonance absorption of microwave or RF radiation by processing magnetic moments (spin waves or magnons) within the material when subjected to an external magnetic field.

In FMR, a high-frequency alternating magnetic field is applied to the sample, inducing the magnetic moments to precess around the field direction. The resonance condition occurs when the frequency of the applied field matches the natural frequency of precession of the magnetic moments, resulting in enhanced absorption or emission of electromagnetic radiation.

FMR also provides insights into spin dynamics, relaxation processes, and damping mechanisms in ferromagnetic materials, contributing to the development of magnonic devices and spin-wave-based technologies. Additionally, FMR spectroscopy enables the measurement of magnetic anisotropy parameters and the study of spin-wave excitations, enhancing our understanding of magnetic domain behavior and magnetic switching processes.

Overall, Ferromagnetic Resonance Spectroscopy is a versatile and powerful technique for probing the dynamic magnetic properties of ferromagnetic materials, providing essential information for fundamental research and technological applications in magnetism, spintronics, and magnetic materials science.

8.2.1 Instrumentation of FMR spectroscopy

The instrumentation of FMR is designed to generate, manipulate, and detect electromagnetic radiation and magnetic fields to study the dynamic magnetic properties of ferromagnetic materials. Here's an overview of the essential components of FMR instrumentation as addressed in Fig. 14.

a) Microwave source: The microwave source generates microwave or radiofrequency radiation at the desired frequency range, typically in the gigahertz to terahertz range. This radiation excites the magnetic resonance in the sample and induces the precession of magnetic moments. Common sources include microwave generators, klystrons, Gunn diodes, or solid-state microwave oscillators.

b) Magnetic field source: The magnetic field source produces a static magnetic field that can be varied in strength and direction to control the resonance condition and study the magnetic properties of the sample. Electromagnets or permanent magnets are commonly used to generate magnetic fields with high stability and precision.

c) Sample holder: The sample holder provides a platform for mounting and positioning the ferromagnetic sample within the magnetic field. It allows for precise alignment and orientation of the sample relative to the applied magnetic field and microwave radiation. Sample holders may include temperature control mechanisms for studying temperature-dependent magnetic properties.

d) Detector system: The detector system measures the absorption or emission of electromagnetic radiation by the sample as a function of magnetic field strength, frequency, and temperature. Various types of detectors can be used, including microwave antennas, cavity resonators, or superconducting quantum interference devices (SQUIDs), depending on the frequency range and sensitivity requirements.

e) Temperature control: Temperature control systems maintain the sample at a controlled temperature during measurements to study the temperature dependence of magnetic properties. Cryogenic systems, resistive heaters, or thermal stages may be employed for precise temperature control over a wide range of temperatures.

f) Data acquisition and analysis system: The data acquisition and analysis system records and analyzes the FMR spectra obtained from the sample. It includes lock-in amplifiers, spectrum analyzers, and computer interfaces for signal processing, data visualization, and quantitative analysis of magnetic resonance data.

g) Shielding and isolation: Shielding and isolation measures are implemented to minimize external electromagnetic interference and noise sources that can affect the accuracy and sensitivity of FMR measurements. Faraday cages, magnetic shields, and low-noise amplifiers are commonly used to ensure high signal-to-noise ratios and reliable data acquisition.

These components work together to enable the generation, manipulation, and detection of electromagnetic radiation and magnetic fields for studying the dynamic magnetic properties of ferromagnetic materials using FMR Spectroscopy.

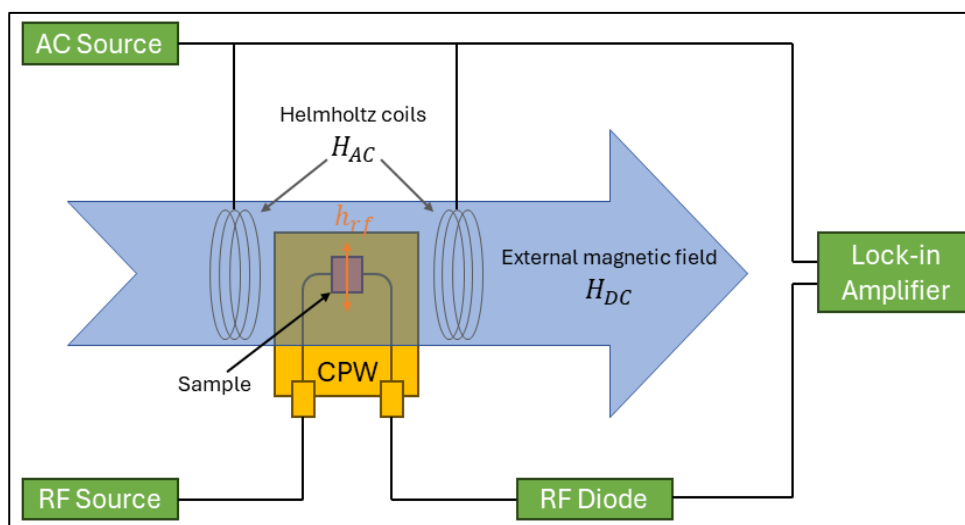


Fig. 14: Schematic of a broadband Ferromagnetic resonance spectroscopy setup.

8.2.2 Principle of FMR Spectroscopy

FMR Spectroscopy is based on the principle of magnetic resonance, where the precession of magnetic moments within a ferromagnetic material is excited and detected using electromagnetic radiation. The principle of FMR spectroscopy can be understood as follows:

a) Magnetic moments precession: In a ferromagnetic material, such as a magnetic thin film or a bulk magnetic sample, the magnetic moments of individual atoms or electron spins align in the same direction, resulting in a net macroscopic magnetization. When an external magnetic field is applied to the material, the magnetic moments align with the field direction and begin to precess around it.

b) Precession frequency: The precession frequency of the magnetic moments, known as the Larmor frequency, is directly proportional to the strength of the applied magnetic field and inversely proportional to the gyromagnetic ratio of the material. The Larmor frequency is given by the equation:

$$f_{\text{Larmor}} = \gamma B \quad (5)$$

where f_{Larmor} is the Larmor frequency, γ is the gyromagnetic ratio, and B is the applied magnetic field strength.

c) Resonance condition: When the frequency of an externally applied electromagnetic radiation matches the Larmor frequency of the magnetic moments, a resonance condition is established. This resonance condition occurs when the energy of the incident radiation matches the energy required to induce transitions between different spin states in the material.

d) Absorption and emission: At resonance, the magnetic moments absorb energy from the incident electromagnetic radiation, causing a decrease in the intensity of the transmitted or reflected radiation. This absorption of energy is detected as a decrease in the intensity of the electromagnetic radiation passing through the sample.

e) Resonance response: The resonance response of the material depends on various factors, including the strength of the applied magnetic field, the frequency of the incident radiation, and the magnetic properties of the material (e.g.,

saturation magnetization and magnetic anisotropy). By varying these parameters, researchers can study the magnetic properties, dynamics, and interactions in ferromagnetic materials.

8.2.3 Applications of introduction to FMR spectroscopy

FMR Spectroscopy finds numerous applications in various scientific and technological fields due to its ability to provide valuable insights into the dynamic magnetic properties of ferromagnetic materials. Some of the applications of FMR spectroscopy are furnished below.^[126-130]

a) Magnetic thin films and multilayers: Magnetic nanomaterials are extensively used in various disciplines of science and technology. Magnetic nanomaterials are used in energy storage devices, imaging instruments, etc. Therefore, it becomes essential to be aware of the magnetic properties of thin films, multilayers, and nanostructures. It provides information about the magnetic anisotropy, saturation magnetization, magnetic damping, and spin dynamics in magnetic thin films used in magnetic recording media, magnetic sensors, and spintronics devices.

b) Magnetic nanoparticles and nanomaterials: FMR spectroscopy is employed to study the magnetic properties of nanoparticles, nanowires, and nanostructures for applications in biomedicine, data storage, and magnetic resonance imaging (MRI). It enables the characterization of magnetic interactions, magnetic relaxation processes, and magnetic switching behavior in nanoscale magnetic materials.

c) Spintronics and magnetic devices: FMR spectroscopy plays a crucial role in spintronics research and device development. It provides insights into spin transport phenomena, spin-wave excitations, and spin dynamics in ferromagnetic materials, essential for the design and optimization of spintronic devices such as magnetic tunnel junctions, spin valves, and magnetic random-access memories (MRAM).

d) Magnetic recording and data storage: FMR spectroscopy

is used in developing and quality control magnetic recording media and data storage devices. It enables the characterization of magnetic properties, recording stability, and thermal stability of magnetic thin films and magnetic recording heads used in hard disk drives, magnetic tapes, and magnetic memory devices.

e) Magnetic resonance imaging: FMR spectroscopy techniques, such as Electron Spin Resonance and Nuclear Magnetic Resonance, are employed in biomedical imaging and diagnostics. FMR-based imaging techniques provide non-invasive and high-resolution imaging of biological tissues, organs, and materials, with applications in medical diagnosis, neuroscience, and materials science.

f) Magnetic materials research: FMR spectroscopy is used in fundamental research on magnetic materials, magnetic phase transitions, and magnetic ordering phenomena. It provides insights into magnetic domain behavior, magnetization dynamics, and magnetic interactions, contributing to advancements in magnetic materials science, magnetism, and condensed matter physics.

g) Magnetic resonance spectroscopy and sensing: FMR spectroscopy techniques are utilized in magnetic resonance spectroscopy and sensing applications. They enable the detection and quantification of magnetic species, paramagnetic impurities, and magnetic defects in materials, as well as the monitoring of magnetic field variations and magnetic resonance phenomena for sensing and detection purposes.

Overall, FMR Spectroscopy offers a versatile and powerful tool for studying the dynamic magnetic properties of ferromagnetic materials, with widespread applications in materials science, spintronic, magnetic recording, biomedical imaging, and magnetic resonance spectroscopy.

8.2.4 Limitations of introduction to FMR spectroscopy

FMR Spectroscopy is a valuable technique for studying the dynamic magnetic properties of ferromagnetic materials, but it also has several limitations. The limitations of Ferromagnetic Spectroscopy are depicted as follows:

a) Sample requirements: FMR spectroscopy requires samples with well-defined magnetic properties and homogeneous magnetization. Irregularities, defects, or inhomogeneities in the sample can lead to broadening of resonance lines and distortion of FMR spectra, affecting the accuracy and reliability of measurements.

b) Sample size and geometry: The size and geometry of the sample can influence the FMR signal and resonance behavior. Thin films, nanoparticles, and nanoscale structures may exhibit size-dependent magnetic properties and complex resonance spectra, requiring careful interpretation and analysis.

c) Frequency range: FMR spectroscopy is typically limited to microwave or radiofrequency ranges, which may not cover all relevant magnetic resonance frequencies in ferromagnetic materials. Some materials may exhibit resonance behavior at higher frequencies or require specialized instrumentation for

high-frequency FMR measurements.

d) Instrumentation complexity: FMR spectroscopy instrumentation can be complex and expensive, requiring specialized equipment such as microwave sources, magnetic field sources, and sensitive detectors. Maintenance, calibration, and operation of FMR instruments may require expertise in electromagnetics, spectroscopy, and materials science.

e) Temperature dependence: The magnetic properties and resonance behavior of ferromagnetic materials can be strongly temperature-dependent. Temperature variations can affect the linewidth, resonance frequency, and intensity of FMR signals, necessitating precise temperature control and thermal stabilization during measurements.

f) Signal-to-noise ratio: FMR spectroscopy measurements may be susceptible to noise sources and electromagnetic interference, particularly at low signal levels or in high-frequency regimes. Achieving high signal-to-noise ratios and minimizing background noise are essential for accurate and reliable FMR measurements.

g) Surface sensitivity: FMR spectroscopy primarily probes the surface region of materials, which may limit its applicability to surface-sensitive systems or thin films. Depth profiling capabilities are limited, and FMR measurements may not provide information about subsurface layers or buried interfaces without additional techniques.

h) Complexity of analysis: Interpretation and analysis of FMR spectra can be challenging, especially for complex materials, multilayers, or nanoscale structures. Extracting quantitative information about magnetic parameters, spin dynamics, and magnetic interactions from FMR data may require sophisticated modeling and theoretical analysis.

Despite these limitations, FMR Spectroscopy remains a valuable tool for studying the dynamic magnetic properties of ferromagnetic materials, providing essential insights into spin dynamics, magnetic interactions, and magnetic phenomena at the atomic scale. Addressing these limitations through methodological improvements and complementary techniques continues to be an active area of research in FMR spectroscopy.

9 Miscellaneous types

9.1 X-ray absorption spectroscopy

X-ray absorption spectroscopy (XAS) is a powerful analytical technique that provides insights into the electronic and structural properties of materials at the atomic scale. The X-Rays were discovered by Wilhelm Conrad Roentgen in 1895. Soon after its discovery, it became very popular among physicians and was used to detect fractures in bone. Then, in the modern period, X-rays are used for various types of imaging like CT-Scan, etc. Apart from this, X-rays are used for stereotactic surgery and in radiation therapy to treat cancer. Apart from medicinal use, X-rays are widely used in material sciences, nanoscience, and technology for confirmation of types and crystallinity of the sample under study.

The historical development of XAS accelerated with the pioneering work of Henry G. J. Moseley in the early 20th

century. Moseley's experiments revealed a systematic relationship between the X-ray absorption edges of elements and their atomic numbers, leading to the concept of atomic number as a fundamental property. This laid the foundation for XAS as a spectroscopic technique for studying the electronic structure of materials.^[131-133]

Modern XAS encompasses two main spectroscopic techniques: X-ray Absorption Near-Edge Structure (XANES) and Extended X-ray Absorption Fine Structure (EXAFS). XANES provides information about the electronic structure and oxidation states of elements, while EXAFS reveals details about the local atomic environment and coordination geometry around the absorbing atoms.

In contemporary XAS experiments, synchrotron radiation sources have become indispensable for generating intense and tunable X-ray beams required for high-resolution spectroscopy. Synchrotron facilities worldwide offer state-of-the-art X-ray beamlines equipped with advanced instrumentation for XAS studies. Sample preparation techniques have also evolved, allowing for the investigation of diverse materials, including solids, liquids, and gases, under various conditions of temperature and pressure.

XAS finds applications across a wide range of scientific disciplines, including chemistry, physics, materials science, environmental science, and biochemistry. It is used to study catalysts, semiconductors, magnetic materials, proteins, minerals, and environmental contaminants, among others. Its versatility makes it an invaluable tool for fundamental research, technological development, and industrial applications.

In summary, X-ray Absorption Spectroscopy has undergone significant historical development since its discovery, evolving into a sophisticated analytical technique with broad scope and diverse applications. With advancements in synchrotron technology and sample preparation methods, XAS continues to play a vital role in advancing our understanding of materials and their properties at the atomic level.

9.1.1 Instrumentation of XAS

XAS is a powerful technique used to investigate the electronic and atomic structure of materials. Fig. 15 demonstrates the diagram of X-ray absorption. The instrumentation for XAS typically involves the following components:

a) X-ray source: XAS requires a source of high-energy X-rays, typically produced by an X-ray tube or synchrotron radiation.

b) Monochromator: X-rays emitted from the source often have a broad energy spectrum. A monochromator is used to select X-rays of a specific energy or wavelength, ensuring that only the desired energy range is used for the experiment.

c) Sample holder: The sample holder holds the material being studied. It should allow for precise positioning and alignment of the sample, as well as control over its orientation relative to the incident X-rays.

d) Detector: X-ray detectors capture the intensity of X-rays transmitted through or scattered by the sample as a function of energy. Different types of detectors, such as ionization chambers, solid-state detectors, or proportional counters, may be used depending on the specific requirements of the experiment.

e) Energy calibration system: Accurate energy calibration is crucial for XAS measurements. This system ensures that the energy scale of the experiment is calibrated precisely, typically using known reference materials with well-defined absorption edges.

f) Data acquisition system: A data acquisition system is used to record the X-ray absorption spectra obtained from the detector. This system collects, processes, and stores the experimental data for further analysis.

g) Control system: A control system coordinates the operation of various components of the XAS instrument, including the X-ray source, monochromator, sample holder, and detector. It allows for precise control over experimental parameters such as X-ray energy, sample positioning, and data acquisition settings.

h) Vacuum system: XAS experiments are often performed under vacuum conditions to minimize X-ray absorption and scattering by air. A vacuum system maintains a controlled environment around the sample during the experiment.

i) Beamline optics: In synchrotron-based XAS experiments, beamline optics such as mirrors, lenses, and slits are used to manipulate and focus the X-ray beam, optimizing its intensity and spatial resolution at the sample position.

j) Safety interlocks: X-ray sources can pose radiation hazards, so safety interlocks are essential to ensure that the instrument operates safely. These interlocks may include radiation shielding, emergency stop mechanisms, and radiation monitoring systems.

Overall, the instrumentation of XAS is designed to provide precise control over experimental parameters and to obtain accurate measurements of XAS for the study of material properties and electronic structure.

9.1.2 Principle of XAS

X-ray absorption spectroscopy (XAS) is a technique used to probe the electronic and atomic structure of materials based on their interaction with X-rays. The principle behind XAS lies in the absorption of X-rays by atoms in the material and the subsequent analysis of how this absorption varies with X-ray energy.

The working of XAS is furnished as below:

a) X-ray absorption: Like other electromagnetic waves, when an X-ray passes through the sample under study, it interacts with inner-shell electrons of the atoms in the sample. Once the energy difference between core energy levels (typically K or L shell matches), the X-rays can be absorbed by the atom, causing the core electron to be ejected from its shell. This process is known as X-ray absorption.

b) Absorption edge: The absorption of X-rays by atoms

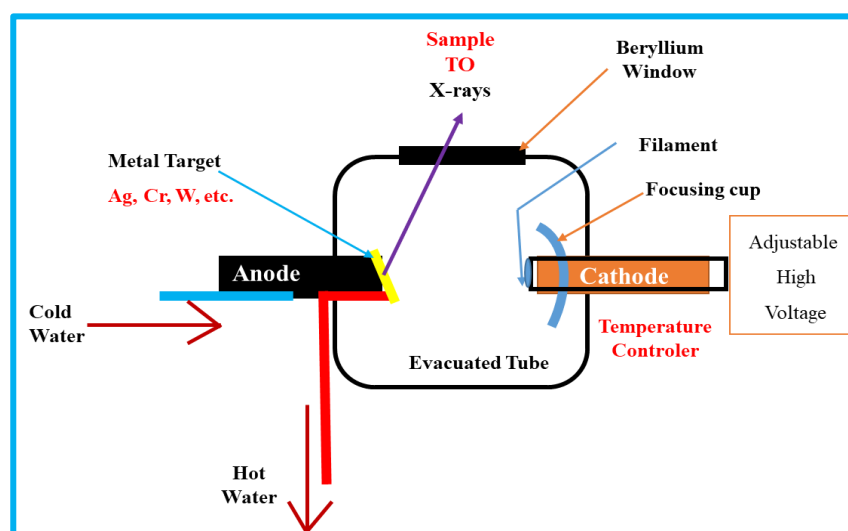


Fig. 15: A pictorial representation of X-ray absorption spectroscopy.

shows a characteristic energy dependence. At specific X-ray energies, known as absorption edges, there is a sharp increase in X-ray absorption due to resonant excitation of core electrons. The absorption edge corresponds to the energy required to promote an electron from a core level to an unoccupied state in the atom.

c) X-ray absorption spectrum: By measuring the intensity of X-rays transmitted through the sample as a function of X-ray energy, one can obtain an X-ray absorption spectrum. The spectrum typically exhibits a series of peaks and valleys, with each peak corresponding to an absorption edge of a specific element in the sample. The energy position, shape, and intensity of these peaks provide information about the electronic and atomic structure of the material, including the oxidation state, coordination environment, and chemical bonding.

d) X-ray fluorescence: In addition to absorption, X-rays can induce fluorescence in the sample, where the absorbed energy is re-emitted as lower-energy X-rays or visible light. The detection of fluorescence emitted by the sample can provide complementary information to X-ray absorption spectroscopy.

e) XAS techniques: There are several variations of XAS techniques, including XANES and EXAFS. XANES focuses on the spectral region near the absorption edge and provides information about the electronic structure and chemical bonding. At the same time, EXAFS analyzes the fine structure beyond the absorption edge and yields details about the local atomic environment and coordination.

In summary, X-ray Absorption Spectroscopy is based on the principle of X-ray absorption by atoms in a material, which results in characteristic absorption edges in the X-ray spectrum. By analyzing the energy dependence of X-ray absorption, valuable insights into the electronic and atomic structure of materials can be obtained.

9.1.3 Applications of XAS

XAS finds applications across various fields due to its

capability to provide detailed information about the electronic and atomic structure of materials. Some of the critical applications of XAS include:

a) Material science and catalysis: Today, material science has become an essential area of research. Researchers from different disciplines like physics, chemistry, biological sciences, engineering, and technology are actively engaged in material science research to analyze materials that require advanced characterization techniques. XAS is extensively used to study the local structure, electronic configuration, and chemical bonding in a wide range of materials, including metals, alloys, oxides, semiconductors, and catalysts. It helps in understanding the mechanisms of catalytic reactions, identifying active sites, and optimizing catalyst performance.

b) Environmental science and engineering: XAS investigates the speciation, distribution, and behavior of elements in environmental samples such as soils, sediments, water, and air. It aids in studying processes like metal ion sorption, mineral dissolution, and redox transformations, contributing to environmental remediation and pollution control efforts.

c) Biological and life sciences: XAS plays a crucial role in structural biology and bioinorganic chemistry by providing insights into the coordination geometry, oxidation states, and metal-binding sites in biomolecules such as proteins, enzymes, nucleic acids, and metalloproteins. It helps understand biological functions, metal ion transport, and metalloenzyme mechanisms.

d) Geosciences and mineralogy: XAS characterizes the composition, structure, and transformation of minerals, rocks, and geological materials. It helps in studying processes like mineral weathering, diagenesis, mineral-microbe interactions, and the behavior of contaminants in natural systems.

e) Energy materials and batteries: XAS is valuable for investigating the electronic and atomic structure of energy-related materials such as electrodes, electrolytes, and catalysts in batteries, fuel cells, and photovoltaic devices. It aids in

improving energy storage and conversion technologies by understanding factors like charge transfer mechanisms, ion diffusion, and electrode-electrolyte interfaces.

f) Nanomaterials and nanotechnology: XAS is used to characterize the local structure, electronic properties, and surface chemistry of nanomaterials including nanoparticles, nanowires, and nanocomposites. It helps in designing and optimizing nanomaterials for applications in electronics, sensors, catalysis, and biomedicine.

g) Art and archaeology: XAS is employed in cultural heritage studies to analyze the composition, degradation, and conservation of artworks, archaeological artifacts, and historical materials. It aids in identifying pigments, detecting forgeries, studying corrosion processes, and developing preservation strategies.

h) Semiconductors and electronics: XAS is utilized in semiconductor research and device fabrication to investigate defects, impurities, doping profiles, and interface properties in semiconductor materials and devices. It contributes to the development of advanced electronic and optoelectronic devices.

Overall, XAS is a versatile analytical technique with applications spanning diverse fields, contributing to fundamental research, technological advancement, and societal challenges.^[134-136]

9.1.4 Limitation of XAS

Although XAS is a technique for probing the electronic and atomic structure of materials, it does have certain limitations:

a) Elemental specificity: XAS is most effective for elements with heavy atoms (*e.g.*, transition metals and rare earth elements) due to their strong absorption of X-rays. Lighter elements such as hydrogen and helium have weaker absorption signals, making their analysis challenging with conventional XAS techniques.

b) Sample requirements: XAS requires samples to be in a suitable form for X-ray analysis, which may involve preparation techniques such as powdering, pelletizing, thin-film deposition, or immobilization on substrates. Sample manipulation and preparation can introduce artifacts or alter the native structure, potentially affecting the interpretation of results.

c) Limited spatial resolution: XAS typically provides information averaged over the entire sample volume or surface region interrogated by the X-ray beam. Spatially resolved XAS techniques exist, but achieving high spatial resolution is challenging, particularly in heterogeneous samples or at the nanoscale.

d) Surface sensitivity: XAS probes the near-surface region of the sample, typically within a few micrometers to tens of nanometers, depending on the X-ray energy and material properties. This surface sensitivity can be advantageous for studying surface phenomena but may limit the depth resolution and penetration depth in bulk or deeply buried layers.

e) Data analysis complexity: Analysis of XAS spectra involves sophisticated data processing and interpretation techniques, including background subtraction, energy calibration, normalization, and fitting to theoretical models or reference spectra. Interpreting XAS results requires expertise in spectroscopy, quantum chemistry, and materials science.

f) Synchrotron access: High-quality XAS experiments often require access to synchrotron radiation facilities, which may be limited in availability and require specialized training and collaboration. Traveling to synchrotron beamlines and allocating beam time can also present logistical challenges for researchers.

g) Radiation hazards: XAS experiments involve exposure to X-rays, which can pose radiation hazards to operators and require appropriate safety precautions and shielding measures to minimize risks.

h) Limited chemical sensitivity: XAS provides valuable information about the local atomic structure and chemical bonding, but it may not always distinguish between different chemical environments or coordination geometries if they have similar X-ray absorption properties.

Despite these limitations, X-ray Absorption Spectroscopy remains a valuable tool for materials characterization, offering unique insights into the electronic and atomic properties of a wide range of materials in diverse fields of science and technology. Advances in instrumentation, data analysis methods, and synchrotron beamline capabilities continue to address some of these limitations and expand the capabilities of XAS techniques.

9.2 Atomic absorption spectroscopy

Atomic absorption spectroscopy (AAS) is a powerful analytical technique used to quantify the concentration of elements in a sample based on their light absorption at specific wavelengths. AAS was developed in 1955 by Alan Walsh, and today, it has become a vital characterization tool in agro-chemistry, environmental science and engineering, food science, nanoscience and technology, chemistry, *etc.* Various elements' presence and concentration can be determined using AAS.^[137-140]

The working principle of AAS depends upon the interaction of electromagnetic radiation with atoms in the sample. By detecting the intensity of electromagnetic radiation at the time of entry into the sample and after exiting from the sample, the concentration of the target element can be determined.

The scope of AAS is vast, encompassing the analysis of a wide range of elements across various sample matrices. It offers high sensitivity, selectivity, and precision, making it suitable for trace analysis and quality control in diverse industries. AAS is particularly valuable for quantifying metals and metalloids, including toxic heavy metals like lead, cadmium, and mercury, at low concentrations.

Modern AAS instruments are sophisticated and user-friendly, incorporating advanced features for enhanced

performance and productivity. They typically consist of a light source (e.g., hollow cathode lamp), monochromator, sample introduction system (e.g., flame or graphite furnace), atomizer, and detector (e.g., photomultiplier tube). Automation and computerization have streamlined the analytical workflow, allowing rapid sample throughput and data processing.

In conclusion, AAS has a rich historical development, from early observations of elemental absorption patterns to the development of sophisticated modern instruments. Its scope spans diverse applications, driven by its sensitivity, selectivity, and precision, making it an indispensable tool in analytical chemistry.

9.2.1 Instrumentation of AAS

Instrumentation typically involves several key components designed to facilitate measuring the absorption of light by atoms in a sample. The key components of AAS are described below, and a schematic diagram is shown in Fig. 16.

a) Light source: AAS essentially requires a suitable light source. In AAS instruments, a hollow cathode lamp or electrodeless discharge lamp is used as the primary light source. These lamps emit specific wavelengths of light corresponding to the resonance lines of the target elements. The choice of lamp depends on the elements to be analyzed.

b) Monochromator: All spectroscopic techniques require a monochromatic source of radiation. Therefore, a suitable monochromator selects a narrow range of wavelengths from the emitted light source. It ensures that only the wavelengths corresponding to the absorption lines of the target element are directed to the sample and subsequently measured.

c) Sample introduction system: Samples are introduced into the AAS instrument in various forms, such as liquid solutions, solid samples, or gases. The sample introduction system typically consists of an atomizer, which converts the sample into atomic vapor for analysis. Common atomization techniques include flame atomization (liquid samples) and graphite furnace atomization (solid samples).

d) Optical path: The optical path in an AAS instrument includes components such as lenses, mirrors, and a flame or graphite furnace. These components direct the light from the monochromator through the sample and onto the detector.

e) Detector: A photodetector, such as a PMT or photodiode array, detects the intensity of light transmitted through the sample. The detector converts the optical signal into an electrical signal, which is then processed for analysis.

f) Signal processing and data Analysis: The detector captures the intensity of light. The resultant electrical signal is processed and analyzed by the instrument's electronics. This includes background correction, signal averaging, and calibration. The resulting absorption spectrum provides information about the concentration of the target element in the sample.

g) Control and data acquisition system: AAS instruments are controlled by software that allows users to set experimental parameters, control instrument functions, and collect data. The data acquisition system records absorption measurements as a function of wavelength or time, facilitating quantitative analysis.

h) Safety features: AAS instruments incorporate safety features to ensure safe operation, especially when dealing with flammable gases, high temperatures, and potentially hazardous chemicals. These features may include interlocks, gas sensors, and automatic shut-off mechanisms.

Overall, the instrumentation of AAS is designed to provide accurate and reliable measurements of elemental concentrations in various sample matrices, making it a widely used technique in analytical chemistry.

9.2.2 Principle of AAS

AAS is based on the absorption of electromagnetic radiation (typically ultraviolet or visible light) by ground-state atoms in a sample. This absorption occurs at specific wavelengths corresponding to the electronic transitions between energy levels within the atoms.

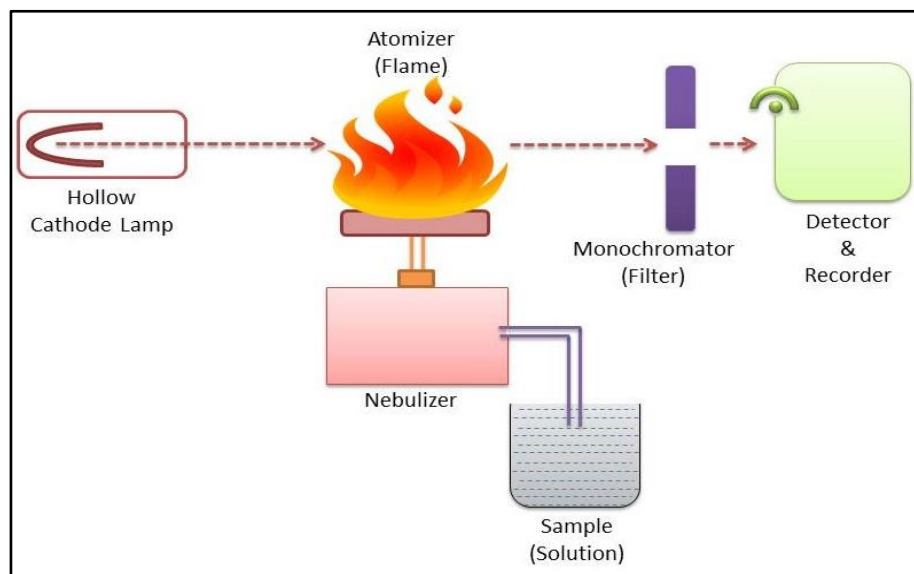


Fig. 16: A pictorial representation of atomic absorption spectroscopy.

Here's a breakdown of the principle of AAS:

a) Atomic absorption: When a beam of light with a range of wavelengths passes through a sample containing atoms of the target element, the atoms absorb light at characteristic wavelengths specific to the element. This absorption occurs when the energy of the photons matches the energy difference between the ground state and an excited state of the atom.

b) Beer-Lambert law: Beer Lambert's law states that 'When a beam of monochromatic light is passed through homogeneous absorbing medium, then the rate of decrease in intensity of radiation with the thickness of absorbing medium is directly proportional to the intensity of light as well as concentration of solution'. Mathematically, this relationship is expressed as equation 4.

$$A = \epsilon bc \quad (4)$$

where A is the absorbance, ϵ is the molar absorptivity (or extinction coefficient), b is the path length, and c is the concentration of the absorbing species.

c) Absorption spectrum: By scanning the wavelength of the light source across a range of wavelengths and measuring the absorbance at each wavelength, an absorption spectrum is obtained. The absorption spectrum typically exhibits one or more absorption lines or bands corresponding to transitions between different electronic energy levels of the atoms.

d) Quantitative analysis: The intensity of the absorption lines in the spectrum is directly related to the concentration of the target element in the sample. By comparing the absorbance of the sample to that of standard solutions with known concentrations, the concentration of the target element in the sample can be quantified.

e) Selectivity and sensitivity: AAS is highly selective because each element has its characteristic absorption lines, allowing for the simultaneous determination of multiple elements in a single sample. Additionally, AAS is sensitive, capable of detecting elements at trace levels (parts per million to parts per billion) due to the sharpness of the absorption lines and the low noise of the instrument.

Thus, Atomic Absorption Spectroscopy provides valuable quantitative information about the concentration of specific elements in a sample based on their absorption of light at characteristic wavelengths, making it a widely used technique in analytical chemistry for elemental analysis.

9.2.3 Applications of AAS

Some of the key applications of AAS include:

a) Environmental analysis: The detection of presence and concentration of pollutants present in water sample; soil sample or air sample is important in environment monitoring. AAS is an effective tool for the same. The presence and concentration of heavy metal ions such as lead, cadmium, mercury, arsenic *etc.* can be determine using AAS.

b) Clinical and pharmaceutical analysis: AAS is employed in clinical laboratories and pharmaceutical industries for the analysis of biological fluids, tissues, drugs, and pharmaceutical products. It facilitates the determination of

essential elements and toxic metals in biological samples, aiding in disease diagnosis, nutritional assessment, and drug quality control.

c) Food and beverage analysis: Food adulteration is a global problem. To control over it Food and Drug department of various countries are working. AAS can be effectively used to determine presence and concentration of various heavy metals in the food sample. It helps in monitoring nutritional elements, assessing food safety, and detecting contaminants such as heavy metals, pesticides and toxic elements that may pose health risks to consumers.

d) Metallurgical and mining industry: AAS plays a crucial role in the metallurgical and mining industry for the analysis of ores, metals, alloys, and metallurgical products. It enables the determination of elemental composition, impurity levels, and quality control parameters in raw materials and finished products, contributing to process optimization and product quality assurance.

e) Agricultural and soil science: AAS is applied in agricultural research and soil science to analyze the elemental content of soils, fertilizers, plant tissues, and agricultural products. It assists in assessing soil fertility, nutrient uptake by plants, and the impact of agricultural practices on soil quality, helping farmers optimize crop yields and sustainable farming practices.

f) Forensic science: AAS is used in forensic laboratories for elemental analysis of forensic samples, including trace evidence, biological fluids, and illicit substances. It aids in forensic investigations, crime scene analysis, and the detection of toxic elements or chemical contaminants in forensic samples.

g) Industrial process control: AAS is employed in industrial process control and quality assurance in manufacturing industries such as metal production, chemical processing, and electronics manufacturing. It enables real-time monitoring of elemental composition, impurity levels, and product quality parameters, ensuring compliance with regulatory standards and specifications.

h) Art and archaeology: AAS is applied in cultural heritage studies to analyze the elemental composition of archaeological artifacts, artworks, and historical materials. It helps in identifying materials, studying manufacturing techniques, detecting forgeries, and assessing the conservation status of cultural artifacts.

9.2.4 Limitations of AAS

While AAS is a widely used and valuable analytical technique, it does have certain limitations:

a) Limited elemental coverage: AAS is primarily limited to elements that have suitable absorption lines within the wavelength range of the instrument. Light elements such as hydrogen and helium, as well as some heavier elements, may not have strong absorption lines in the ultraviolet or visible region, restricting the elemental coverage of AAS.

b) Interference effects: AAS measurements can be affected

by interference from matrix components or coexisting species in the sample. Interference can arise from spectral overlap with other elements, chemical interactions, or physical processes such as scattering or background absorption, leading to inaccurate results if not properly addressed.

c) Sample matrix effects: The composition and properties of the sample matrix can influence the accuracy and precision of AAS measurements. Matrix effects may cause signal suppression or enhancement, spectral interferences, or changes in atomization efficiency, particularly in complex sample matrices or high-concentration samples.

d) Sample preparation requirements: AAS often requires samples to be prepared in a specific form suitable for analysis, such as dissolution, digestion, or atomization. Sample preparation can be time-consuming, labor-intensive, and may introduce potential sources of error or contamination if not performed carefully.

e) Sensitivity and dynamic range: While AAS is capable of detecting elements at trace levels (parts per million to parts per billion), its sensitivity and dynamic range may be limited compared to other analytical techniques such as inductively coupled plasma mass spectrometry (ICP-MS) or atomic fluorescence spectroscopy (AFS). High-concentration samples may saturate the detector or require dilution, limiting the measurement range.

f) Single-element analysis: AAS typically measures one element at a time, requiring separate measurements for each element of interest. This can be time-consuming for multi-element analysis and may not be suitable for high-throughput screening or simultaneous analysis of multiple elements in complex samples.

g) Instrument complexity and cost: AAS instruments can be complex and require skilled operators for operation, maintenance, and data interpretation. Additionally, the initial cost of AAS instrumentation and the ongoing expenses for consumables, calibration standards, and maintenance may be significant, particularly for high-performance instruments.

9.3 Photoluminescence spectroscopy

Photoluminescence spectroscopy is a powerful analytical technique used to study the emission of light (photoluminescence) from materials upon excitation by photons. Its development traces back to the late 19th and early 20th centuries when scientists observed luminescence phenomena in certain materials. However, the modern understanding and application of photoluminescence spectroscopy began to take shape in the mid-20th century with advancements in quantum mechanics and solid-state physics.

The discovery of photoluminescence dates back to the 19th century when researchers observed the phenomenon of fluorescence and phosphorescence in certain materials. Notable milestones include Sir George Stokes' discovery of fluorescence in 1852 and Henri Becquerel's observation of phosphorescence in uranium salts in 1896. These early observations laid the groundwork for the study of

luminescence phenomena, leading to the development of photoluminescence spectroscopy techniques.^[141-143]

Photoluminescence spectroscopy encompasses a wide range of applications in materials science, physics, chemistry, biology, veterinary sciences, nanoscience, environmental sciences, and engineering. It is used to investigate the electronic structure, energy levels, and optical properties of materials, including semiconductors, quantum dots, organic molecules, nanomaterials, and biological specimens. Photoluminescence spectroscopy is valuable for characterizing materials in fields such as optoelectronics, photonics, photovoltaics, sensors, and biomedical imaging.

Modern photoluminescence spectroscopy instrumentation typically includes a light source to excite the sample, optical components for wavelength selection and light focusing, a sample holder or stage for positioning the sample, and a detector to measure the emitted light. Excitation sources may include lasers, lamps, or LEDs, while detectors range from PMT to CCD cameras or spectrometers. Advanced instrumentation may incorporate time-resolved techniques to study luminescence dynamics on nanosecond to femtosecond timescales, as well as spatially resolved techniques for mapping luminescence properties across a sample surface.

In summary, photoluminescence spectroscopy has a rich historical development, from early observations of luminescence phenomena to the modern understanding and application of spectroscopic techniques. Its scope encompasses diverse fields of research and technology, with applications ranging from fundamental studies of materials to practical applications in optoelectronic devices, sensors, and biomedical imaging.

9.3.1 Instrumentation of photoluminescence spectroscopy

The instrumentation of photoluminescence spectroscopy typically includes several key components designed to excite a sample with light and analyze the emitted photoluminescence. The instrumentation diagram is presented in Fig.17.

Here's an overview of the main components:

a) Light source: Photoluminescence spectroscopy requires a light source to excite the sample with photons of specific energy or wavelength. Common light sources include lasers, lamps, light-emitting diodes (LEDs), or arc lamps, depending on the spectral range and intensity required for the experiment.

b) Monochromator or filters: A monochromator or optical filters are used to select a narrow range of wavelengths from the excitation light source. This ensures that only photons of the desired energy are used to excite the sample, minimizing background noise and interference.

c) Sample holder or stage: The sample holder or stage holds the sample securely in place during the experiment. It may allow for precise positioning and alignment of the sample relative to the excitation light source and the detection system.

d) Detector: A photodetector detects the emitted photoluminescence from the sample. Common detectors

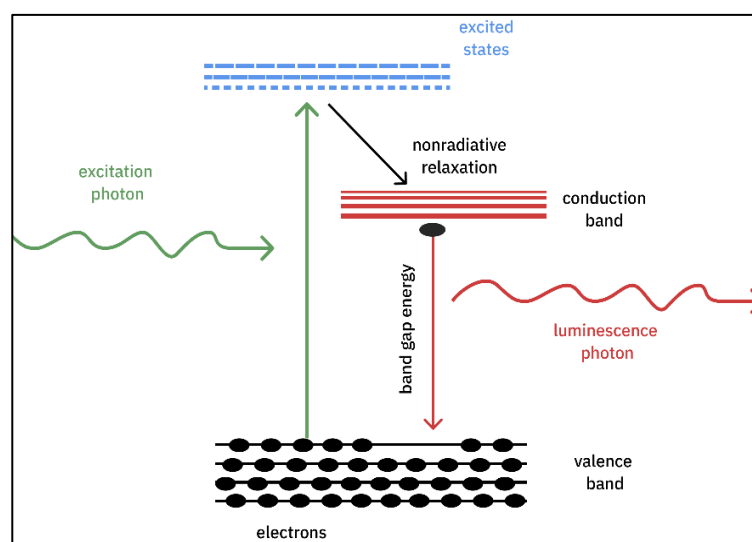


Fig. 17: A pictorial representation of supercontinuum laser photoluminescence spectroscopy.

include PMTs, CCDs, or photomultiplier arrays. The choice of detector depends on factors such as sensitivity, spectral range, and time resolution required for the experiment.

e) Optical components: Various optical components such as lenses, mirrors, and filters are used to direct, focus, and manipulate the excitation light and emitted photoluminescence. These components optimize the signal-to-noise ratio and spatial resolution of the spectroscopic measurements.

f) Excitation and emission optics: Excitation and emission optics are designed to separate the excitation and emission light paths, minimizing the excitation light that reaches the detector and maximizing the collection efficiency of the emitted photoluminescence.

g) Data acquisition system: A data acquisition system records and processes the photoluminescence signal detected by the photodetector. This system may include ADCs, signal amplifiers, data acquisition cards, and computer software for data analysis and visualization.

h) Control and automation: Many photoluminescence spectroscopy instruments are equipped with control and automation features to facilitate experiment setup, parameter adjustment, and data acquisition. This may include motorized stages, computer-controlled shutters, and software interfaces for user interaction.

9.3.2 Principle of photoluminescence spectroscopy

Photoluminescence spectroscopy is based on the principle of analyzing the emission of light (photoluminescence) from materials upon excitation by photons. The technique provides valuable insights into the electronic structure, energy levels, and optical properties of materials.

Here's a breakdown of the principle of photoluminescence spectroscopy:

a) Excitation of sample: Photoluminescence spectroscopy begins with the excitation of a sample using photons of specific energy or wavelength. These photons are typically

provided by a light source such as a laser, lamp, or LED. The energy of the excitation photons must match or exceed the energy required to promote electrons in the material from the ground state to higher-energy excited states.

b) Electronic transitions: Upon absorption of the excitation photons, electrons in the material are promoted to higher-energy electronic states. This excitation creates electron-hole pairs (excitons) or other excited states within the material's electronic band structure. The energy difference between the ground state and the excited state determines the wavelength of the emitted photoluminescence.

c) Emission of photons: After a brief period of time, the excited electrons relax back to lower-energy states, releasing energy in the form of photons. This process, known as photoluminescence, can occur through different mechanisms depending on the material and its electronic structure. Common types of photoluminescence include fluorescence, phosphorescence, and photoluminescence from semiconductor band-edge or defect states.

d) Spectral analysis: The emitted photoluminescence is collected and analyzed spectroscopically to characterize the material's optical properties. A spectrometer or monochromator is typically used to separate the emitted light into its constituent wavelengths or energies. The resulting photoluminescence spectrum provides information about the energy levels, bandgap, excitonic transitions, and other optical properties of the material.

e) Quantitative analysis: Photoluminescence spectroscopy can also be used for quantitative analysis by correlating the intensity of the photoluminescence signal with properties of the sample such as concentration, purity, or defect density. Calibration curves or standards with known properties are often used for quantitative measurements.

Thus, photoluminescence spectroscopy is a powerful technique for studying the optical properties and electronic structure of materials. It provides valuable information about excitonic processes, bandgap energies, defect states, and

luminescent behavior, making it widely used in materials science, semiconductor physics, optoelectronics, and other fields of research and technology.

9.3.3 Applications of photoluminescence spectroscopy

Photoluminescence spectroscopy finds diverse applications across various fields due to its ability to provide valuable insights into the optical properties and electronic structure of materials. Some of the key applications of photoluminescence spectroscopy include:

a) Semiconductor characterization: Photoluminescence spectroscopy is extensively used in semiconductor research and device fabrication to characterize the electronic structure, bandgap energies, and defect states in semiconductor materials such as silicon, gallium arsenide, and quantum dots. It helps in optimizing the performance of semiconductor devices such as light-emitting diodes (LEDs), solar cells, and lasers by studying carrier dynamics and recombination processes.

b) Nanomaterials and nanotechnology: Photoluminescence spectroscopy plays a crucial role in the characterization of nanomaterials such as nanoparticles, nanowires, and nanocomposites. It provides insights into size-dependent optical properties, quantum confinement effects, and surface states, facilitating the design and engineering of nanomaterials for applications in electronics, photonics, sensing, and biomedicine.

c) Optical Materials and photonics: Photoluminescence spectroscopy is used to study the optical properties of materials such as phosphors, luminescent materials, and optical fibers. It aids in the development of advanced optical materials for applications in lighting, displays, sensors, telecommunications, and fiber-optic communication systems.

d) Biosensing and biomedical imaging: In biomedical research and diagnostics, photoluminescence spectroscopy is employed for fluorescence imaging, biomolecular sensing, drug discovery, and molecular imaging of biological tissues and cells. It offers non-invasive, high-resolution imaging capabilities for studying biological processes, disease mechanisms, and drug interactions in living organisms.

e) Materials science and chemistry: Photoluminescence spectroscopy is used in materials science and chemistry to study the luminescent properties of organic and inorganic compounds, polymers, and dyes. It facilitates the investigation of chemical structure-property relationships, photophysical processes, and luminescent behavior in solution or solid-state materials.

f) Environmental monitoring: Photoluminescence spectroscopy is applied in environmental science and engineering for pollution monitoring, trace metals and organic compounds in air, water and samples. It aids in identifying contaminants, assessing environmental health risks, and monitoring remediation efforts.

g) Art and archaeology: Photoluminescence spectroscopy is utilized in cultural heritage studies to analyze the luminescent

properties of archaeological artifacts, artworks, and historical materials. It helps in identifying pigments, detecting forgeries, studying degradation mechanisms, and assessing the conservation status of cultural artifacts.

Thus, photoluminescence spectroscopy is a versatile analytical technique with broad applications in materials science, nanotechnology, photonics, biomedicine, environmental science, and cultural heritage studies, providing valuable insights into the optical properties and electronic structure of materials across diverse fields of research and technology.

9.3.4 Limitations of photoluminescence spectroscopy

While photoluminescence spectroscopy is a powerful analytical technique, it does have certain limitations. The limitations of photoluminescence spectroscopy are furnished below:

a) Sample preparation requirements: Photoluminescence spectroscopy often requires samples to be prepared in specific forms, such as thin films, powders, or solutions, depending on the experimental setup and the properties of the material under study. Sample preparation requires special skills and also time consuming process.

b) Surface sensitivity: Photoluminescence spectroscopy is sensitive for the upper layer, while the internal bulk may not be accurately determined. Therefore, it cannot be used effectively for determination of internal structure of the materials.

c) Fluorescence quenching: In some cases, the presence of impurities, defects, or quenchers in the sample can lead to fluorescence quenching, where the intensity or lifetime of the photoluminescence signal is reduced. This can affect the accuracy and reliability of the spectroscopic measurements, particularly in samples with high concentrations of impurities or reactive species.

d) Spectral overlap: Photoluminescence spectra can exhibit spectral overlap with other luminescent species or background signals, making it challenging to resolve individual components or identify specific emission features. This can complicate data analysis and interpretation, particularly in complex samples with multiple emitting species.

e) Temperature sensitivity: Photoluminescence properties, such as emission intensity, wavelength, and lifetime, maybe temperature-dependent, particularly in materials with energy transfer or thermal quenching effects. Controlling and maintaining sample temperature during measurements may be necessary to minimize temperature-induced variations in photoluminescence behavior.

f) Limited quantitative analysis: While photoluminescence spectroscopy can provide qualitative information about the optical properties and electronic structure of materials, quantitative analysis may be challenging due to factors such as sample heterogeneity, instrumental limitations, and calibration requirements. Establishing accurate calibration curves or standards for quantitative measurements can be

difficult, particularly for complex samples or dynamic processes.

g) Instrumentation complexity: High-performance photoluminescence spectroscopy setups may require sophisticated instrumentation, including lasers, spectrometers, detectors, and optical components. The complexity of the instrumentation can make experimental setup, calibration, and data analysis challenging, requiring expertise and specialized training.

Despite these limitations, photoluminescence spectroscopy remains a valuable tool for studying the optical properties and electronic structure of materials in various fields of research, offering unique insights into luminescent phenomena and enabling advances in materials science, nanotechnology, photonics, and biomedicine.^[144,145]

9.4 Localized surface plasmon resonance

Localized surface plasmon resonance (LSPR) is a captivating optical phenomenon that arises when noble metal nanoparticles interact with electromagnetic radiation, resulting in strong absorption and scattering of light at specific wavelengths. The historical development of LSPR can be traced back to early investigations into the optical properties of metal colloids in the late 19th and early 20th centuries. However, the modern understanding and exploitation of LSPR emerged in the late 20th century with advancements in nanotechnology and the development of novel fabrication techniques for controlling the size, shape, and composition of metal nanoparticles.

The discovery of LSPR as a distinct phenomenon occurred through a combination of experimental observations and theoretical insights. In the 1960s and 1970s, researchers observed the unique optical properties of metal nanoparticles and their ability to absorb and scatter light with remarkable efficiency. Theoretical models based on the principles of electromagnetism, quantum mechanics, and classical electrodynamics were developed to explain these observations, leading to the recognition of

LSPR as a fundamental aspect of nanoparticle optics.

LSPR has since emerged as a highly versatile and valuable tool with a wide scope of applications across various disciplines. In sensing and detection, LSPR-based sensors offer unparalleled sensitivity and specificity for detecting analytes in biological, chemical, and environmental samples. In biomedical imaging, LSPR nanoparticles serve as contrast agents for visualizing biological structures and tracking molecular processes with high resolution and sensitivity. In catalysis and photonics, LSPR nanoparticles exhibit enhanced optical and catalytic properties, enabling new opportunities in energy conversion, photocatalysis, and plasmonic device fabrication.

Modern manufacturing techniques enable precise control over the size, shape, and composition of LSPR nanoparticles, allowing researchers to tailor their optical properties for specific applications. Methods such as colloidal synthesis,

lithography, and chemical vapor deposition facilitate the fabrication of LSPR nanoparticles with precisely engineered characteristics at scales ranging from individual nanoparticles to large-area arrays.

In summary, Localized Surface Plasmon Resonance represents a fascinating intersection of nanoscience, optics, and materials engineering, offering unprecedented opportunities for exploring and harnessing the unique properties of metal nanoparticles for a diverse range of applications in science, technology, and industry.

9.4.1 Instrumentation of LSPR

The LSPR spectroscopy typically involves several key components designed to measure the optical response of noble metal nanoparticles to incident light, as shown in Fig.18.

The key component of the instrument is described below:

a) Light source: Like all other spectroscopic techniques LSPR technique requires a light source which can provide a broad spectrum of wavelengths, typically in the visible or near-infrared range. Light sources may include light-emitting diodes, lasers, or broadband lamps.

b) Optical setup: The optical setup consists of lenses, mirrors, and optical fibers used to direct and focus the incident light onto the sample containing the noble metal nanoparticles. The setup may also include polarization optics to control the polarization state of the incident light.

c) Sample holder: The sample holder securely holds the sample containing the noble metal nanoparticles. It may be designed to accommodate different sample formats, such as liquid suspensions, thin films, or nanoparticle-deposited substrates.

d) Detector: A detector measures the optical response of the sample, typically in terms of absorbance, scattering, or intensity changes, as a function of wavelength or angle. Detectors may include photodiodes, PMTs, or CCDs.

e) Spectrometer or monochromator: A spectrometer or monochromator is used to separate the incident light into its constituent wavelengths, allowing the measurement of the LSPR response across a range of wavelengths. This component is crucial for obtaining the LSPR spectrum of the nanoparticles.

f) Data acquisition system: A data acquisition system records and processes the signals from the detector, allowing the measurement of the LSPR spectrum and the analysis of the optical response of the nanoparticles. This system may include analog-to-digital converters, signal amplifiers, and computer software for data analysis.

g) Temperature control system: Some LSPR spectroscopy setups include temperature control systems to maintain the sample at a constant temperature during measurements. This is particularly important for studies involving temperature-dependent changes in the optical response of the nanoparticles.

h) Alignment and calibration tools: Alignment and calibration tools are used to ensure the proper alignment of optical components and to calibrate the instrument for

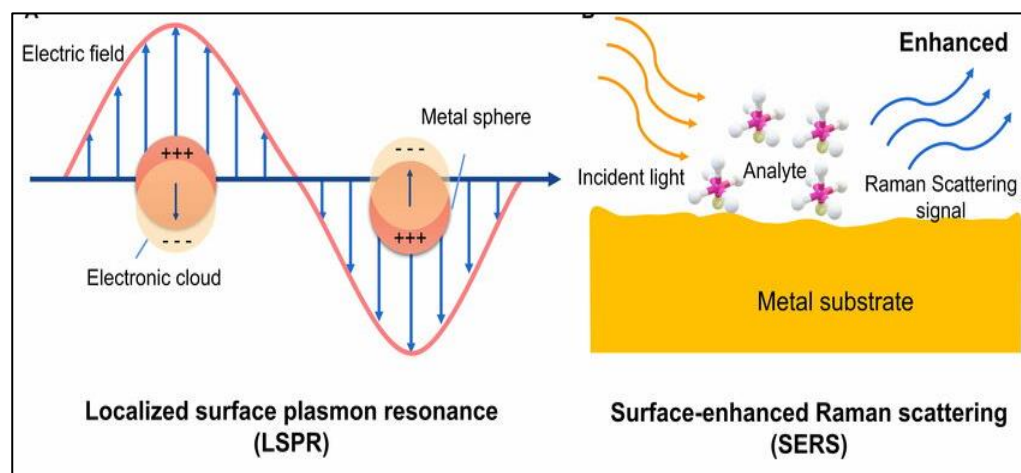


Fig. 18: A pictorial representation of localized surface plasmon resonance spectroscopy.

accurate measurements. This may involve the use of reference materials with known optical properties.

Overall, the instrumentation of LSPR spectroscopy is designed to provide precise control over experimental parameters, such as incident light intensity, wavelength, and polarization, enabling the measurement and analysis of the optical response of noble metal nanoparticles and their LSPR characteristics.

9.4.2 Principle of LSPR

LSPR arises from the collective oscillations of conduction electrons in noble metal nanoparticles when they interact with electromagnetic radiation, particularly light. The principle of LSPR is rooted in the interaction between the incident electromagnetic field and the free electrons in the metal nanoparticles, leading to resonant absorption and scattering of light at specific wavelengths.

The LSPR phenomenon can be understood through classical electrodynamics and quantum mechanics. At the heart of LSPR is the excitation of surface plasmon, which are collective oscillations of conduction electrons at the surface of a metal nanoparticle when the frequency of the incident electromagnetic field matches the natural frequency of these surface plasmons, resonance occurs, resulting in enhanced electromagnetic fields near the surface of the nanoparticle.

The resonance condition for LSPR can be described by the following principles:

a) Resonance condition: The resonance condition for LSPR occurs when the frequency of the incident light matches the natural frequency of the surface plasmon in the metal nanoparticles. This resonance condition is influenced by the size, shape, composition, and dielectric environment of the nanoparticles.

b) Enhanced electromagnetic fields: At the resonance wavelength, the interaction between the incident electromagnetic field and the surface plasmon leads to the concentration of electromagnetic energy near the surface of the nanoparticles. This results in enhanced absorption and scattering of light by the nanoparticles.

c) Wavelength dependence: The resonance wavelength of LSPR is highly sensitive to changes in the size, shape, and composition of the nanoparticles, as well as to changes in the refractive index of the surrounding medium. This wavelength dependence enables the tuning of LSPR by controlling these parameters.

d) Localized nature: Unlike surface plasmon polaritons in extended metal films, which propagate across the surface, LSPR involves localized oscillations of electrons confined to the nanoparticle surface. This localized nature of LSPR makes it particularly sensitive to the size and shape of the nanoparticles and allows for the manipulation of plasmonic properties at the nanoscale.

Thus, the principle of Localized Surface Plasmon Resonance involves the excitation of surface plasmons in noble metal nanoparticles by incident electromagnetic radiation, leading to resonant absorption and scattering of light at specific wavelengths. Understanding and controlling LSPR enables the design of plasmonic nanostructures with tailored optical properties for a wide range of applications in sensing, imaging, catalysis, and photonics.

9.4.3 Application of LSPR

LSPR has a wide range of applications across various fields due to its unique optical properties and sensitivity to changes in the local environment. Some of the key applications of LSPR are furnished below:

a) Sensing and detection: LSPR-based sensors offer high sensitivity and specificity for detecting analytes in biological, chemical, and environmental samples. LSPR sensors operate by monitoring changes in the resonance wavelength or intensity of plasmonic nanoparticles upon binding of target molecules to their surface. This enables label-free detection of biomolecules, DNA/RNA, proteins, small molecules, toxins, pathogens, and environmental pollutants with applications in medical diagnostics, food safety, environmental monitoring, and homeland security.

b) Biomedical imaging: LSPR nanoparticles serve as contrast agents for various biomedical imaging techniques, including

SERS, photoacoustic imaging, and dark-field microscopy. LSPR-based imaging enables high-resolution visualization of biological structures, detection of disease biomarkers, tracking of molecular processes, and monitoring of drug delivery in living organisms. Applications include cancer diagnosis, drug development, and regenerative medicine.

c) Catalysis and photocatalysis: LSPR nanoparticles exhibit enhanced catalytic activity due to their unique optical and electronic properties. They are employed as catalysts or photocatalysts for various chemical reactions, including hydrogen generation, pollutant degradation, and organic synthesis. LSPR-based catalysis offers opportunities for efficient and selective conversion of chemical feedstocks, energy production, and environmental remediation with applications in green chemistry and sustainable energy technologies.

d) Plasmonic devices and photonics: LSPR-based devices, such as plasmonic waveguides, nanoantennas, and metamaterials, are used to manipulate light at the nanoscale for applications in optical communications, data storage, and photonic circuits. These devices exploit the strong light-matter interactions enabled by LSPR to achieve novel functionalities, including waveguiding, light trapping, and spectral modulation, with potential applications in telecommunications, information processing, and integrated photonics.

e) Surface-enhanced spectroscopy: LSPR nanoparticles are employed as substrates for surface-enhanced spectroscopic techniques such as SERS and surface-enhanced fluorescence spectroscopy. These techniques enhance the vibrational or fluorescence signals of molecules adsorbed on the nanoparticle surface, enabling ultrasensitive detection and identification of trace analytes in complex samples with applications in chemical analysis, forensic science, and materials characterization.

Therefore, Localized Surface Plasmon Resonance offers a versatile platform for a wide range of applications in sensing, imaging, catalysis, photonics, and spectroscopy, providing unprecedented opportunities for exploring and harnessing the unique properties of plasmonic nanoparticles for diverse technological and scientific advancements.

9.4.4 Limitation of LSPR

While LSPR offers numerous advantages for various applications, it also has several limitations as mentioned below:

a) Spectral overlap and multiplexing: In multiplexed sensing applications where multiple analytes are simultaneously detected, spectral overlap between different LSPR peaks can complicate data interpretation and limit the number of analytes that can be distinguished. This can hinder the scalability and specificity of LSPR-based sensing platforms.

b) Sensitivity to environmental changes: LSPR is highly sensitive to changes in the local environment, including variations in temperature, pH, salt concentration, and refractive index. These environmental factors can affect the

resonance wavelength, intensity, and linewidth of LSPR spectra, leading to potential measurement errors or inaccuracies in sensing applications.

c) Nonspecific binding and interference: LSPR-based sensors may experience nonspecific binding of molecules to the nanoparticle surface, leading to false positive signals or reduced sensitivity. Additionally, interference from background signals or matrix effects in complex samples can obscure the detection of target analytes, limiting the reliability and accuracy of LSPR measurements.

d) Signal enhancement and reproducibility: Achieving consistent and reproducible LSPR signals across different batches of nanoparticles or experimental conditions can be challenging due to variations in nanoparticle size, shape, surface chemistry, and aggregation state. Controlling these parameters to ensure uniform signal enhancement and reproducibility is critical for reliable sensing and detection.

e) Biocompatibility and toxicity: Some noble metal nanoparticles used in LSPR applications, such as gold and silver nanoparticles, may exhibit cytotoxicity or induce immune responses in biological systems. Ensuring the biocompatibility and safety of LSPR nanoparticles for biomedical applications is essential to avoid adverse effects on cells, tissues, or organisms.

f) Cost and scalability: Fabricating and functionalizing LSPR nanoparticles with tailored properties can be expensive and time-consuming, particularly for large-scale or commercial applications. Improving the cost-effectiveness and scalability of LSPR-based sensing platforms is necessary for widespread adoption in industrial and clinical settings.^[146-148]

Despite these limitations, ongoing research efforts are focused on addressing these challenges and advancing LSPR technology to enhance its sensitivity, specificity, reliability, and applicability across diverse fields of research and technology.

9.5 Mössbauer spectroscopy

Mössbauer spectroscopy, named after its discoverer Rudolf Mössbauer, revolutionized the field of nuclear physics and solid-state chemistry by providing a precise tool to probe the behavior of atomic nuclei in materials. The technique emerged in the late 1950s and early 1960s, driven by Mössbauer's pioneering work. For this wonderful discovery, he was awarded the Nobel Prize in Physics in 1961. About a third of known elements, especially the heavier elements, when formed by radioactive decay of an isotope of the same or different elements, are initially produced in an excited nuclear state, after a very short delay, of the order of micro-second, the excited nucleus reverts to the ground state and emits energy of a very high frequency, usually in the γ region of spectrum. It is the study of γ emission and subsequently reabsorption which constitutes Mössbauer Spectroscopy.

The discovery of the Mössbauer effect occurred in 1958 when Mössbauer, then a young physicist, observed a peculiar phenomenon while studying the gamma-ray emission of ^{191}Ir

nuclei. He found that when these nuclei were embedded in a crystal lattice and subjected to gamma-ray radiation, they emitted gamma rays with extremely narrow energy linewidths, contrary to the expected Doppler broadening due to thermal motion. This groundbreaking discovery laid the foundation for Mössbauer spectroscopy, revealing the unique ability of certain atomic nuclei to absorb and re-emit gamma rays without recoil energy.^[149,150]

Mössbauer spectroscopy has since become an indispensable tool in various scientific disciplines, including physics, chemistry, materials science, and geoscience. Its scope encompasses the study of nuclear properties, crystallography, chemical bonding, oxidation states, magnetic interactions, and phase transitions in a wide range of materials, from metals and alloys to minerals, ceramics, and biological systems. Modern Mössbauer spectroscopy enables researchers to obtain high-resolution spectra with sub-meV energy resolution, allowing for precise determination of hyperfine interactions and electronic environments of nuclei.

The manufacture of Mössbauer spectroscopy instruments involves sophisticated instrumentation and technological advancements. Modern Mössbauer spectrometers typically consist of a gamma-ray source, a sample holder, a high-resolution gamma-ray detector, and specialized electronics for data acquisition and analysis. Gamma-ray sources such as ^{57}Co , ^{119}Sn , and ^{191}Ir provide monochromatic gamma rays for the spectroscopic measurements. The sample holder allows for the precise positioning and alignment of the sample concerning the gamma-ray source and detector. Advanced detectors, including high-purity germanium (HPGe) or scintillation coupled with multichannel analyzers, are employed to record the gamma-ray spectra with high sensitivity and energy resolution. Additionally, sophisticated data acquisition systems and computer software enable the acquisition, processing, and interpretation of Mössbauer spectra, facilitating the study of nuclear and solid-state phenomena with unprecedented precision and accuracy.

9.5.1 Instrumentation of Mössbauer spectroscopy

The instrumentation of Mössbauer spectroscopy involves several key components designed to generate, manipulate, and detect gamma rays emitted by Mössbauer-active isotopes in a sample. The pictorial image is displayed in Fig 19. The main components of the instrument are described as follows:

a) Gamma-ray source: Mössbauer spectroscopy requires a gamma-ray source that emits gamma rays with a narrow energy linewidth. Commonly used gamma-ray sources include radioactive isotopes such as ^{57}Co , ^{119}Sn , and ^{191}Ir , which undergo Mössbauer transitions. These sources are typically encapsulated in a matrix to prevent radiation exposure and ensure safety.

b) Sample holder: The sample holder securely holds the sample containing the Mössbauer-active nuclei in the path of the gamma rays. The sample holder allows for precise positioning and alignment of the sample with respect to the

gamma-ray source and detector. It may include provisions for temperature control to study temperature-dependent phenomena.

c) Gamma-ray detector: A gamma-ray detector measures the intensity and energy of gamma rays emitted by the sample. High-resolution detectors such as high-purity germanium or scintillation detectors coupled with multichannel analyzers are commonly used in Mössbauer spectroscopy to achieve excellent energy resolution and sensitivity. These detectors can distinguish between gamma rays of different energies, enabling the recording of Mössbauer spectra.

d) Collimation system: A collimation system is used to collimate the gamma rays emitted by the source and transmitted through the sample. Collimators help to define the geometry of the gamma-ray beam and minimize background noise, improving the signal-to-noise ratio and energy resolution of the spectroscopic measurements.

e) Mechanical drive system: Some Mössbauer spectrometers include a mechanical drive system to induce motion in the sample relative to the gamma-ray source or detector. This allows for the measurement of time-dependent phenomena such as recoilless nuclear resonance absorption, enabling the observation of spectra and hyperfine interactions.

f) Data acquisition system: A data acquisition system records and processes the signals from the gamma-ray detector. This system may include ADCs, signal amplifiers, and computer software for data analysis. Mössbauer spectroscopy software is used to acquire, analyze, and interpret spectra, extracting information about nuclear properties, chemical environments, and magnetic interactions in the sample.

Overall, the instrumentation of Mössbauer spectroscopy is designed to provide precise control over experimental parameters, generate and detect gamma rays with high sensitivity and energy resolution, and facilitate the acquisition and analysis of Mössbauer spectra for the study of nuclear and solid-state phenomena in materials.

9.5.2 Principle of Mössbauer spectroscopy

Mössbauer spectroscopy is based on the Mössbauer effect, a phenomenon observed when gamma rays are emitted and absorbed by atomic nuclei in a solid-state environment. The principle of Mössbauer spectroscopy revolves around the interaction between incident gamma rays and Mössbauer-active isotopes in a sample, resulting in characteristic energy shifts that provide detailed information about the nuclear and electronic environments of the nuclei.

The Mössbauer effect is observed in certain atomic nuclei with low-energy nuclear transitions and specific energy levels. These nuclei, referred to as Mössbauer-active isotopes, include isotopes of elements such as iron ^{57}Fe , tin, ^{119}Sn , and ^{57}Co , among others. When a Mössbauer-active isotope is embedded in a crystal lattice and subjected to gamma-ray radiation, it can undergo resonant absorption and re-emission processes without recoil energy.

The key principles of Mössbauer spectroscopy include:

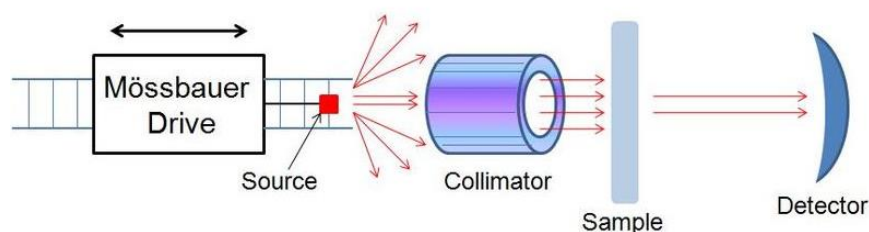


Fig. 19: A pictorial representation of Mossbauer spectroscopy.

a) Resonant absorption: In Mössbauer spectroscopy, the incident gamma rays have energies that match the energy difference between the ground state and excited state of the Mössbauer-active nucleus. When a gamma ray is resonantly absorbed by the nucleus, it excites the nucleus to an excited state without imparting any recoil energy to the lattice.

b) Recoilless emission: After a Mössbauer-active nucleus absorbs a gamma ray and transitions to an excited state, it can re-emit the gamma ray with high probability without experiencing significant recoil energy. This is due to the fact that in a solid-state environment, the recoil energy is shared among many lattice atoms, leading to a negligible change in momentum for the emitting nucleus.

c) Energy shifts: The energy of the gamma rays emitted by the Mössbauer-active nuclei is influenced by the local environment of the nuclei, including factors such as chemical bonding, lattice structure, oxidation state, and magnetic field effects. As a result, the Mössbauer spectrum obtained from the sample consists of characteristic peaks corresponding to different nuclear energy states and their interactions with the surrounding environment.

By analyzing the Mössbauer spectrum, researchers can obtain detailed information about the hyperfine interactions, electronic environments, oxidation states, and magnetic properties of the nuclei in the sample. Mössbauer spectroscopy is widely used in various scientific disciplines, including physics, chemistry, materials science, and geoscience, for studying nuclear and solid-state phenomena in a wide range of materials and systems.^[151,152]

9.5.3 Applications of Mössbauer spectroscopy

Mössbauer spectroscopy finds diverse applications across multiple scientific disciplines due to its ability to provide detailed information about the nuclear and solid-state properties of materials. Some key applications include:

a) Material characterization: Mössbauer spectroscopy is extensively used in materials science to characterize the structural, chemical, and magnetic properties of materials. It provides valuable insights into the oxidation states, coordination environments, and local symmetries of atoms in crystalline, amorphous, and nanoscale materials. Applications include the analysis of metals, alloys, minerals, ceramics, catalysts, and nanomaterials.

b) The chemical shift: It is known that a nucleus in chemical surroundings different from those of the source does not

absorb at the same frequency; this effect is referred to as a chemical shift or sometimes as an isomer shift, and an observed shift is usually reported in cm s^{-1} . Such a shift is illustrated in the Mössbauer Spectrum of ferro-cyanide ions. The main factor affecting the magnitude of the chemical shift is the electron density at the nucleus concerned, and since p, d, *etc.* orbital have zero density at their nucleic, it is s orbital density which, in turn, gives an estimate of the bond character of atoms or ions chemically attached to the Mössbauer nucleus. The main factor affecting the magnitude of chemical shift is the electron density at the nucleus concerned and since p and d orbitals have zero density at their nuclei, therefore s orbital density is important. Observation of the chemical shift then allows the measurement of relative electron density, which, in turn, gives an estimate of the bond character of atoms or ions chemically attached to the Mossbauer nucleus. An important in the determination of chemical shift is that the nuclei in the excited state have different radii from those of the ground state. They may be either smaller or larger. Measurement of change in nuclear size on excitation clearly has great importance for theories of nuclear structure. For instance, it is observed fact that in I^{129} , the excited nucleus is larger, whereas in the case of another isotope Ir^{127} , the ground state nucleus is larger in size. The nuclei are formed in an excited state by decay from the radioactive tellurium nuclei Te^{129} and Te^{127} .

c) Mineralogy and geochemistry: Mössbauer spectroscopy is employed in Earth and planetary sciences to study the composition, oxidation states, and magnetic properties of minerals, rocks, soils, meteorites, and lunar samples. It aids in the identification of mineral phases, the determination of mineralogical compositions, and the investigation of geological processes such as weathering, metamorphism, and impact cratering.

d) Quadrupole effect: This happens because the majority of Mossbauer nuclei have a non-zero spin and, further, most of them have a half-integral rather than integral spin. The spin of the excited state is invariably different from that of the ground state, and so it follows that either or both of the nuclear states must involve a spin greater than $\frac{1}{2}$, i.e., one or both will have a quadrupole moment, and this will interact with electric field gradient in the vicinity. The quadrupole effect can be described as an interaction between a nucleus with an asymmetric charge distribution and the surrounding electric field gradient. The interaction can be described by Spin Hamiltonian term. Some effects of quadrupole effects can be ascribed as (1) Nuclear

precession: The quadrupole interaction causes nuclear precession, which has multiple precession frequencies because the energy levels are not equally placed; (2) Nuclear quadrupole resonance: The nuclear quadrupole resonance is a technique that can characterize phase transition in solids. It is very sensitive to the nature and symmetry of bonding around the nucleus; (3) Quadrupole coupling in NMR: Quadrupole coupling imparts the broad resonance lines in NMR spectra. This is because a non-uniform electric field the technique that can characterize phase transition in solids. It is very sensitive to nature and symmetry of bonding around the nucleus; (4) Quadrupole Magnetic Field Effect: When the rotation is slow, electrons follow magnetic field lines. If the plasma rotation is fast, the plasma smears out of the cylinder.

e) Catalysis and materials chemistry: Mössbauer spectroscopy is used in catalysis research to study the coordination environments and electronic structures of catalytic centers in solid catalysts. It provides insights into catalytic mechanisms, active sites, and reaction intermediates, facilitating the design and optimization of catalyst materials for industrial applications such as petroleum refining, chemical synthesis, and environmental remediation.

f) Magnetism and spin dynamics: Mössbauer spectroscopy is a powerful tool for investigating magnetic properties and spin dynamics in materials. It enables the determination of magnetic ordering, magnetic phase transitions, and magnetic relaxation processes, providing fundamental insights into the behavior of spins in magnetic materials such as ferromagnets, antiferromagnets, and spin glasses.

g) Biological and environmental sciences: Mössbauer spectroscopy is applied in biological and environmental research to study metalloproteins, metalloenzymes, and metal-containing biomolecules. This spectroscopy is widely used in bioimaging chemistry, especially for the study of iron-containing proteins and enzymes such as iron-sulfur proteins, ferritins and heme including cytochrome. The oxidation state of iron can be determined by this technique. It elucidates the coordination chemistry, redox reactions, and catalytic activities of metal ions in biological systems, as well as their interactions with environmental pollutants, toxicants, and biomolecules. The vibrational spectra of Fe-enriched biomolecules can be acquired using nuclear resonance vibrational spectroscopy in which sample is scanned through a range of synchrotron-generated X-Rays, centered at the Mossbauer absorbance frequency.

h) Archaeometry and cultural heritage: Mössbauer spectroscopy is used in archaeology and cultural heritage studies to analyze archaeological artifacts, ancient pigments, and historical materials. It provides information about the composition, provenance, and degradation processes of cultural heritage objects, aiding in their conservation, restoration, and authentication.

Overall, Mössbauer spectroscopy offers a versatile and non-destructive tool for investigating the nuclear and solid-state properties of materials across a wide range of scientific

disciplines, contributing to advancements in materials science, earth sciences, chemistry, biology, and cultural heritage research.

9.5.4 Limitations of Mössbauer spectroscopy

While Mössbauer spectroscopy is a powerful analytical technique, it does have certain limitations as furnished below:

a) Isotope availability: Mössbauer spectroscopy requires the availability of Mössbauer-active isotopes, which have specific nuclear properties suitable for Mössbauer transitions. Not all elements have isotopes that exhibit the Mössbauer effect, limiting the range of elements that can be studied using this technique.

b) Isotope sensitivity: The sensitivity of Mössbauer spectroscopy is influenced by the natural abundance and concentration of the Mössbauer-active isotope in the sample. Samples with low concentrations of the Mössbauer-active isotope may require longer acquisition times or higher activities of the gamma-ray source to achieve sufficient signal-to-noise ratios.

c) Limited depth profiling: Mössbauer spectroscopy provides information about the nuclear and solid-state properties of materials near the surface of the sample. However, it has limited depth profiling capabilities and may not be suitable for studying buried interfaces, subsurface layers, or three-dimensional structures in materials.

d) Temperature dependence: Mössbauer spectroscopy measurements are often conducted at cryogenic or elevated temperatures to study temperature-dependent phenomena such as phase transitions, magnetic ordering, and lattice dynamics. However, temperature variations can affect the stability of the sample, alter the hyperfine interactions, and introduce thermal broadening in the Mössbauer spectra, complicating data interpretation.

e) Environmental sensitivity: Mössbauer spectroscopy is sensitive to changes in the local environment of the Mössbauer-active nuclei, including factors such as chemical bonding, lattice strain, and magnetic fields. External influences such as pressure, humidity, and irradiation can perturb the nuclear and electronic environments, affecting the accuracy and reproducibility of the spectroscopic measurements.

f) Complex data analysis: Interpretation of Mössbauer spectra often requires sophisticated data analysis techniques and theoretical models to extract meaningful information about hyperfine interactions, oxidation states, and magnetic properties of the nuclei in the sample. Analyzing complex spectra with overlapping peaks or multiple components may require advanced computational methods and experimental validation.

Despite these limitations, Mössbauer spectroscopy remains a valuable tool for studying the nuclear and solid-state properties of materials across various scientific disciplines, providing unique insights into atomic-scale interactions, electronic structures, and magnetic behaviors in materials.

10. Conclusion

Spectroscopic techniques play a pivotal role in the comprehensive analysis of nanomaterials, offering invaluable insights into their structural, optical, electronic, and magnetic properties. This review has highlighted the diverse array of spectroscopic methods, which are available for nanomaterial as well as bulk material characterizations.

Each spectroscopic technique offers unique advantages and capabilities, enabling researchers to probe different aspects of nanomaterials with high precision and sensitivity. UV-Vis provides valuable information about the optical properties and bandgap characteristics of nanomaterials, while FT-IR spectroscopy offers insights into their chemical composition and molecular structure. Raman spectroscopy is particularly useful for identifying vibrational modes and detecting structural defects in nanomaterials, while XPS and EELS provide detailed information about their elemental composition, chemical bonding, and electronic structure. Additionally, FMR spectroscopy offers a powerful means of investigating the magnetic properties of nanomaterials, including their magnetic anisotropy, spin dynamics, and magnetic interactions.

Moreover, the synergistic combination of multiple spectroscopic techniques allows for a more comprehensive and multi-dimensional analysis of nanomaterials, facilitating a deeper understanding of their fundamental properties and behavior. Furthermore, recent advancements in spectroscopic instrumentation, data analysis algorithms, and computational modeling techniques continue to enhance the capabilities and versatility of spectroscopic methods for nanomaterial analysis.

Thus, the spectroscopic techniques represent indispensable tools for characterizing nanomaterials across a wide range of disciplines, including materials science, chemistry, physics, biology, and engineering. As nanotechnology continues to revolutionize various fields and drive innovation in diverse applications, the continued advancement and integration of spectroscopic methods are essential for unlocking the full potential of nanomaterials and realizing their societal impact.

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Conflict of Interest

There is no conflict of interest.

Supporting Information

Not applicable.

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